Embracing Respiration-Induced Motion in Cardiovascular MRI

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Abstract

Respiratory motion is a well-known complicating factor in Magnetic Resonance Imaging (MRI), especially when imaging elements of the cardiovascular system. Periodic motion of the lungs during the respiratory cycle affects the torso and abdomen, which impacts imaging of the heart, aorta, liver, and other organs and vessels. The resulting motion artifacts, which appear in traditional Cartesian imaging as distinct replicas in the phase-encoding direction, negatively impact the quality of diagnostic information that can be gleaned from MR images. One common way to mitigate motion artifacts in clinical imaging is to instruct subjects to hold their breath for the duration of the acquisition. However, breath-holds can be infeasible for long acquisitions or for subjects who have difficulty holding their breaths. Alternatively, the acquisition can be gated such that data are only acquired during a single phase of respiration, but this can lead to considerable reductions in scan efficiency. Additionally, respiration has important physiological consequences, and there is merit to exploring the impacts of respiration and respiration-induced motion on the cardiovascular system.

In this thesis, I seek to implement and optimize methods for not only mitigating the effects of respiration in cardiovascular MRI, thereby enabling otherwise impossible imaging, but also to utilize information from the respiratory cycle to improve data efficiency, reduce scan time, and extract valuable diagnostic information. This will be performed in three different scientific contexts, each focusing on a different anatomical region: (1) validation and improvement of radial 2D Phase Contrast (2DPC) MRI for measuring pulse wave velocity in the aorta with sequential and simultaneous multi-slice acquisitions; (2) acceleration of

the acquisition of radial, free-breathing 4D flow MRI in the portal venous system; and (3) implementation of a 5D MRI approach for resolving cardiac and respiratory motion of the heart in radiation therapy treatment planning.

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Chapter 1: Introduction

1.1. Motivation

Magnetic Resonance Imaging (MRI) is a highly flexible tool for non-invasively acquiring diagnostic information that can be used to guide medical decision-making and treatment. However, there are a number of technical challenges to solve in the application of MRI, including the management of respiratory motion in cardiovascular MRI. Cardiovascular MR acquisitions are often time-resolved and typically require the collection of data over multiple cardiac cycles. As a result, periodic motion of the diaphragm and lungs introduces complex, non-rigid displacement of thoracic and abdominal structures, including the heart, aorta, and liver. These motion effects manifest as blurring, ghosting, and loss of spatial fidelity, ultimately compromising both qualitative visualizations and quantitative measurements derived from MRI data.

Traditional approaches to mitigate respiratory motion have largely relied on breath-holding techniques or respiratory gating. While breath-holding can effectively eliminate motion artifacts in short acquisitions, scan duration is fundamentally limited by the length of the breath hold and patient compliance. Many advanced cardiovascular MRI techniques, such as 4D flow MRI, require acquisition times that exceed practical breath-hold durations. This limitation is further exacerbated in patient populations with compromised respiratory function or inability to comprehend instructions. Retrospective gating strategies offer an effective alternative by recording a respiratory waveform throughout the acquisition and retroactively reconstructing only data acquired within the desired respiratory phase (typically end-exhalation). However, retrospective gating has its own trade-offs, including increased scan time and reduced data efficiency during reconstruction.

Even then, respiration is not merely a source of artifacts to be suppressed; it is a physiologically meaningful process that influences cardiovascular dynamics. Rather than solely attempting to eliminate respiration-induced artifacts, there is growing interest in investigating dynamics affected by respiratory motion. Achieving this requires advanced acquisition and reconstruction strategies capable of capturing motion-resolved information for both cardiac and respiratory motion.

The overarching motivation of this dissertation is threefold. First, I aim to optimize and implement MRI acquisition and reconstruction techniques that mitigate effects of respiratory motion through free-breathing acquisitions, enabling imaging in scenarios where traditional approaches may not be feasible. Second, I intended to improve and accelerate acquisitions by increasing data efficiency in respiratory gating. Third, I seek to enable respiratory-resolved imaging for investigating respiration-induced motion and physiological effects. This will be performed in three different scientific contexts, each focusing on a different anatomical region: (1) validation of improved radial, free-breathing 2D Phase Contrast (2DPC) and simultaneous multi-slice (SMS) sequences for measuring pulse wave velocity in the aorta; (2) acceleration of the acquisition of radial, free-breathing 4D flow MRI in the portal venous system with phase unwrapping and motion correction strategies; and (3) implementation of a 5D MRI approach for resolving cardiac and respiratory motion of the heart in radiation therapy treatment planning.

1.2. Chapter Summary

The chapters of this dissertation are organized as follows:

Chapter 2: Background

The fundamentals of MRI and its applications to cardiovascular imaging are reviewed. This includes the basics of MR data acquisitions and image reconstructions with velocity mapping based on phase contrast imaging.

Chapter 3: Validation of Free-Breathing Radial 2D Phase Contrast MRI

for Pulse Wave Velocity Measurements in an Aortic Flow Phantom

Radially undersampled single slice and SMS 2DPC MRI is compared to standard Cartesian 2DPC and ground truth ultrasound methods for assessing PWV in a controlled flow phantom study.

Chapter 4: Feasibility of Free-Breathing Radial 2D Phase Contrast MRI for In Vivo Aortic Pulse Wave Velocity Measurements

The radial, free-breathing 2DPC sequence is applied to the Longitudinal Impact of Fitness and Exercise (LIFE) study, which explores the effects of habitual fitness on cardiovascular health in elderly subjects at risk for Alzheimer's Disease, to measure PWV. I perform an updated analysis of acquired data and discuss the challenges of 2DPC-based PWV measurement methods.

Chapter 5: Respiratory Effects on Aortic Pulse Wave Velocity Measurements with a Radial Simultaneous Multi-Slice 2D Phase Contrast Acquisition

A feasibility study for assessing PWV in different modes of respiration with the radial, free-breathing SMS 2DPC sequence is performed in a small cohort of healthy volunteers. PWV is assessed for the inhalation and exhalation periods of free-breathing as well as under metronome-paced tachypnea.

Chapter 6: Optimizing Venc in Radial Free-Breathing 4D Flow MRI for the Portal Venous System

Velocity encoding (venc) optimization strategies are explored in the application of 4D flow MRI for imaging the hepatic portal vein in subjects at risk for development of gastroesophageal varices. A 4D Laplacian method for unwrapping aliased velocity data is applied to multiple datasets acquired with different vencs to identify the optimal venc to use for future acquisitions.

Chapter 7: Iterative Motion Correction for Improved Efficiency of Radial Free-Breathing 4D Flow MRI

An iterative motion correction (iMoCo) method is implemented for improving the efficiency of radial, free-breathing 4D flow MRI of the portal venous system. The iMoCo method is applied to retrospectively subsampled datasets to demonstrate its ability to improve image quality while reducing scan time.

Chapter 8: A Flexible 5D Cardiac MRI Reconstruction Framework for Motion Characterization in Radiation Therapy

A 5D MRI reconstruction framework for assessing cardiac substructure motion in radiation therapy is implemented. Cardiac- and respiratory-resolved images are acquired in a small cohort of healthy volunteers and left ventricular motion is analyzed.

Chapter 9: Conclusion

The conclusions and contributions of this work are summarized and future avenues of research that can be explored are discussed.

Chapter 2: Background

This chapter discusses the fundamentals of Magnetic Resonance Imaging and its applications to cardiovascular imaging, particularly the use of phase contrast sequences for dynamic 2D and 3D velocity mapping of blood flow. Additionally, techniques for improving and accelerating image acquisition are discussed.

2.1. Magnetic Resonance Imaging

Magnetic Resonance Imaging (MRI) is a noninvasive imaging technique that produces detailed images of the human body by exploiting the magnetic properties of atomic nuclei. All elementary particles have an intrinsic property known as spin that is restricted to discrete values of positive or negative half-integers. When subatomic particles combine to form nuclei, conservation of angular momentum mandates that the resulting nuclei have a spin equal to the sum of the spins of the component particles. If the resulting spin is non-zero, then the nuclei will have a net magnetic moment. In the absence of an external magnetic field, the magnetic moments of individual nuclei will be randomly oriented, resulting in an overall magnetically neutral state. However, when a magnetic field B_0 is applied, the magnetic moments partially align with or against the field creating an equilibrium state with the net magnetization vector, M_0, in the direction of B_0. This causes the individual magnetic moments to precess around the direction of B_0 at the Larmor frequency ω_0:

$$\omega_0 = \gamma B_0 \tag{2.1}$$

where γ is the gyromagnetic ratio. The gyromagnetic ratio is specific to every material and for hydrogen, which is the primary nucleus imaged with MRI due to its abundance in both water and fat, is equal to 42.58 MHz/T. The full description of the mechanisms that govern MRI excitation and signal generation can only be described by quantum mechanics. However, most of the mechanisms relevant for hydrogen imaging can also be described with classical electromagnetism, so I limit the review to a basic introduction. A more detailed description of MR physics can be found elsewhere [1, 2].

To generate a measurable signal, MRI systems apply a radiofrequency (RF) pulse, B_1, at the Larmor frequency for hydrogen which perturbs the net magnetization vector M_0 and tips it away from the direction of the B_0 field [1]. The degree to which M_0 is tipped is known as the flip angle α and depends on the duration and strength of the RF pulse. After excitation, the individual magnetic moments begin to precess coherently in the transverse plane which induces a voltage in receiver coils that can be read out as the MR signal. After the RF pulse ends, M_0 returns to its equilibrium state through two decay mechanisms, namely longitudinal and transverse relaxation [1]. Longitudinal relaxation represents the transfer of energy from the precessing spins to the external environment over time which results in the component of M_0 parallel to B_0, M_z, decaying according to Equation 2.2 where $T1$ is a tissue-specific time constant that reflects the time it takes for $1 - \frac{1}{e}$ (roughly 63%) of this energy transfer to occur [1].

$$M_z(t) = M_0 \left(1 - e^{\frac{-t}{T_1}}\right) \tag{2.2}$$

On the other hand, transverse relaxation describes the spontaneous dephasing of the individual magnetic moments due to interactions with the local environment. This results in the component of M_0 perpendicular to B_0, M_{xy}, decaying according to Equation 2.3 where $T2$ is another tissue-specific time constant that reflects the time it takes for $\frac{1}{e}$ (roughly 37%) of this dephasing to occur [1].

$$M_{xy}(t) = M_0 \left(e^{\frac{-t}{T_2}}\right) \tag{2.3}$$

However, the transverse decay is also affected by additional factors such as B_0 field inhomogeneities and magnetic susceptibility differences which result in M_{xy} decaying faster than expected with $T2$ effects alone. The contributions from these additional effects are described with the time-constant $T2_i$ resulting in a new time constant $T2^*$ which sums the contributions from both $T2$ and $T2_i$ [1].

$$\frac{1}{T2^*} = \frac{1}{T2} + \frac{1}{T2_i} \tag{2.4}$$

Once transverse decay has begun, it can actually be reversed via the application of additional RF pulses or magnetic gradient fields, which alter the strength of the main B_0 field. This has the effect of inverting the dispersion of phases, causing them to re-converge at a later time which generates a secondary echo of the signal [1]. The time at which this rephasing occurs is defined as the echo time (TE). This can then be repeated multiple times to generate multiple echoes. The time between repeated excitations is defined as the repetition time (TR). By varying the flip angle α, TE, and TR, different image contrasts can be generated based on the relative contributions of $T1$, $T2$, and $T2^*$ effects from each tissue type. This series of RF pulses, gradients, and signal readouts is known as a pulse sequence.

The two most important pulse sequences that form the basis of most other sequences are the spin echo (SE) and gradient recalled echo (GRE) sequences [1, 3]. The SE sequence consists of a 90° excitation pulse followed by a 180° pulse at time $\frac{TE}{2}$. This generates an echo at time TE which can then be read out as a signal. In contrast, GRE sequences consist of only a single 90° excitation pulse. The echo is instead generated via a gradient field that dephases and then rephases the spins (see Figure 2.1). While this approach is more sensitive to T_2^* effects than the SE sequence, it is also faster and more versatile. All of the work presented in this thesis is performed with sequences based on this fundamental GRE sequence [1–3].

Once a signal has been generated, spatial encoding is required to localize generated RF signals in space. This is accomplished through the use of time-varying gradient fields in each direction—$G(x, t)$, $G(y, t)$, and $G(z, t)$—which add a spatial dependence to the B_0 field [1]. Typically, a slice selection gradient is played out in the z direction with $G(z)$ such

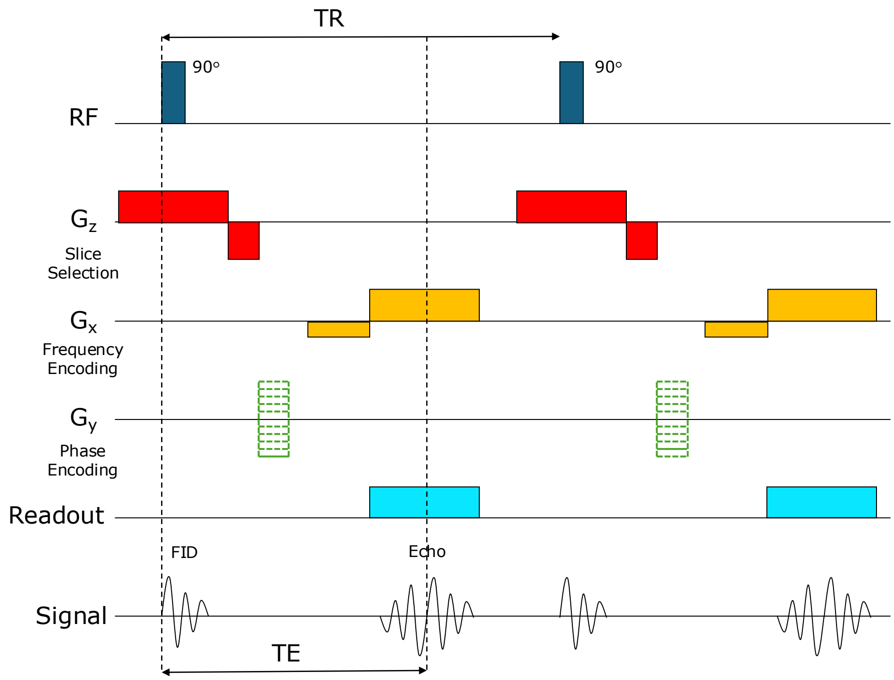

Figure 2.1: Diagram of a gradient recalled echo (GRE) pulse sequence. A 90° radiofrequency (RF) pulse is used to tip the net magnetization vector into the transverse plane generating a free induction decay (FID) signal. A slice selection gradient (G_z) is played to excite a slice of the imaging volume. The phase encoding gradient (G_y) is incremented with each repetition time (TR) to fill in lines of k-space. The frequency encoding gradient (G_x) generates an echo at the echo time (TE) and reads out the signal

that only a narrow range of frequencies that fall within the bandwidth of the B_1 pulse are excited. The excited region can either be a narrow slice (2D acquisition) or a large slab (3D acquisition). Next, frequency encoding and phase encoding steps are applied via $G(x,t)$ and $G(y,t)$ to localize signals during readout within the x-y plane [1]. Frequency encoding applies an additional location dependent gradient which causes frequencies of the spins along that axis to vary spatially. Phase encoding is then applied such that the phase of the spins vary spatially along that specific axis. As there are a limited number of phases available for a given frequency, the phase encoding step is repeated multiple times for each frequency encode. Thus, spins can be localized in the x-y plane according to the unique frequency and phase of the readout signal (note that signal acquisitions need not necessarily be in the x-y plane; slice select gradients can be played along any axis, even double oblique orientations which is particularly common in cardiac imaging).

The output of the signal from each frequency and phase encoding step is a complex number representing a specific frequency. Image reconstruction is most easily accomplished using the properties of the Fourier Transform which maps frequencies in Fourier space (k-space) to real space [2]. By filling k-space with repeated frequency and phase encoding steps, a Fourier transform can be applied to reconstruct the signal representation in image space, which is a spatial map of the MR signal that depends on a combination of parameters including the proton density, T1 and T2 relaxation times, and sequence parameters. The resulting image is also complex valued with a magnitude and phase, though the phase image is often discarded as the magnitude image contains most of the anatomical information.

2.1.1 Radial Acquisitions

During acquisition, the order and direction in which k-space is sampled can be chosen arbitrarily, though limited by the gradient performance. The most common method of sampling k-space is the Cartesian approach, where k-space is sampled line-by-line in a rectilinear fashion (Figure 2.2A). This approach has many benefits, one of which is that the data are conveniently organized in an evenly spaced grid-like format that is easy for computers to process and apply the 2D Fast Fourier Transform (FFT) [1]. However, there are alternative acquisition trajectories that carry various pros and cons compared to Cartesian sampling. Radial sampling is one of these methods.

With radial sampling, frequency and phase encoding are applied simultaneously along every axis such that k-space is traversed along spokes that pass through the center of k-space (Figure 2.2B) [4]. Each spoke represents a projection of the object at a different angle, and successive spokes are rotated to achieve full k-space coverage. Because the center of k-space is sampled repeatedly, radial acquisitions inherently emphasize low spatial frequency information, which governs overall image contrast and signal intensity [4].

One of the primary advantages of radial sampling is its resilience to motion. In Cartesian

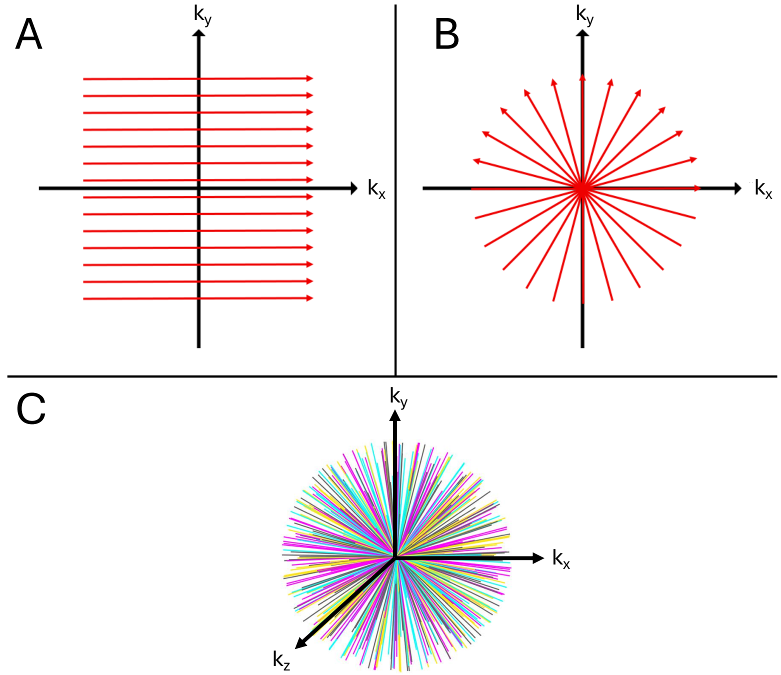

Figure 2.2: (A) 2D Cartesian sampling of k-space. Frequency encode direction is left to right while phase encode direction is top to bottom. (B) 2D radial sampling of k-space. Frequency and phase encoding directions are rotated simultaneously to acquire projections through the origin. (C) 3D radial sampling. Slice select, frequency, and phase encode directions are rotated simultaneously to acquire projections through the origin.

imaging, motion typically results in coherent ghosting artifacts along the phase-encoding direction [5]. In contrast, motion in radial acquisitions leads to more diffuse, incoherent streaking artifacts and blurring due to signal averaging of the low spatial frequencies as seen in Figure 2.3 [4–7]. As such, radial sampling is particularly well suited for applications involving respiratory or cardiac motion. This in turn also enables better support for constrained reconstruction techniques as the incoherent motion artifacts enable improved sparsity constraints (see Section 2.3) [7]. Radial sampling also allows for greater flexibility in how data are sorted during reconstruction compared to Cartesian imaging. If projections are acquired in pseudo-random order, such as with a golden angle or bit-reversed view ordering [7, 8], projections can be arbitrarily grouped into dynamic frames with reasonable coverage of k-space and reconstructed as images. This allows arbitrary gating based on physiological signals (see Section 2.2), as well as retrospective subsampling of full datasets to simulate shorter scan times or to remove projections compromised by bulk motion [4, 7, 9–11].

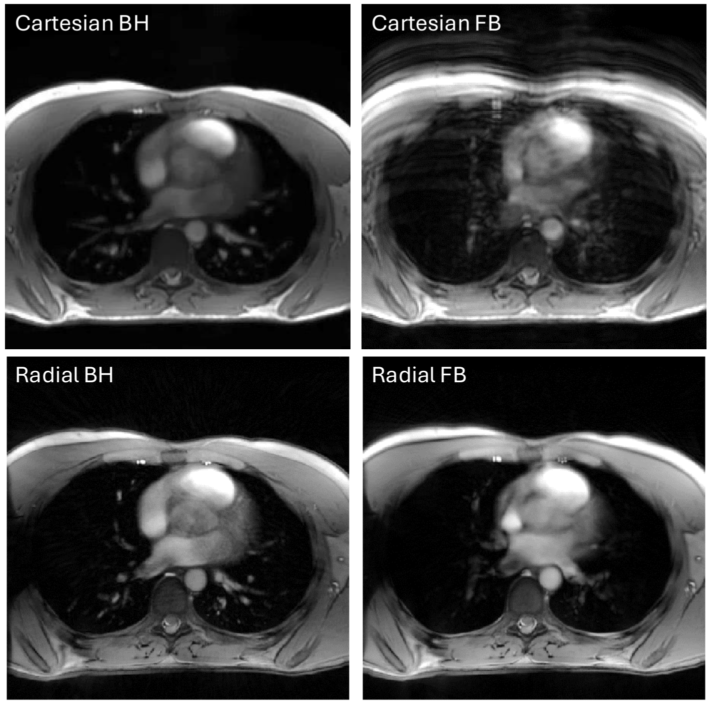

Figure 2.3: Breath-hold and free-breathing acquisitions with Cartesian and radial sampling. Free-breathing results in motion artifacts that manifest as ghosting artifacts in the phase encode direction (top-bottom) in the Cartesian acquisition and an almost imperceptible increase in pseudo-noise all across the image in the radial acquisition

Despite these advantages, radial sampling also presents some technical challenges. The non-uniform sampling density—characterized by oversampling at the center of k-space and undersampling at higher spatial frequencies—requires the use of density compensation during reconstruction to prevent the so-called "halo artifact" where low frequency signals dominate the image [12]. Furthermore, radial data cannot be directly reconstructed using a standard FFT and instead require interpolation (gridding) onto a Cartesian grid which increases computational complexity and thus reconstruction time [12]. Streaking artifacts may also become prominent in cases of severe undersampling, the presence of high-contrast structures, or low signal-to-noise ratio (SNR) [7]. Additionally, eddy currents induced by fast gradient switching can introduce phase errors for radial acquisitions necessitating the use of trajectory correction schemes [13].

2.1.2 Parallel Imaging

Parallel imaging (PI) is an advanced acquisition strategy that accelerates data acquisition and reconstruction by exploiting the spatial sensitivity profiles of multichannel RF receiver coils. Typically, full sampling of k-space according to the Nyquist criterion is required to reconstruct an image without aliasing artifacts, which can result in relatively long acquisition times [1]. PI addresses this limitation by utilizing information from multiple receiver coils in parallel, each of which sees a portion of the full FOV with different sensitivity profiles. This means that k-space can be undersampled and the missing spatial information can be recovered by combining information from each coil.

Two widely used PI techniques are Sensitivity Encoding (SENSE) and Generalized Autocalibrating Partially Parallel Acquisitions (GRAPPA) [14–16]. In SENSE-based approaches, image reconstruction is performed in the image domain by solving a system of equations that relate the aliased pixel intensities to coil sensitivity maps. This requires accurate estimation of coil sensitivities, which are typically obtained through separate calibration scans or by reconstructing low-pass filtered, low-resolution images such that the contributions from each coil are spread out [15]. In contrast, GRAPPA operates in k-space by synthesizing missing data points from acquired neighboring samples using calibration data, thereby avoiding the need for explicit sensitivity maps during reconstruction [17]. Both approaches enable substantial reductions in scan time, depending on the number of coils and the geometry of the acquisition. Additionally, PI can be combined with other acceleration techniques such as constrained reconstructions [14].

While PI is highly useful for accelerating acquisitions, it also has a few drawbacks. High undersampling leads to noise amplification, commonly quantified by the geometry factor (g-factor), which depends on coil configuration, acceleration factor, and imaging geometry [14]. Higher acceleration rates generally result in increased noise and potential residual aliasing artifacts if coil sensitivities are insufficiently distinct [14]. Additionally, PI may reduce SNR, as fewer data points are acquired overall [14]. Lastly, PI with non-Cartesian trajectories can also be computationally intensive. These limitations necessitate careful optimization of acquisition parameters and coil design to balance scan time with image quality.

2.1.3 Simultaneous Multi-Slice

An alternate application of PI is in simultaneous multi-slice (SMS) acquisitions, which aim to accelerate imaging by exciting and acquiring multiple slices at the same time. In conventional 2D MRI, if multiple slices are desired, they must be excited and read out sequentially, with total scan time scaling linearly with the number of slices. SMS overcomes this limitation by exciting multiple slices at the same time and then recovering each image in a method similar to regular PI.

To describe this further, let $A(t)$ denote a standard complex RF waveform with a slice selective gradient and $P(t)$ be a modulation function that determines the slice position $\Delta\omega$ and phase ϕ [18]. A single slice-selective RF pulse is then described by

$$RF(t) = A(t)P(t) = A(t)e^{i(\Delta\omega+\phi)} \tag{2.5}$$

For a multiband excitation with N slices (also referred to as slice acceleration or SMS factor), individual RF pulses can be summed together to form [18].

$$RF_{MB}(t) = A(t)\sum_{n=0}^{N_s} e^{i(\Delta\omega_n+\phi)} \tag{2.6}$$

However, with this excitation, the received signal from a multichannel coil array now represents a superposition of signals from all excited slices. If reconstructed without modification, this leads to complete slice aliasing, where anatomical structures from different slices overlap indistinguishably [18]. To resolve this, SMS reconstructions leverage the same principle underlying traditional PI reconstructions. Because each coil exhibits a unique spatial sensitivity pattern, the combined signal encodes spatial information that can be used to separate the aliased slices during reconstruction [18]. Methods such as SENSE or GRAPPA can be extended to the slice dimension, effectively treating the slice direction as an additional encoding axis [14, 18].

A key development that enabled more robust SMS reconstruction is the Controlled Aliasing in Parallel Imaging Results in Higher Acceleration (CAIPIRINHA) technique [19]. Named for a Brazilian cocktail, this method is performed by adding a slice dependent phase modulation ϕ_n to the composite RF pulse which allows for easier disentangling of adjacent slices. This reduces the local overlap of aliased voxels, improving the conditioning of the reconstruction problem. The final images can then be recovered by multiplying the data by the complex conjugate phase pattern [18, 19].

The CAIPIRINHA approach was then extended to radial acquisitions by Yutzy et al [20]. This was done by taking the slice-specific phase offset applied to alternating lines in the Cartesian acquisition and applying them instead to alternating radial projections [19, 20]. In this approach, the phase modulation that results in a spatial shift in the image in Cartesian acquisitions instead results in constructive and destructive interference in a radial acquisition [20]. Multiplication by the complex conjugate phase pattern modifies which slice is recovered (constructive interference) and which slices become incoherent (destructive interference) [20].

Further refinement of the CAIPIRINHA technique led to the development of blipped-CAIPI [21]. In blipped-CAIPI, gradient blips with additional blipped rewinder gradients are applied along the slice-select direction during the phase-encoding steps. These blips introduce the same phase shifts as the RF phase modulations used in the CAIPIRINHA method but without the large overall phase accrual, which enables higher SMS factors with fewer overlap artifacts [21]. This method can also be applied to radial SMS acquisitions.

While SMS acquisitions enable high slice accelerations with no SNR penalties, there is a limit to the number of slices that can be acquired. Summation of multiple RF pulses results in the total power of the pulse increasing [18]. At high SMS factors, this can exceed specific absorption rate (SAR) limits which limit the amount of RF energy that can be deposited in an object. However, there are methods to reduce peak amplitude scaling and enable higher SMS factors [22, 23].

2.2. Physiological Motion

As discussed previously, motion during acquisition can result in artifacts. While bulk motion can be avoided by instructing the subject not to move, other types of motion, such as physiological motion, are unavoidable for imaging certain anatomy. Respiratory motion is the most common type of physiological motion encountered in thoracic and abdominal imaging. It is most commonly managed via breath-holding, however this places an upper limit on the acquisition time. Most sequences have to be accelerated to fit within the time span of an average breath-hold (often 10–20 seconds). Multiple sequential breath-holds are used for longer multi-slice 2D scans with increased volumetric coverage, but breath-hold inconsistencies can result in slice misalignment [24]. However, breath-hold length varies between subjects and can be reduced (<10 seconds) or even impossible in certain populations with breathing difficulties.

The other common type of physiological motion is cardiac motion and comes into play when imaging the heart and vascular system. Unlike respiratory motion, cardiac motion cannot be suspended even temporarily. MR is a relatively slow modality and most types of acquisitions do not have the temporal resolution necessary to generate full, real-time images in the space of a single R-R interval [25]. As such, alternate methods of acquiring data without motion artifacts must be utilized.

The most common method is a segmented acquisition with physiological gating, which spreads the acquisition of each image over multiple cardiac cycles and requires synchronization of the acquisition with a physiological gating signal (see Figure 2.4) [26]. In the case of cardiac imaging, an electrocardiogram (ECG) or photoplethysmogram (PPG) can be utilized for obtaining a cardiac signal; for respiratory motion, abdominal bellows and 1D navigator acquisitions have been used to acquire a respiratory signal [27]. Gating an acquisition can be performed two ways: prospectively, where the acquisition triggers off of the physiological waveform to acquire data during the desired phase of motion, or retrospectively, where data are acquired continuously alongside the physiological signal and then sorted during image reconstruction [28]. Prospectively gated acquisitions have the advantage of improved data efficiency during reconstruction, however, they can only acquire data during a single predefined phase of a physiological cycle and considerable amounts of time are spent not acquiring data. On the other hand, retrospectively-gated reconstructions acquire data for all phases of a physiological cycle and offer increased flexibility for reconstruction. Yet, they are more sensitive to irregular signals such as those caused by arrhythmias and have poorer data efficiency if only a single motion state is desired (respiratory gated acquisitions are often reconstructed only in end-exhale resulting in $> 50\%$ of data being discarded).

2.3. Constrained Reconstruction

Traditionally, MR acquisitions have been viewed as limited by the Nyquist sampling criterion to generate images free from undersampling errors, i.e., if insufficient k-space samples are acquired to cover the whole FOV, artifacts will occur [1]. However, full sampling, especially for 3D acquisitions, often requires clinically infeasible long scan times. The solution to this challenge is in the application of constrained reconstruction techniques, which utilize a priori knowledge of image sparsity to reconstruct images at sub-Nyquist sampling rates.

More explicitly, a constrained reconstruction takes the form of the inverse problem

$$\min_{x} \; \|F \cdot S \cdot x - y\|_2^2 + \lambda \, R(x) \tag{2.7}$$

where F is the nonuniform Fourier transform, S are coil sensitivity maps, x is the estimated image, y is the acquired k-space data, λ is a regularization parameter, and R is the enforced constraint [29]. The form of R in Equation 2.7 varies depending on the type of constraint

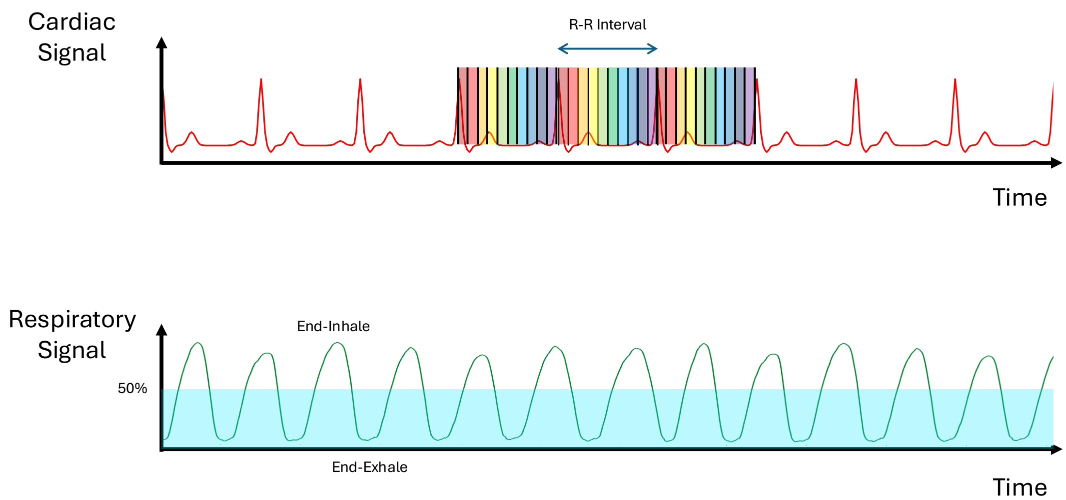

Figure 2.4: (Top) A cardiac waveform over time. The cardiac waveform is made up of multiple R-R intervals. Each individual R-R interval can be divided into separate bins and then data from each bin (denoted by color) can be combined to form a single cardiac frame averaged over multiple R-R intervals. (Bottom) A respiratory waveform over time, where the peaks are end-inhale and troughs are end-exhale. Shaded region indicates a threshold operation such that only data below the threshold is used during reconstruction. Data from each respiratory period is combined in this operation to form an image averaged over multiple respiratory periods in the thresholded respiratory phase.

being enforced.

One example of a constrained reconstruction technique is compressed sensing (CS) [30]. In CS, the constraint is that the data are sparse in some transformed domain and redundancies can be exploited to recover the original image, i.e., $R(x) = \|\Psi x\|_1$ where Ψ is a sparsifying transform (e.g., wavelets) [30]. Another type of constrained reconstruction is a local low-rank (LLR) approximation, which exploits the rank of the acquired dataset [31, 32]. For an LLR reconstruction, the assumption is that there are spatial and temporal redundancies across local patches of the dataset, which can be approximated as the product of lower-rank matrices, thereby reducing the complexity of the original matrix and approximating a solution, i.e., $R(x) = \sum_p \|\mathrm{Mat}(x_p)\|_*$ where Mat is a tensor unfolding operation along either a spatiotemporal dimension and $*$ is the nuclear norm. With either method, an iterative reconstruction can be utilized to solve Equation 2.7 and recover a higher-quality image from a dataset undersampled well below the Nyquist criterion, reducing the amount of data needed to be acquired and the time taken to acquire it [29]. However, the trade-off is that the reconstruction times for these iterative methods can be considerably longer than the traditional FFT-based reconstructions. Additionally, care must be taken to select the optimal regularization parameter. Too much regularization can lead to spatial or temporal smoothing [30].

2.4. Phase Contrast MRI

Phase Contrast MRI (PC) is a well-established for MRI-based motion encoding, particularly for the quantitative assessment of blood velocity [33]. As mentioned previously, the phase of the images is often discarded as it does not contain anatomical information. However, PC imaging is an exception as it utilizes bipolar gradient pulses to encode motion information into the phase of the complex signal [33].

The phase accumulated by a spin due to a time-varying gradient $\vec{G}(t)$ is given by

$$\phi(\vec{r},t) = \int_0^{\mathrm{TE}} \gamma \Delta B \, dt + \int_0^{\mathrm{TE}} \gamma \vec{G}(t) \cdot \vec{r}(t) \, dt \tag{2.8}$$

where $\phi(\vec{r},t)$ is the phase, TE is the echo time, ΔB is the contribution of local magnetic field inhomogeneities, and $\vec{r}(t)$ is the position of the spin as a function of time [34, 35]. The first term evaluates to a constant that can be described as the background phase ϕ_0. For the second term, the position function can be expanded with a Taylor series expansion

$$\vec{r}(t) = \vec{r}_0 + \vec{v}t + ... \tag{2.9}$$

where the ellipsis are higher order motion terms that can typically be neglected in blood velocity mapping due to minimal effects [33]. This can then be substituted back into Equation 2.8 to produce

$$\phi(\vec{r}, t) = \phi_0 + \gamma \vec{r}_0 \int_0^{\mathrm{TE}} \vec{G}(t)\, dt + \gamma \vec{v} \int_0^{\mathrm{TE}} \vec{G}(t) t\, dt \tag{2.10}$$

The second and third terms of Equation 2.10 represent the 0^{th} and 1^{st} moments of the gradient represented as M_0 and M_1. If the gradient is designed such that total area under the gradient waveform is 0, i.e., a bipolar gradient with equal but opposite lobes, then M_0 is nulled which eliminates position dependence. The phase then becomes sensitive only to velocity of the spins in the direction of the spatial gradient via the first gradient moment M_1 [34, 35]:

$$\phi(\vec{r}, t) = \phi_0 + \gamma \vec{v} M_1 \tag{2.11}$$

The background phase term ϕ_0 poses a problem, however. It is impossible to distinguish between phases due to motion and unknown background phases caused by acquisition imperfections, field inhomogeneities, magnetic susceptibility, physiological effects, etc., so a second acquisition must be recorded such that there are two measurements for the two unknown phase contributions [34]. Most commonly, this is done by playing the bipolar gradients in reverse order immediately after the first set and then subtracting (see Figure 2.5) [35]. The resulting phase difference is then given by

$$\Delta\phi = \gamma \vec{v}(M_{1,2} - M_{1,1}) = \gamma \vec{v} \Delta M_1 \tag{2.12}$$

Solving for velocity gives

$$\vec{v} = \frac{\Delta\phi}{\gamma \Delta M_1} \tag{2.13}$$

The value of ΔM_1 functions as a controllable parameter via the gradient strength and time duration that scales the value of the phase to the velocity [34]. More specifically, the maximum velocity that can be encoded in the phase, or velocity encoding parameter (venc),

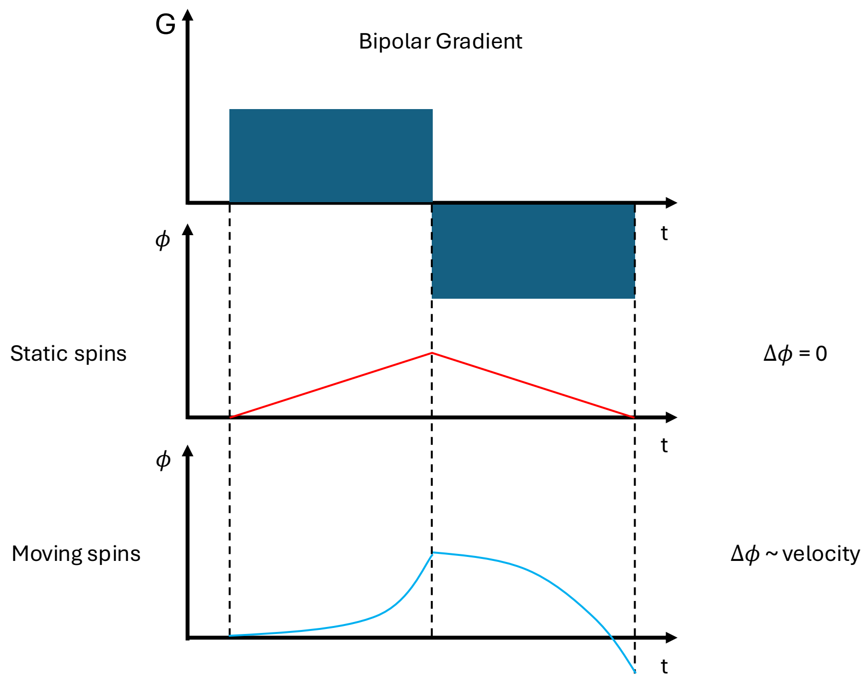

Figure 2.5: (Top) The design of a time-varying bipolar gradient for phase contrast imaging. (Middle) The phase evolution over time, ϕ, for a static spin. As the gradient has equal but opposite order and the position of a static spin does not change, the net phase $\Delta\phi$ is 0. (Bottom) Phase evolution over time for a spin moving with constant velocity in the spatial direction that the bipolar gradient is played out. Exposure to the bipolar gradient will result in a position dependent phase accumulation that linearly grows with gradient strength and time duration.

is

$$\text{venc} = \frac{\pi}{\gamma \Delta M_1} \tag{2.14}$$

However, since phases are constrained to values of $[-\pi, \pi]$, if the measured velocities are higher than the set venc, the phase will wrap around to the minimum velocity (Figure 2.6). This is known as velocity aliasing [34]. The simplest solution to this problem is to set the venc higher than the maximum potential velocity. However, this is typically not known a priori, especially in diseased states such as vessel stenoses or valve insufficiencies, where blood velocity may be abnormal [36]. High vencs come with the disadvantage of a reduced velocity-to-noise ratio (VNR) so it is desirable to set a venc just above the maximum velocity to optimize VNR.

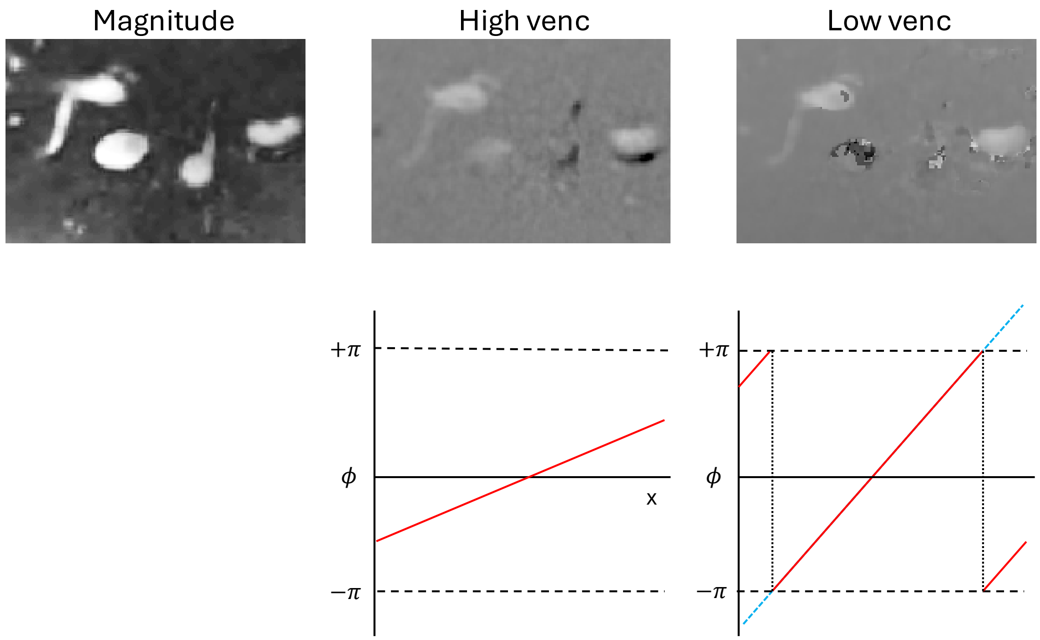

Figure 2.6: Depictions of velocity aliasing in a phase contrast (PC) MRI acquisition. (Left) Magnitude image shows the anatomical region with blood vessels. (Middle) Phase image acquired with a high velocity encoding setting (venc). Measured velocities are well below the venc, so no aliasing is present. However, velocity-to-noise ratio is low which makes distinction of some vessels difficult. (Right) Phase image acquired with a low venc. True velocity exceeds the venc, so phase values wrap beyond $\pm\pi$ resulting in aliasing artifacts. However, VNR is improved relative to the high venc image.

2.4.1 2D Phase Contrast Imaging

The method described previously for generating velocity-encoded images is known as 2-point encoding and is primarily used for PC imaging [35]. Through-plane velocity maps are acquired with bipolar gradients played out in the slice encoding direction and dynamic images acquired with cardiac gating [25]. Both magnitude and phase images are reconstructed: magnitude images provide the anatomical context and are often used for segmentation, while phase images contain the velocity information. Additionally, complex difference images or angiograms are often derived from the magnitude and phase images according to

Equation 2.15

$$CD = M\left|\sin\left(\frac{\Delta\phi}{2}\right)\right| \tag{2.15}$$

where CD and M are the complex difference and magnitude images, respectively [35]. This produces a representation of vascular structures where static tissue is suppressed, which is ideal for segmentation. Once the vessel lumen is segmented, velocity waveforms can be extracted across time, and volumetric flow rates can be calculated by integrating velocity over the cross-sectional area (Figure 2.7).

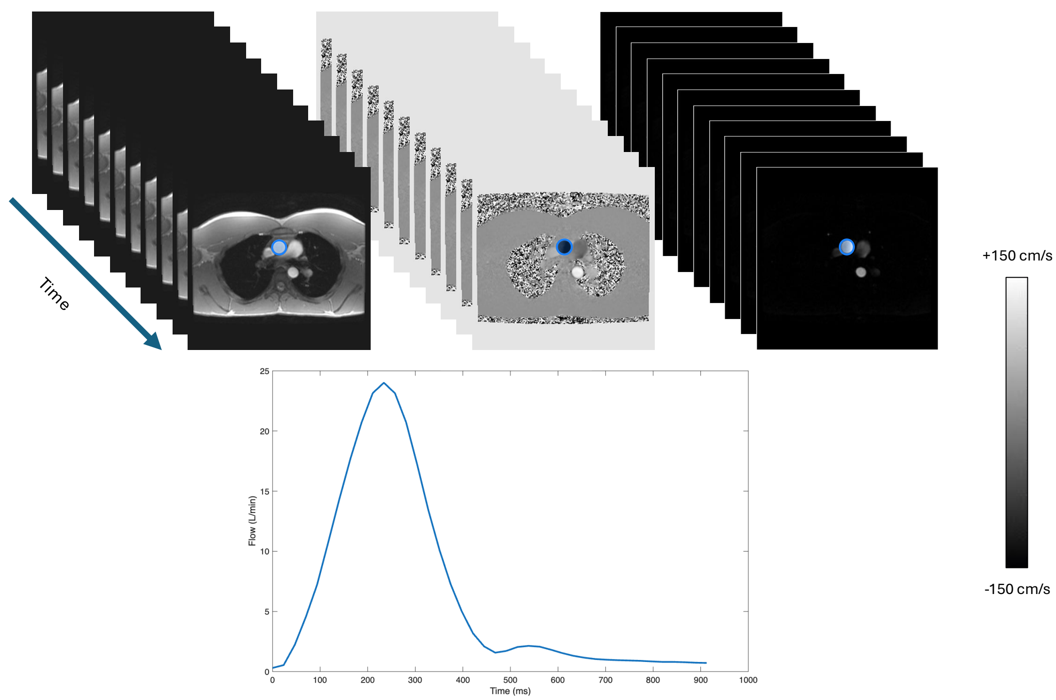

Figure 2.7: (Top) Magnitude (left), phase (middle), and complex difference (right) images from a time-resolved 2D phase contrast acquisition. Pixel intensity of the phase image indicates velocity, where dark pixels are motion into the image and bright pixels are motion out of the image. (Bottom) Flow curves can be generated by integrating velocity over the area of the cross-section. Ascending aorta is segmented in blue and the measured flow curve is plotted.

2.4.2 4D Flow Imaging

Four-dimensional flow Magnetic Resonance Imaging (4D flow MRI) is an extension of phase contrast MRI that enables comprehensive, three-dimensional assessment of blood flow [36, 37]. In this technique, velocity is encoded in all three spatial directions in one acquisition through the use of multiple bipolar gradients repeated across each encoding direction to reconstruct full velocity vectors (see Figure 2.8) [36, 37]. A flow compensated reference encoding must also be acquired, but there are alternate encoding schemes that do not require flow compensation in each individual direction such as 4-point balanced encoding and 5-point encoding [38].

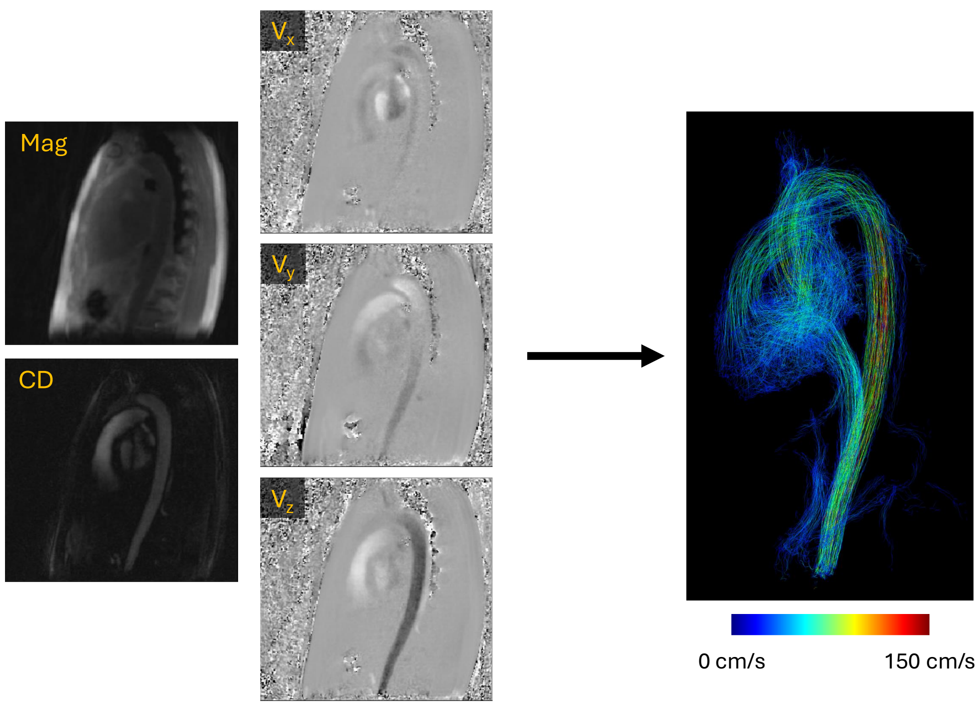

Figure 2.8: Example images of a PCVIPR 4D flow acquisition in the heart. Images shown are a magnitude image (Mag), complex difference angiogram (CD), and phase images for velocity in x, y, and z (V_x, V_y, V_z). Images are used to generate a 3D hemodynamic visualization with calculated streamlines showing the path of the spins.

A key advantage of 4D flow MRI is its ability to characterize complex hemodynamics that cannot be adequately captured with single-plane techniques. Because the velocity field is captured for an entire volume, flow can be retrospectively analyzed at any location without the need for precise slice prescription during scanning. It also enables detailed visualization of multidirectional flow phenomena such as vortices, helices, and turbulent jets. In addition to qualitative visualization, the technique supports quantitative analysis of a wide range of hemodynamic parameters, including volumetric flow rates, peak velocities, wall shear stress, vorticity, and pulse wave velocity [36, 37].

However, despite its versatility, 4D flow MRI is associated with several technical challenges. The acquisition is inherently more time-intensive than 2D methods due to the need for volumetric coverage and multiple velocity encodings, especially if performing cardiac gating [36]. Furthermore, spatial and temporal resolution are often lower than those achieved with 2D acquisitions, potentially affecting the accuracy of derived measurements. Similar to the 2D approach, it is also vulnerable to velocity aliasing and background phase errors [37].

Many of the acceleration techniques described above can be applied to 4D flow MRI to speed up acquisitions. The Cardiovascular and Hemodynamics Imaging Research Program (CHIRP) at the University of Wisconsin-Madison has developed a 4D flow sequence called Phase Contrast Vastly undersampled Isotropic PRojection imaging (PCVIPR) which incorporates radial undersampling, parallel imaging, iterative and constrained reconstructions to acquire high resolution data in clinically feasible scan times [10]. This sequence forms the basis for a substantial portion of the work presented in this thesis.

Chapter 3: Validation of Free-Breathing Radial 2D Phase Contrast MRI for Pulse Wave Velocity Measurements in an Aortic Flow Phantom

Recently, the CHIRP group at UW Madison introduced a novel, free-breathing radial single slice and simultaneous multi-slice (SMS) phase contrast sequence for measuring aortic pulse wave velocity (PWV) [39]. However, the SMS implementation had several shortcomings, including a limitation to the number of slices that could be acquired, a hard-coded slice spacing, and limited reconstruction framework. In this chapter, I present an improved radial SMS PC sequence that overcomes these limitations. Additionally, I aim to employ an improved phantom design to simulate more realistic flow dynamics in the aorta and perform more accurate ground truth measurements to better validate measurements made with the two radial 2DPC sequences. This work resulted in 1 first author abstract [40].

3.1. Introduction

One of the most clinically relevant biomarkers for assessing cardiovascular health is arterial stiffness, as it is a vital early predictor of cardiovascular disease [41]. In the early stages of cardiovascular diseases, vessel wall thicknesses increase without necessarily altering vessel diameters via the Glagov phenomenon leading to increased blood Pulse Wave Velocity (PWV) [42]. This relationship is described by the Moens-Korteweg equation

$$PWV = \sqrt{\frac{Eh}{2r\rho}} = \sqrt{\frac{1}{\rho D}} \tag{3.1}$$

where E is the elastic modulus of the vessel wall for a given wall thickness h, vessel radius r, and fluid density ρ, and D is the distensibility coefficient of the vessel [43, 44]. As vessel distensibility is difficult to measure directly, it can be inferred from the inverse relationship with PWV which is more easily measured. The clinical standards for measuring PWV are through the application of arterial pressure wires or applanation tonometry, but the former is an invasive procedure and hence rarely used while the latter is prone to measurement errors [45, 46].

In recent years, MRI has emerged as a promising avenue for measuring aortic PWV. MR-based PWV measurement methods typically rely on time-shift calculations where flow curves are measured at different points along the vessel and the PWV is found by utilizing Equation 3.2

$$PWV = \frac{\Delta d}{\Delta t} \tag{3.2}$$

where Δd is the path length along the aortic centerline between two points and Δt is the temporal difference between the flow curves [42]. Various approaches for MR-based PWV measurements have been proposed, most notably 1D Fourier velocity-encoded M-mode measurements and 4D flow MRI [47, 48]. However, there are trade-offs with both approaches. 1D Fourier methods have very high temporal resolutions but require matched angiograms for vessel length calculations and are unable to handle tortuous vessels. 4D flow provides inherent anatomical information, but has poor temporal resolution and long scan times. One commonly used research approach is the use of 2D Phase Contrast (2DPC) imaging [47, 49]. Yet, there are challenges associated with 2DPC PWV measurements, such as limited temporal resolution due to the need for breath-hold acquisitions. This challenge is particularly amplified in some elderly or diseased patient demographics who may be incapable of holding their breath [50]. The CHIRP group at the University of Wisconsin-Madison recently introduced a novel radial free-breathing 2DPC sequence to address these issues [39]. This sequence enables the capture of flow information without breath-holds which enables PWV measurements in subjects who are unable to perform breath-holds. However, 2DPC-based PWV measurements also typically require the acquisition of 2 or more planes along the vessel path which are sensitive to physiological inconsistencies between sequential acquisitions such as changes in heart rate. For this reason, the CHIRP group also introduced a simultaneous multi-slice (SMS) 2DPC sequence to allow acquiring multiple slices under identical physiological conditions [39]. However, while these sequences have been used in proof of concept studies in human volunteers, thorough validations of this approach have not been conducted [40, 51–54].

Initial work on validation of the introduced sequences was performed by Grant Roberts via construction of an aortic flow phantom for in vivo testing [39]. The goal was to simulate realistic flows through the aorta and use an ultrasound flow probe to measure the flow as ground truth and then calculate ultrasound-derived ground truth PWVs for comparison with the MR-derived PWVs. However, there were a number of challenges with the initial model that compromised validation efforts. For one, the properties of the phantom material and diameter of the vessel made it impossible to attach the ultrasound probe to the locations at which MR flow measurements were to be made. As such, the only method to measure ground truth ultrasound values was to attach additional tubing to the inlet and outlet of the phantom. It was observed that there were considerable differences in the shape and temporal shifts between MR and ultrasound-derived waveforms leading to large differences in PWVs, and it was hypothesized that this construction choice was responsible. Additionally, all arterial branches originating from the aorta in the original segmentation were removed to simplify model construction. This had the effect of modifying the shape and amplitudes of the flow waveforms and potentially increasing the effect of wave reflections as the pulse wave hit the plastic connector at the outlet. Due to these confounding factors, considerable upgrades were made to the construction of the model to improve ground truth measurements with the ultrasound flow probe.

The primary motivation of this work is to validate the developed retrospectively-gated radial, 2D phase contrast single slice and SMS sequences in this updated, more physiologically realistic aortic flow phantom. Traditional, sequential Cartesian 2DPC images are acquired along with sequential single-slice radial images and a radial SMS image. Flow rates and

PWVs are calculated for each MR acquisition type with a custom analysis framework. Max flow rate and calculated PWVs are then compared to PWVs calculated with ground truth flow measurements collected with an ultrasound flow probe.

3.2. Materials & Methods

Simultaneous Multi-Slice Sequence

The radial 2D phase contrast sequence was implemented in the EPIC pulse sequence programming environment (GE Healthcare, Waukesha, WI) with support for golden-angle sampling, iterative reconstructions, and 2-point velocity encoding [7]. The radial simultaneous multi-slice sequence then extended this sequence with support for multi-slice excitations and slice-dependent phase modulation. However, in the preliminary version of this SMS sequence, the number of acquired slices, N_s, and slice separation distance, d_s, were hard coded to values of 2 and 158 mm, respectively, to simplify acquisition and reconstruction. This proved to be infeasible for more complex studies, so the sequence was updated to allow for acquiring an arbitrary N_s and d_s, defined through the use of research control variables. The location of excited slices were arranged evenly around isocenter parallel to the direction of B_0 (the z-axis) according to Equation 3.3

$$\begin{gathered} Z(n) = n \cdot d_s - \frac{d_s}{2(N_s - 1)} \\ \text{for} \quad n \in \{0, 1, \dots, N_s - 1\} \end{gathered} \tag{3.3}$$

where $Z(n)$ is the location of slice n in mm relative to isocenter. Radial CAIPI phase blipping as described by Setsompop et al. [21] was implemented with the slice-specific phase blips ϕ_n constructed according to Equation 3.4

$$\phi_n = \frac{2n - N_s + 1}{N_s}\pi \tag{3.4}$$

and applied to every nth projection to induce the desired phase cycling patterns. A multiband radiofrequency excitation (RF) pulse was constructed by summing N_s complex-valued excitation bands with the slice-specific phase blips ϕ_n as described by Equation 3.5

$$RF(t) = A(t) \sum_{n=0}^{N_s-1} e^{i(\Delta\omega_n + \phi_n)} \tag{3.5}$$

where $A(t)$ is a complex RF waveform and $\Delta\omega_n$ is the frequency of excited slice n. As the SMS acceleration factor utilized for this study was limited to 3, RF amplitude reduction methods as described in Section 2.1.3 were not required, and standard summation was utilized to produce the multiband RF pulse.

To reconstruct SMS images, a custom Python-based reconstruction framework with support for GPU-acceleration based on the SigPy package was developed (available at: `https://github.com/uwmri/flow_recon/tree/sms`) [55]. The original reconstruction framework used for this SMS sequence utilized a standard parallel imaging based reconstruction (PILS) with slices solved for independently by applying the conjugate of the CAIPI phase blip pattern during reconstruction [16]. To improve upon this and enable constrained reconstructions for highly undersampled acquisitions, an iterative conjugate-gradient SENSE (CG-SENSE) reconstruction was implemented [15]. This reconstruction framework was modified according to Yutzy et al. to simultaneously solve for all slices in each iteration [20]. Equation 3.6 describes the inverse problem to be solved where

$$\min_x \frac{1}{2}\|DCFSx - y\|_2^2 + \frac{\lambda}{2}\|x\|_2^2 \tag{3.6}$$

where D is the density compensation factor, C are the conjugate phase blips for each slice, F is the nonuniform Fourier transform, S are multichannel coil sensitivity maps, x is the estimated image, y is the acquired k-space data, and λ is the regularization parameter. Coil sensitivity maps were generated by reconstructing low-pass filtered, low-resolution images with conjugate phase blips applied per slice.

Phantom Construction

The anatomical basis for the phantom model came from a contrast-enhanced MR angiogram of a 27-year-old male volunteer. The angiogram was segmented in Mimics (Materialise Inc., Leuven, Belgium) from the ascending aorta to the renal arteries using a manual intensity threshold. In this updated model, the arterial branches were re-added to simulate a more physiologically realistic splitting of the flow and reduce the effect of wave reflections. The resulting isosurface was manually smoothed before being imported into 3–Matic software (Materialise Inc., Leuven, Belgium), where it was meshed and wrapped to further smooth the surface. Additionally, the thickness of the phantom wall was increased to 2.75 mm from the original 2 mm to improve the optical properties of the material and allow for direct ultrasound flow measurements at the regions of interest (ROIs) which was not possible on the original model. Figure 3.1 shows a comparison between the original phantom and the updated model.

The resulting model was fabricated using stereolithography on a Formlabs Form 4 3D printer with Elastic 50A V2 Resin as the base material (Formlabs, Utrecht, Netherlands). The phantom was cured under ultraviolet light at 70 °C for 30 minutes to enhance mechanical stability. The model was then connected to the rest of the flow circuit using plastic connectors at the inlet (40 mm length), outlet (60 mm length), and at the arterial branches (30 mm), all of which were sealed with Dowsil 732 silicone (Dow Inc, Midland MI, USA). Segments of distensible Tygon tubing were attached to the inlet and outlet connectors (S3 E-3603, 0.75-inch diameter) which served as locations for reference flow measurement into and out of the model. In the original model, these segments were the only locations where the ultrasound flow probes could be placed which compromised the validation effort (locations 1 and 3 in Figure 3.1B).

A pulsatile displacement pump (Model PD-1100 BDC Laboratories, Wheat Ridge, CO, USA) was connected to the model using 30 ft of 0.75-inch PVC tubing (MSC Industrial Supply, MO, USA). Additionally, compliance chambers were added downstream of both

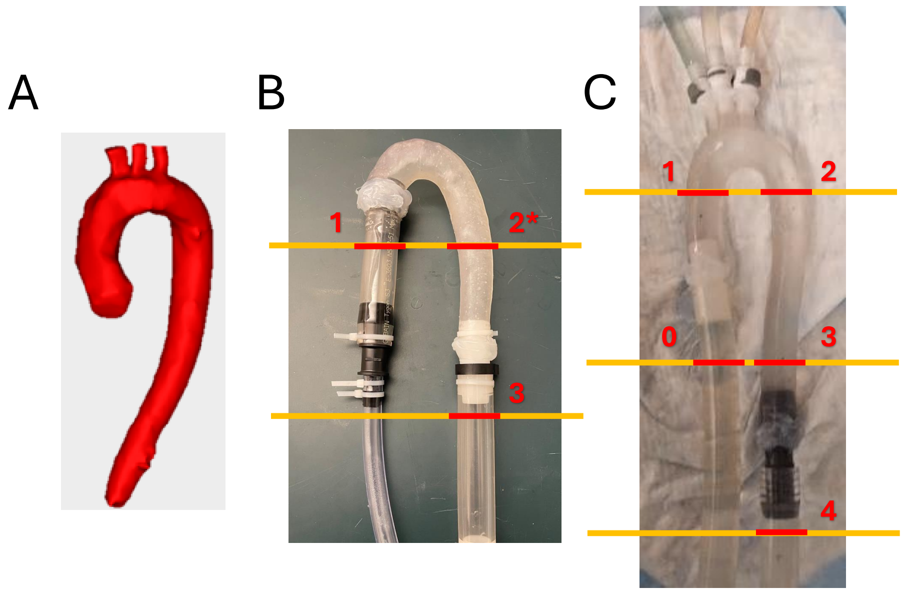

Figure 3.1: (A) Rendered 3D model of the aorta segmented from a contrast-enhanced MRA from a 27-year-old, male volunteer. (B) The original phantom model. (C) The updated phantom model with re-added arterial branches and extended aortic root and abdominal aorta. This model allowed for direct ultrasound measurements at any desired location along the aorta except the regions where connectors were located within the lumen. Yellow lines indicate the locations of MRI scan prescriptions. Numbers indicate locations of flow measurements performed with MRI and ultrasound (Note: location 2* in B was measured with only MRI as the ultrasound probe could not be attached directly.)

the arterial branches and abdominal outlet to modulate flow wave amplitudes. Resistors were connected in sequence with the compliance chambers to further adjust the properties of the flow waves. A full diagram of the flow circuit set up can be seen in Figure 3.2. Approximately 10 liters of tap water doped with 7 mL of gadobenate dimeglumine (Bracco Diagnostics, Milan, Italy) were added to the flow circuit.

Ground truth testing of the flow phantom was performed in the MRI control room before being moved into the MRI bore. Ultrasound flow measurements were taken with a clamp-on 16-PXL ultrasound flow probe and TS410 acquisition module (Transonic Systems Inc., Ithaca, NY, USA) that samples at a rate of 5000 Hz. This flow probe was connected directly to the flow pump allowing for simultaneous measurement and recording of the flow data

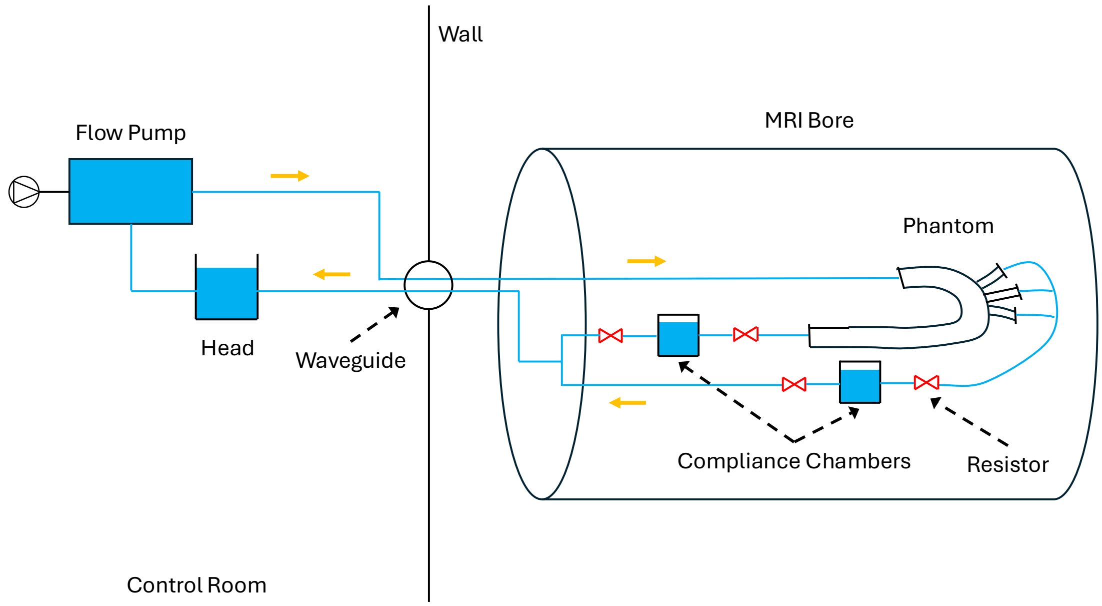

Figure 3.2: Diagram of the flow circuit used with the developed phantom. Flow pump and compliance head were placed in the MRI control room while the tubing was fed through the waveguide and connected to the phantom on the scanner table. Red hourglasses indicate locations of resistors to manage flow waveform.

and motor signal. Additionally, the flow probe was calibrated to the custom material of the phantom by measuring multiple flow rates on a cylindrical test section and performing linear regression with the known reference flow values. The pump was set to produce a waveform with a 30% systolic duration and flow rate of approximately 3 L/min and ensuring a roughly 60/40 flow split between the main aorta and the arterial branches, respectively. For ground truth flow measurements, 3 different heart rates were simulated with the flow pump: 60 bpm. 75 bpm, and 90 bpm. Flow measurements were recorded sequentially with a single ultrasound probe at 5 ROIs around the phantom where ROI 0 is the phantom inlet, ROI 1 is the ascending aorta (AAo), ROI 2 is the descending aorta (DAo), ROI 3 is the abdominal aorta (AbdAo), and ROI 4 is the outlet of the phantom (see Figure 3.1C). Data were recorded for 10 seconds at each ROI.

MR Imaging Protocol

After ground truth measurements were performed with the ultrasound flow probe, the phantom was moved onto the MRI table along with the auxiliary compliance chambers and resistors. All scans were acquired on a 3T SIGNA Premier system (GE Healthcare, Waukesha, WI) with a 30-channel chest coil plus a 40-channel posterior table coil. Saline bags were arranged around the phantom to allow for background phase correction. The cardiac gating signal for the acquisition was generated via a custom device built to translate the pump motor signal into electrical pulses that could be read by the scanner's ECG leads. A series of ungated anatomical balanced steady state free precession (bSSFP) product sequences (axial, coronal, sagittal) were acquired to aid with aortic centerline tracing and vessel path length calculation. Next, a series of Cartesian 2DPC product sequences, radial 2DPC research sequences, and a radial SMS 2DPC research sequence were acquired. For the Cartesian and radial sequences, 3 slices were prescribed to match the locations of the ultrasound measurements with the superior slice (aortic arch) capturing ROIs 1 and 2, the intermediate slice (abdominal aorta) capturing ROIs 0 and 3, and the inferior slice (outlet) capturing ROI 4 (Figure 3.1). For the SMS sequence, a single acquisition with 3 slices was acquired with the spacing between slices calculated to match the locations of the prescribed Cartesian and single slice radial sequences. Additionally, the number of projections acquired in the SMS acquisition was increased compared to the single slice radial to reduce the effect of slice aliasing that occurs with high undersampling. These set of scans were repeated three times with simulated heart rates of 60 bpm, 75 bpm, and 90 bpm. Scan parameters for each acquisition type are listed in Table 3.1.

Cartesian acquisitions were automatically reconstructed on the scanner post-acquisition by the GE Healthcare product reconstruction engine. The radial 2DPC and SMS 2DPC scans were reconstructed offline with custom reconstruction frameworks built in C++ and Python respectively. Both offline reconstructions were performed using an iterative CG-SENSE reconstruction for 20 iterations to reduce undersampling artifacts with included Maxwell

Table 3.1: Acquisition and reconstruction parameters for Cartesian, radial, and SMS 2D phase contrast scans as well as anatomical images (axial, sagittal, coronal) used for centerline tracing.
*sagittal acquisition had slice thickness of 3.2 ms
+axial acquisition had reconstructed spatial resolution of 0.70×0.70 mm^2

Parameter	**Anatomical**	**Cartesian 2DPC**	**Radial 2DPC**	**SMS 2DPC**
TR / TE (ms)	3.0 / 1.3	5.1 / 3.1	7.5 / 4.3	8.4 / 5.0
Flip angle	45°	25°	25°	10°
Views per segment	1	4	N/A	N/A
Sampling	Cartesian	Cartesian	Golden-Angle	Golden-angle
Acquired FOV (cm)	48	36	32	32
Slice thickness (mm)	8^*	6	5	5
Recon. matrix	512×512	256×256	256×256	256×256
Recon. spatial res. (mm^2)	$0.94 \times 0.94^+$	1.41×1.41	1.40×1.40	1.40×1.40
Recon. cardiac frames	40	40	40	40
Venc (cm/s)	N/A	150	150	150
Velocity encoding	N/A	2-point	2-point	2-point
Number of projections	N/A	N/A	2,000	5,000
Scan time (s)	8–13	12–20 s	30	75

term phase offset corrections and gradient calibration to correct trajectory errors [10, 15, 56]. Time-resolved and time-averaged velocity and magnitude datasets were obtained for both reconstructions.

Pulse Wave Velocity Analysis

Acquired ultrasound and MRI data were analyzed in MATLAB (Mathworks, Natick, MA) to compare ultrasound-derived vs MRI-derived PWVs. MRI data was analyzed using a custom-built analysis platform developed by Grant Roberts and myself in MATLAB (available at: `https://github.com/uwmri/PWV_2DPC`) [57]. This tool consists of two different Graphical User Interfaces (GUIs), one for aortic centerline tracing to calculate Δd and the second for time-shift calculations between velocity waveforms for Δt.

In the first step, axial and sagittal balanced steady-state free precession (bSSFP) images are loaded into the centerline tracing GUI. The GUI then guides users through placing fiducial points along the aorta across all slices for both axial and sagittal image sets (see Figure 3.3). Next, 2DPC images of the aortic arch and abdominal aorta are imported, and

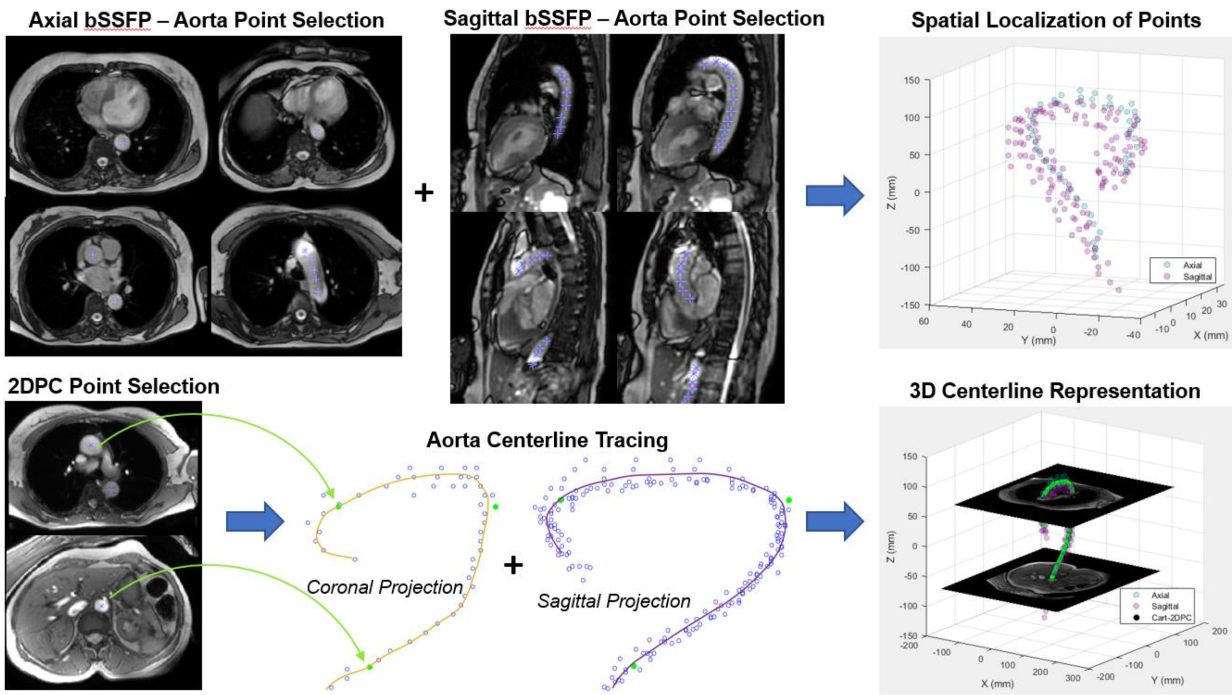

Figure 3.3: Example of aortic centerline tracing with the custom-built MATLAB GUI for pulse wave velocity analysis. Points are placed along the aorta in axial and sagittal anatomical images. Vessel region of interests (ROIs) are chosen on 2D phase contrast images. A curve tracing through coronal and sagittal projections of placed points is then drawn by the user. Interpolation is then performed to generate the final centerline. Images shown are from in vivo data acquired in volunteer studies used to test and develop the GUI. Anatomical images of the phantom were utilized for in vitro centerline tracing. Figure courtesy of Grant Roberts.

additional markers are placed at the centers of the ascending, descending, and abdominal aorta corresponding to the flow measurement ROIs. All marker coordinates are transformed from image space into a physical coordinate system referenced to the scanner isocenter using DICOM header information. Sagittal and coronal projections of these coordinates are then generated, and the GUI guides users through tracing an aortic centerline passing through the fiducial points and vessel ROI locations. The resulting centerline is interpolated to 150 evenly spaced points using modified Akima interpolation, after which distances between the flow measurement ROIs are automatically calculated.

Once the centerline is exported from the centerline tracing GUI, it can then be imported into the time-shift calculation GUI. 2DPC images are imported into the time-shift GUI

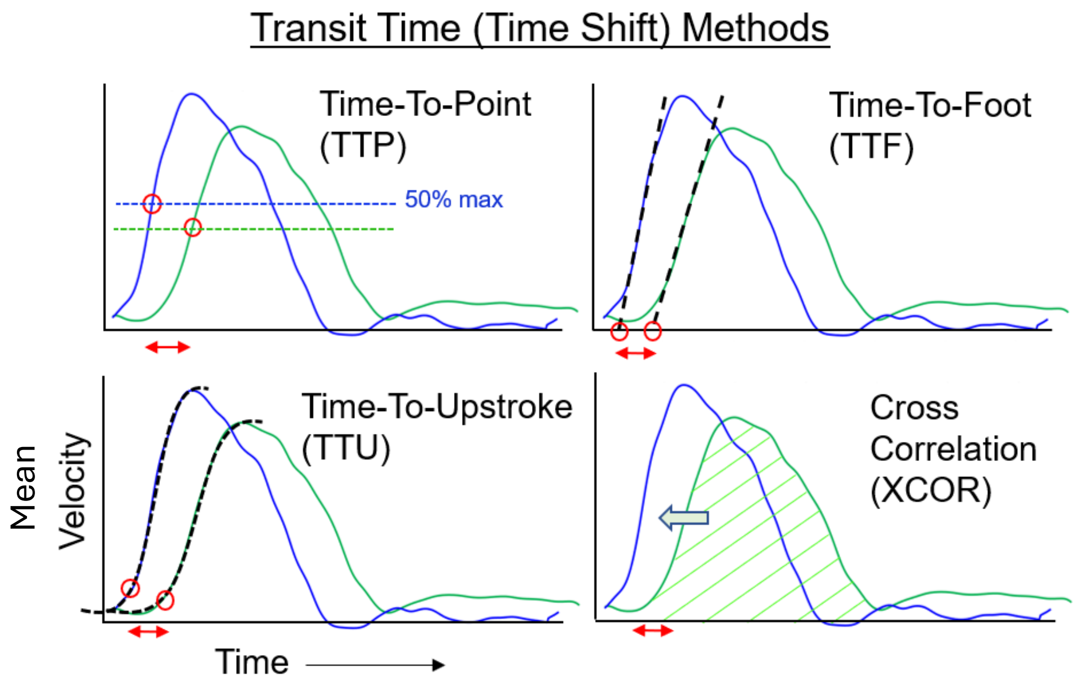

Figure 3.4: Four commonly used time-shift algorithms in the literature for calculating PWV: Time-to-Point (TTP), Time-to-Foot (TTF), Time-to-Upstroke (TTU), and Cross-Correlation (XCOR). All four algorithms were included in the custom-built MATLAB GUI for pulse wave velocity analysis. Figure courtesy of Grant Roberts.

and users are guided through placing circular ROIs around the ascending, descending, and abdominal aorta. Each ROI is propagated across all cardiac time frames, and temporal velocity waveforms are generated by computing the mean velocity within the ROI at each time point. The GUI also allows for shifting of the velocity waveforms by a user-defined number of frames such that systole occurs near the beginning of the cardiac cycle to account for situations where physiological delays in the waveform are present, as is the case when photoplethysmogram (PPG) gating is used. The waveforms are then temporally upsampled to 1000 points to match timestamps between acquisitions and smoothed using a Gaussian-weighted moving-average filter to reduce velocity noise. Flow waveforms are calculated by multiplying the velocity curve by the area of ROI. As there is no consensus on the optimal transit-time algorithm to use for determining temporal offsets between flow waveforms, the four most commonly used transit-time algorithms in the literature were implemented. These methods include time-to-point (TTP) [58, 59], time-to-foot (TTF) with baseline offset correction [60], time-to-upstroke (TTU) [59], and cross-correlation (XCOR) [61] (Figure 3.4). For each method, the measured transit times are plotted against the corresponding centerline distances along the aorta. A simple linear regression is then performed using the three measurement locations where the inverse of the fitted slope is the global PWV [42]. In this study, 3 aortic ROIs are acquired (ROIs 0 and 4 are utilized only for reference measurements) which gives 2 path lengths: AAo → DAo and DAo → AbdAo. The global PWV is then the overall regional PWV from AAo → AbdAo.

For ultrasound data, as only one ultrasound probe was available, each set of sequential recordings was aligned together using the sinusoidal motor signal recorded from the flow pump to determine temporal shifts relative to the first ROI (location 1 in Figure 3.1C) in a method based off an ECG alignment procedure [62]. Once aligned, 8 R-R intervals were extracted from the 10-second acquisition and averaged together to form a composite ultrasound waveform to match the MRI acquisition. Centerline distances between ROIs were taken from the same centerline created from the anatomical MRI data. PWVs were calculated using the same 4 transit time-algorithms used for the MRI analysis. Lastly, flow rates were plotted for the inlet and outlet of the phantom to check agreement between modalities.

3.3. Results

Figure 3.5 plots the flow values determined from both MRI and ultrasound data at inlet (ROI 0) and outlet (ROI 4) of the phantom. Agreement in max flow was high at the inlet (errors of 9.0%, 3.9%, and 1.6% for the Cartesian, radial, and SMS acquisitions, respectively, relative to ultrasound) while less so at the outlet (errors of 2.2%, 53.0%, and 24.0% for the Cartesian, radial, and SMS acquisitions, respectively, relative to ultrasound). The Cartesian flow curves also exhibited some temporal shifts relative to the ultrasound at both the inlet and outlet while the radial and SMS slices appeared to be well aligned.

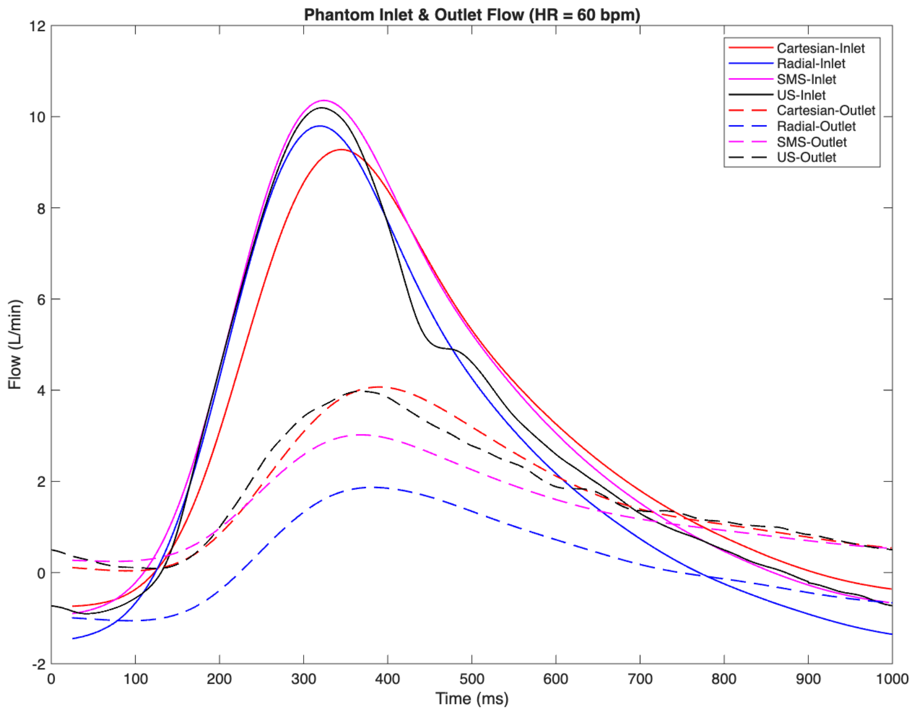

Figure 3.5: Flow curves for the inlet and outlet of the phantom from Cartesian, radial, simultaneous multi-slice (SMS), and ultrasound (US) modalities. Roughly 70% of flow is shunted to the arterial branches. Flow waveforms measured with each method show strong agreement at the inlet (< 10% error in max flow), but much poorer agreement at the outlet. In particular, the radial sequence shows a 53% error in max flow while the SMS sequence shows a 24% error. The Cartesian acquisition showed close agreement at the outlet.

Figure 3.6 depicts reconstructed time-averaged magnitude images of the 2D phase contrast images from the Cartesian, radial, and SMS acquisitions for the 60 bpm acquisition along with the corresponding ROI numbers. Notably, there are some artifacts present in the radial outlet image which are also present in the 75 bpm and 90 bpm radial outlet images. No artifacts were observed in any of the slices of the SMS acquisition.

Table 3.2 shows global PWV values for all combinations of heart rate, transit time method, and modality including the ground truth ultrasound values. Calculated PWVs showed varied agreement with ultrasound, with modalities such as the SMS acquisitions showing <14.1% error for all transit time algorithms and heart rates, while the Cartesian and radial acquisitions had large percent errors for certain combinations of heart rates and transit time algorithms. Additionally, mean PWVs across transit time algorithms showed a

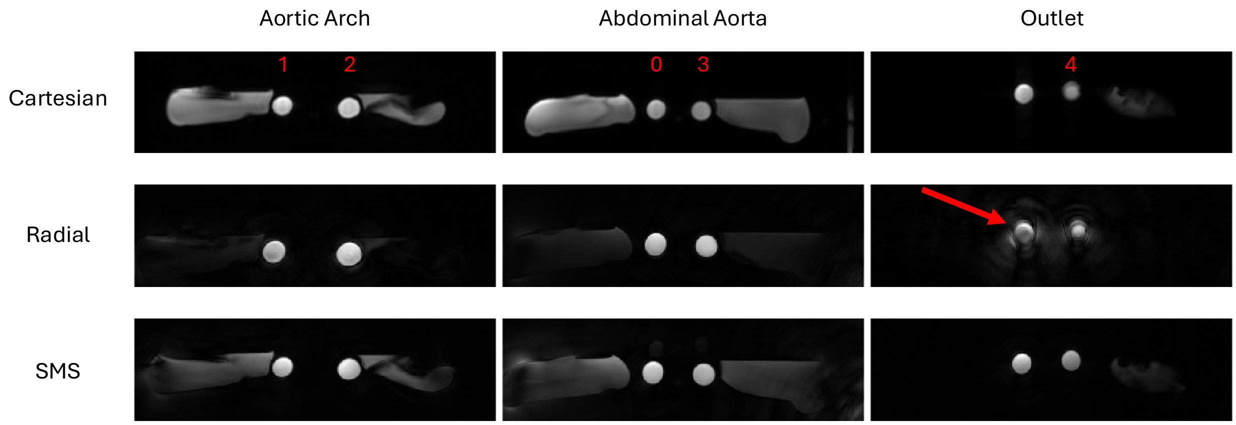

Figure 3.6: Magnitude images from the Cartesian, radial, and simultaneous multi-slice (SMS) 2D phase contrast acquisitions at a heart rate of 60 bpm. Red numbers correspond to locations of flow measurements. Image quality was comparable between acquisitions, with the exception of the Radial Outlet image where a noticeable artifact is present (red arrow). This same artifact was present in the 75 and 90 bpm radial outlet images and did not appear in the SMS images.

positive increase with heart rate.

3.4. Discussion

In this study, we updated the construction of a pulsatile aortic flow phantom to validate the performance of novel free-breathing, radial 2D Phase Contrast (PC) sequences, including both single-slice and simultaneous multi-slice (SMS) acquisitions. By incorporating arterial branches and modifying the structural properties, the updated phantom enabled simulation of more physiologically realistic flow dynamics of the aorta. These modifications enabled a more robust ground truth ultrasound measurement for evaluating radial 2DPC sequences. Additionally, the functionality of the SMS sequences was improved, allowing for more slices to be acquired with adjustable slice separation and a more robust iterative reconstruction.

Updating the construction of the phantom was one of the primary goals of this study as the previous model lacked a direct method of comparison with the developed MR sequences which hindered validation [39]. The updated material and thickness of the model were critical for allowing ultrasound probe placement directly on the phantom to measure flow at the same ROIs where MRI flow measurements were performed. The re-addition of the arterial branches, while making it more complex to construct and set up the flow circuit, aided the accuracy of the hemodynamics by shunting part of the flow away and reducing wave reflections from the outlet. While in this study we utilized normal tap water doped with gadolinium contrast agent, future work could aim to utilize a fluid that better mimics blood in terms of viscosity and MR relaxation times, such as the Non-Newtonian fluid used in Moravia et al [63].

For confirmation of the agreement in MRI and ultrasound-derived flow values, both modalities were used to measure flow at the inlet and outlet of the phantom (ROIs 0 and 4 according to Figure 3.1C); however, while the inlet showed strong agreement between MRI and ultrasound, the outlet showed much poorer agreement, particularly with the radial acquisition. Based on the artifact visible in the radial outlet image in Figure 3.6, it is possible that gradient-induced phase errors affected the calculated velocities in the most inferior radial slice leading to the sharp decrease in max flow [63].

Regarding agreement with the ground truth ultrasound values, the SMS radial acquisition showed closer agreement than either of the sequential single-slice radial or Cartesian 2DPC acquisitions. Both sets of sequential acquisitions showed larger variability, particularly for certain heart rates and time-shift methods. The improved performance of the SMS acquisition compared to the multiple sequential acquisitions is likely attributable to the simultaneous capture of multiple slices under identical physiological conditions, which mitigates errors arising from temporal misalignment. Both the Cartesian and the single-slice radial sequences may have minor delays in trigger detection times between slices due to the scanner's method of detecting the peak of the R-wave. This can result in small offsets in the start of cardiac waveform, which in turn affects the calculated median R-R interval which is used for retrospectively binning the radial acquisitions. It should also be noted that the intrinsic Cartesian temporal resolution is calculated differently than the radial acquisition and depends on the TR and acquired views per segment. While both acquisition types are interpolated to match the desired temporal resolution, the discrepancy could also potentially explain the temporal shift in the Cartesian flow waveform as seen in Figure 3.5. However, if the delay is consistent between acquisitions, as it was in this phantom study, then the calculated PWV is likely unaffected. Lastly, the observed increase in mean PWV with heart rate across all modalities aligns with prior physiological studies demonstrating heart-rate-dependent arterial stiffening and reduced vascular compliance [63].

While this study confirms the feasibility and accuracy of radial SMS 2DPC acquisitions for PWV measurement, several limitations remain. The phantom, although more anatomically realistic than the previous version, does not fully replicate the complex physiological properties of native aortic tissue. Additionally, the working fluid used does not match the non-Newtonian behavior of blood, so modifications to the fluid composition may enable more accurate PWV measurement. Finally, although the ground truth ultrasound measurements provide a robust reference, standard clinical methods for assessing PWV utilize pressure wires which could prove a better reference value [59].

3.5. Conclusion

In conclusion, the updated aortic phantom provides a physiologically relevant platform for validating MRI-based PWV measurement techniques. The radial SMS 2DPC sequence demonstrated improved accuracy and temporal consistency relative to conventional Cartesian acquisitions, while still enabling free-breathing acquisitions through the use of retrospective gating, similar to the single-slice radial acquisitions. Future work should focus on further evaluation of transit time algorithms and assessing performance across a broader range of hemodynamic conditions.

Acknowledgements

Thank you to both Grant Roberts and Kevin Johnson for their contributions to the development of these sequences and to James Rice for his help with constructing the phantom model and performing the flow experiments.

Table 3.2: Pulse Wave Velocities at multiple heart rates (HR) for four acquisition types: Cartesian, radial, radial simultaneous multi-slice (SMS), and ultrasound, calculated from four different transit time algorithms (TTP = time-to-point; TTF = time-to-foot; TTU = time-to-upstroke; XCOR = cross-correlation). Average values are reported ± standard deviations. Values in parentheses indicate percent error from ground truth ultrasound values

Parameter	**HR=60 bpm**	**HR=75 bpm**	**HR=90 bpm**
Aortic Length (mm)	245.8	245.8	245.8
Ultrasound			
TTP (m/s)	9.4	9.8	10.4
TTF (m/s)	9.2	8.1	11.5
TTU (m/s)	9.2	10.2	11.9
XCOR (m/s)	7.5	7.6	10.9
Average (m/s)	8.8 ± 0.9	8.9 ± 1.3	11.1 ± 0.6
Cartesian			
TTP (m/s)	8.7 (7.5%)	8.3 (15.1%)	11.9 (14.0%)
TTF (m/s)	9.7 (5.5%)	11.2 (38.5%)	16.5 (43.7%)
TTU (m/s)	11.0 (20.1%)	11.9 (16.5%)	11.9 (1.9%)
XCOR (m/s)	6.2 (16.9%)	7.2 (5.2%)	10.0 (8.2%)
Average (m/s)	8.9 ± 2.0	9.6 ± 2.2	12.6 ± 2.8
Standard Radial			
TTP (m/s)	10.6 (12.4%)	10.6 (8.1%)	13.3 (28.0%)
TTF (m/s)	10.8 (18.4%)	12.2 (51.7%)	15.4 (33.7%)
TTU (m/s)	11.2 (21.9%)	9.7 (4.4%)	9.6 (17.6%)
XCOR (m/s)	9.2 (23.1%)	9.6 (27.3%)	8.3 (23.4%)
Average (m/s)	10.4 ± 0.9	10.5 ± 1.2	11.7 ± 3.2
SMS			
TTP (m/s)	9.6 (2.0%)	10.6 (8.0%)	11.6 (10.3%)
TTF (m/s)	9.0 (1.9%)	9.2 (12.8%)	11.0 (4.2%)
TTU (m/s)	10.7 (14.1%)	9.9 (2.9%)	12.2 (4.1%)
XCOR (m/s)	7.0 (6.2%)	8.7 (12.4%)	11.1 (1.6%)
Average (m/s)	9.1 ± 1.5	9.6 ± 0.8	11.5 ± 0.6

Chapter 4: Feasibility of Free-Breathing Radial 2D Phase Contrast MRI for In Vivo Aortic Pulse Wave Velocity Measurements

In this chapter, I assess the feasibility of free-breathing 2D radial PC MRI for PWV measurements in a cohort of middle-to-late age subjects. This work is an extension of prior work in which I incorporate more rigorous inclusion criteria based on data quality to provide more reliable comparisons and analyze the reliability of pulse wave velocity transit-time algorithms in vivo [39]. The updated simultaneous multi-slice acquisition described in Chapter 3 was implemented after the LIFE study had completed and as such was not included in this chapter. However, it is explored in greater detail in Chapter 5. This work resulted in 1 first author abstract [53] and a first author manuscript currently under preparation.

4.1. Introduction

Arterial stiffness is a well-established early predictor for cardiovascular risk. In particular, increased aortic stiffness has been linked with numerous cardiovascular diseases such as stroke, hypertension, and coronary artery disease [64]. Aging is strong contributor to arterial stiffening, largely due to biochemical and structural alterations that thicken the vessel wall and reduce the aorta's ability to buffer pulsatile blood flows [41, 65]. Recent studies have indicated that increased aortic stiffness may also be linked to cranial pathologies such as vascular dementia and Alzheimer's Disease (AD) [66, 67]. Many of the risk factors associated with neurodegeneration such as smoking, hypertension, and the apolipoprotein E4 (APOE4) gene are also associated with cardiovascular disease [68–70]. As such, there is high clinical interest in accurately assessing arterial stiffness.

Direct measurement of arterial stiffness is challenging; however, pulse wave velocity (PWV) provides a feasible alternative for assessing arterial stiffness through its relationship with arterial stiffness via the Moens-Korteweg equation (see Equation 3.1) [43, 44]. Measurements of PWV rely on transit time measurements which are performed by recording flow or pressure waveforms at multiple locations along a vessel and calculating temporal shifts between them. This requires accurate flow measurements with high temporal resolutions as well as vascular path length measures (see Equation 3.2) [42]. Multiple algorithms exist to calculate these time-shifts, some of which are described in Section 3.2, yet there is no consensus on which method is optimal for PWV calculation. This is in part due to how different methods are more or less sensitive to different sources of noise and variability in the acquired waveforms, which can be influenced by the acquisition method and parameters [71]. For example, the 4 transit time methods utilized in Section 3.2—time-to-point (TTP) [58, 59], time-to-foot (TTF) [60], time-to-upstroke (TTU) [59], and cross-correlation (XCOR) [61]—each operate on different features of the waveforms (see Figure 3.4) and thus can result in different values of PWV despite operating on the same waveforms.

As discussed in Chapter 3, Magnetic Resonance Imaging (MRI) has emerged as a promising avenue for non-invasively measuring time-shifts through 2D Phase Contrast (PC) imaging which enables quantification of blood velocity and flow through a plane [42]. These can be combined with anatomical images for determining accurate vessel path lengths, which methods such as ultrasound or pressure wires are unable to provide, leading to more robust PWV calculations. Standard cine 2DPC cardiac imaging is acquired with Cartesian sequences which require breath-holds to avoid respiratory motion artifacts. However, many elderly, senile, or health-compromised patients such including those with AD, are incapable of holding their breath or following breathing directions which can result in severely compromised image quality. The radial free-breathing 2D phase contrast sequence introduced in Chapter 3 provides a promising alternative that can enable PWV measurements in these patient populations. Additionally, the radial acquisitions, plus a longer acquisition time enabled by free-breathing, allow for retrospectively increasing the temporal resolution which can be beneficial for capturing faster transit times.

The motivation of this chapter is to assess the feasibility of a free-breathing, radially undersampled 2DPC sequence for aortic PWV measurements by comparing to a standard, Cartesian 2DPC breath-hold sequence. Additionally, constrained reconstruction techniques are utilized to increase the temporal resolution of the acquired radial data to investigate the effects of increased temporal resolution on PWV. Aortic PWV measurements are obtained in a cohort of middle-aged, cognitively unimpaired adults with three 2DPC methods (Cartesian, standard radial, and high temporal resolution radial) to assess the relationship between aortic PWV and age and compare the impact of 4 different transit time algorithms on calculated PWVs.

4.2. Materials & Methods

Data for this study comes from the Longitudinal Impact of Fitness and Exercise (LIFE) study, which consists of a subset of the greater Wisconsin Alzheimer's Disease Research Center (ADRC) Clinical Core and the Wisconsin Registry for Alzheimer's Prevention (WRAP) observation cohorts. The LIFE study seeks to characterize the longitudinal relationship between cardiorespiratory fitness (CRF) —– a physiological indicator for habitual physical activity —– and the cognitive features of AD. Furthermore, it aims to assess vascular and glucoregulatory function as potential mechanisms underlying the beneficial effects of CRF and so PWV measures were obtained as a marker for vascular health.

The subjects included in this study consist of cognitively healthy middle-aged adults as well as older individuals at elevated risk for AD, characterized in part by a higher prevalence of the APOE4 genotype and a parental history of AD. Eligibility criteria for the LIFE study required participants to be at least 45 years of age and free from cognitive impairment or deficits in decision-making capacity at enrollment, as determined by standardized neuropsychological assessments [72, 73]. Exclusion criteria included current use of antipsychotic medications and general contraindications to MRI, such as pregnancy or the presence of magnetically activated implants. Written informed consent was obtained from all participants and investigations were conducted in compliance with HIPAA regulations and the University of Wisconsin-Madison Health Sciences Institutional Review Board. In total, 150 subjects were recruited for the study.

MR Imaging Protocol

As part of the longitudinal assessment, participants were imaged twice over the course of the study with a baseline exam at the start of the study and a follow-up exam two years later to assess changes in CRF and cerebrovascular health due to habitual exercise. During the course of the study, the scanner model, imaging coil, and location were changed from a 3T Discovery MR750 with an 8-channel chest coil (GE Healthcare, Waukesha, WI) located at the Waisman Center (Madison, WI) to a 3T SIGNA Premier with a 30-channel chest coil (GE Healthcare, Waukesha, WI) located at the Wisconsin Institutes of Medical Research (WIMR) (Madison, WI). To ensure data consistency, only the follow-up visit data acquired at the WIMR location collected between January 2023 and December 2024 were included in the study resulting in 83 cases (see Figure 4.2).

For the chest scans, ungated anatomical balanced steady state free precession (bSSFP) product sequences (axial, coronal, sagittal) were acquired to guide aortic centerline tracing and vessel path length calculation. Parameters for the scans are identical to the ones acquired and described in Section 3.2. For the Cartesian and radial 2DPC sequences, two axial planes were placed along the aorta. The superior slice was prescribed in the aortic arch at the level of the pulmonary bifurcation while the inferior slice was prescribed just above the radial arteries. This resulted in the acquisition of 3 total regions of interest (ROIs) in the aorta: the ascending and descending aortas in the superior slice and the abdominal aorta in the inferior slice (Figure 4.1A). Table 4.1 lists the scan parameters for both types of acquisitions. Photoplethysmography (PPG) data from a peripheral pulse oximeter attached to the index

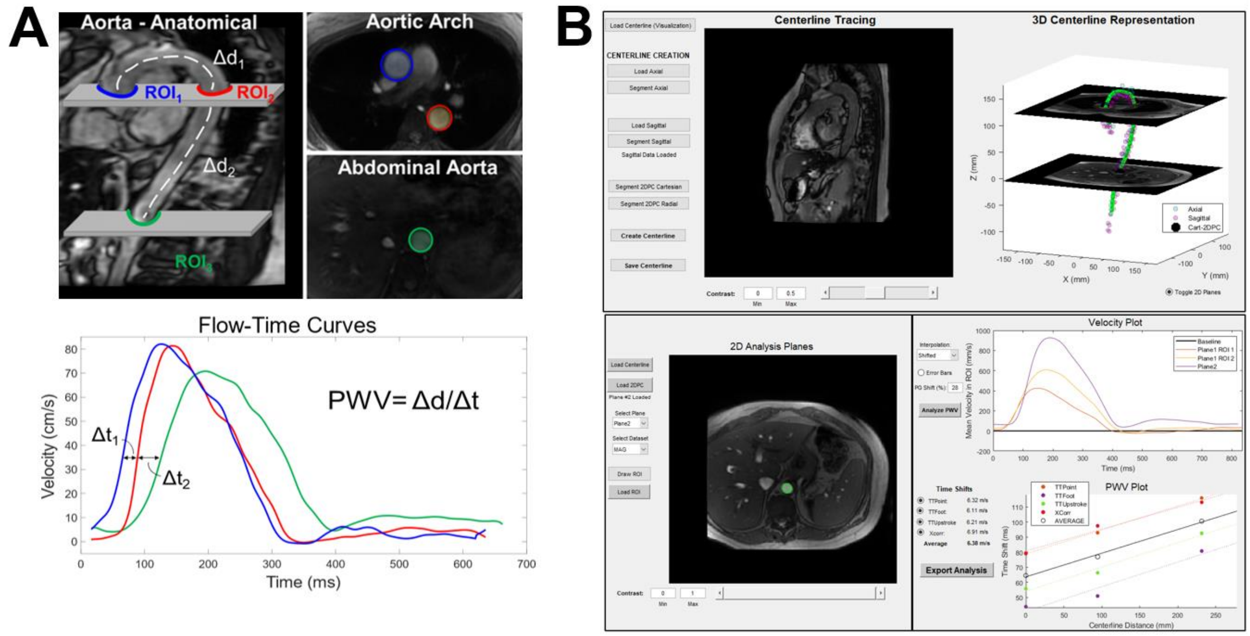

Figure 4.1: Overview of the 2D Phase Contrast (2DPC) acquisition and pulse wave velocity (PWV) analysis framework. (A) Two axial planes were acquired for each PC scan, one bisecting the aortic arch and the other in the abdomen, for a total of three aortic ROIs. (B) A custom software platform was developed to segment the aorta ROIs, trace aortic centerlines, analyze flow curves, and calculate PWV. Figure courtesy of Grant Roberts.

finger and respiratory data from respiratory bellows attached to the chest were recorded during 2DPC acquisitions for retrospective physiological gating.

Cartesian acquisitions were automatically reconstructed on the scanner post-acquisition by the GE Healthcare product reconstruction engine. For the radial acquisition, offline reconstructions were performed with a custom reconstruction framework developed in C++. Two different types of radial reconstructions were employed: a standard reconstruction designed to match the Cartesian acquisition using Parallel Imaging with Localized Sensitivities (PILS) (standard radial) and a constrained reconstruction using a local low rank (LLR) algorithm to increase spatial and temporal resolution (hi-res radial) [16, 31, 74]. Both reconstructions utilized ESPIRiT coil sensitivity estimation for parallel imaging and included Maxwell term phase offset corrections, and gradient calibration to correct trajectory errors [10, 56, 75]. The iterative LLR reconstruction was performed with nuclear norm regularization using a local

Table 4.1: Acquisition parameters for 2D Cartesian and radial 2D phase contrast scans.

Parameter	**Cartesian 2DPC**	**Radial 2DPC**
TR / TE (ms)	5.1 / 3.1	7.5 / 4.3
Flip angle	25°	25°
Views per segment	4	N/A
Sampling	Cartesian	Golden-Angle
Acquired FOV (cm)	36	32
Acquired spatial resolution (mm^2)	1.41×1.41	1.40×1.40
Slice thickness (mm)	6	5
Venc (cm/s)	150	150
Velocity encoding	2-point	2-point
Number of projections	N/A	10,000
Scan time	12–20s	150 s

block size of $4 \times 4 \times 4$ and an empirically determined regularization parameter of $\lambda = 0.003$ for 55 iterations. All data were reconstructed in end-exhalation using a 50% threshold of the respiratory waveform performed with a sliding 10-second window to account for respiratory drift. Table 4.2 lists the reconstructed parameters for all image types.

Table 4.2: Reconstruction parameters for Cartesian, radial low-res, and radial hi-res 2D phase contrast scans. PILS = Parallel Imaging with Localized Sensitivities, LLR = Local Low Rank constrained reconstruction
*Temporal resolutions are dependent on heart rate and are reported as mean ± standard deviation for the cohort.

Parameter	**Cartesian 2DPC**	**Standard Radial**	**Hi-Res Radial**
Reconstruction type	Vendor default	PILS	LLR
Reconstructed matrix	256×256	256×256	320×320
Reconstructed spatial resolution	1.41×1.41	1.40×1.40	1.00×1.00
Reconstructed cardiac frames	40	40	80
Temporal resolution* (ms)	26.2 ± 3.6	25.2 ± 3.6	12.5 ± 1.8

Analysis

PWV calculations for each subject and statistical analyses were performed in MATLAB (version 2020b, Mathworks, Natick, MA) using the GUI tool described in detail in Section 3.2 by a single observer (T.N. with 5 years experience in 2DPC MRI analysis). Images were assessed prior to analysis for general image quality, proper plane placement, and presence of velocity aliasing. Physiological gating signals were also assessed by plotting respiratory waveforms and R-R intervals. Datasets that failed to meet quality standards for any of those reasons were excluded from analysis. The 4 time-shift methods introduced and utilized in Section 3.2—TTP, TTF, TTU, and XCOR—were used to calculate PWVs for each subject with each acquisition type (Cartesian, standard radial, hi-res radial).

Aortic centerline distances and pulse wave velocities (PWVs) were summarized for the full cohort according to age groups (50–59, 60–69, and 70–79 years) and sex. For each variable, the mean, standard deviation, and sample size (N) were reported. Differences in PWV across acquisitions for each time-shift method were evaluated using the Friedman test. PWV distributions are visualized with box plots, showing medians, variability, and outliers, with paired measurements connected by line segments.

Associations between acquisition and time-shift method pairs were assessed using Pearson correlation coefficients (r) and visualized with a correlation matrix. To evaluate the relationship between age and aortic PWV, simple linear regression models were constructed for each acquisition and time-shift method. Regression coefficients (β), coefficients of determination (R^2), and corresponding p-values were reported, with statistical significance defined as $p < 0.05$.

4.3. Results

Of the 83 subjects examined in this study, 21 were excluded due to: poor plane prescription (8), insufficient image quality or velocity aliasing (11), or poor cardiac or respiratory gating (2) in one or more of the three acquisition types. An additional 17 subjects were excluded due to calculated PWVs with one or more of the implemented time-shift algorithms resulting in a physiologically impossible value (PWV $<$ 1 m/s or PWV $>$ 20 m/s). This resulted in a final total of 45 cases included in this analysis (Figure 4.2). Demographics information for these 45 subjects are shown in Table 4.3.

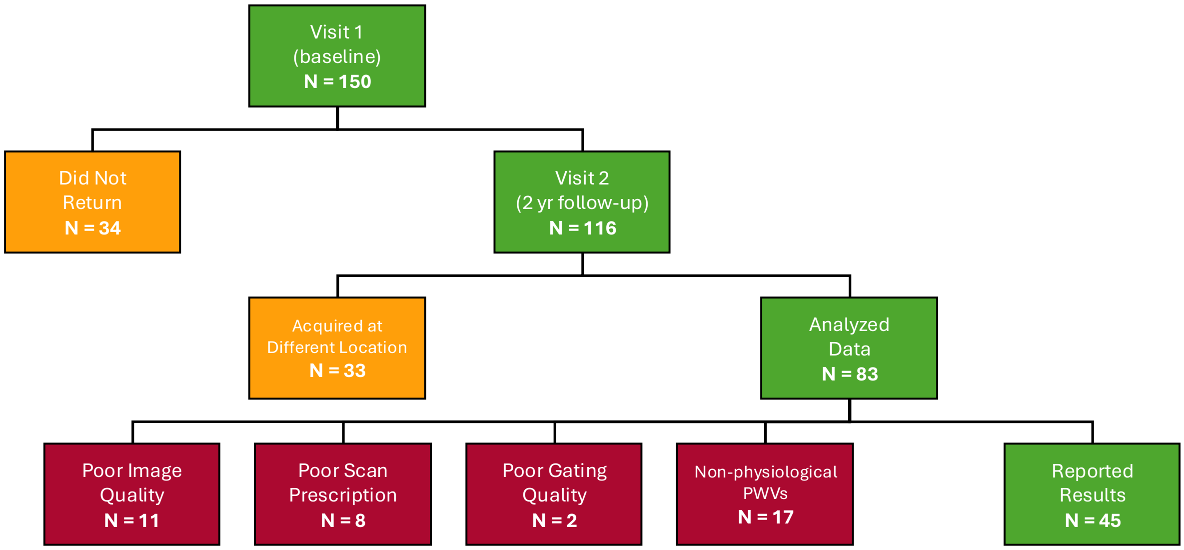

Figure 4.2: The LIFE study recruited 150 subjects for 2 visits two years apart. Of the 116 subjects who returned for the follow-up visit, 33 were excluded due to being scanned at a different location and scanner system. Of the remaining 83, 11 were excluded due to poor image quality (from to various imaging artifacts), 8 due to poor scan prescription (imaging planes placed too high or low), and 2 due to poor gating quality (noisy cardiac or respiratory signals). Furthermore, an additional 17 subjects were excluded due to calculated PWVs with one or more algorithms being non-physiological (PWV $<$ 1 m/s or PWV $>$ 20 m/s). Data for the resulting 45 subjects are presented here.

Average $\pm$ standard deviation temporal resolutions for the Cartesian, standard radial, high-res radial reconstructions were 26.2 $\pm$ 3.6 ms, 25.2 $\pm$ 3.6 ms, and 12.5 $\pm$ 1.8 ms, respectively. Plane distances (in the superior-inferior direction) between the aortic arch and abdominal aorta 2D slices were 131 $\pm$ 18.3 mm. Representative images from each of the 3 acquisition methods are shown in Figure 4.3. Image quality in all acquisitions was sufficient for flow quantification and PWV analysis. Radial images contained some undersampling streak artifacts around the edges of the torso. Table 4.4 provides aortic lengths between each of the 3 measurement locations and PWV values for each acquisition and time-shift method by the whole group, age category, and by sex. Average PWV across the entire cohort ranged from 7.1–7.7 m/s for each time-shift method.

Table 4.3: Patient demographics for analyzed study subjects (N=45). For continuous data, values are expressed as mean $\pm$ SD and data ranges are in parentheses. For categorical data, values are expressed as numerators with percentage in parentheses.
*Race was self-reported and obtained as required by federal funding agencies
**APOE4 = apolipoprotein E4

Variable	**Value**
Age (years)	63.1 ± 7.41 (50–77)
Sex	
Female	34 (76%)
Male	11 (24%)
Race*	
White	44 (97.8%)
Unknown	1 (2.2%)
Education (years)	16.8 ± 1.98 (12–20)
Parental history of dementia	
Yes	36 (80.0%)
No	9 (20.0%)
Parental history of AD	
Yes	23 (51.1%)
No	9 (20.0%)
Unknown	13 (28.9%)
APOE4 carrier**	
Yes	30 (66.7%)
No	14 (31.1%)
Unknown	1 (2.2%)

Friedman test results demonstrated a significant difference in PWV across acquisition methods for the TTP algorithm ($\chi^2 = 8.62$, $p = 0.01$) and the XCOR algorithm ($\chi^2 = 7.20$, $p = 0.03$), whereas no significant differences were observed for the TTF and TTU algorithms. Box plots illustrating PWV distributions by MRI acquisition method (Figure 4.4) indicate that Cartesian-derived PWV values frequently deviate from those obtained using radial acquisitions, particularly at higher PWV ranges.

Pearson correlation coefficients for each combination of time-shift method and acquisition are presented as a correlation matrix in Figure 4.5A, with all pairwise correlations reaching statistical significance. Mean correlations averaged across acquisitions (i.e., within each transit time method) are shown in Figure 4.5B. The highest mean correlations were observed for TTF (mean $r = 0.73$) and XCOR (mean $r = 0.72$) when compared across acqui-

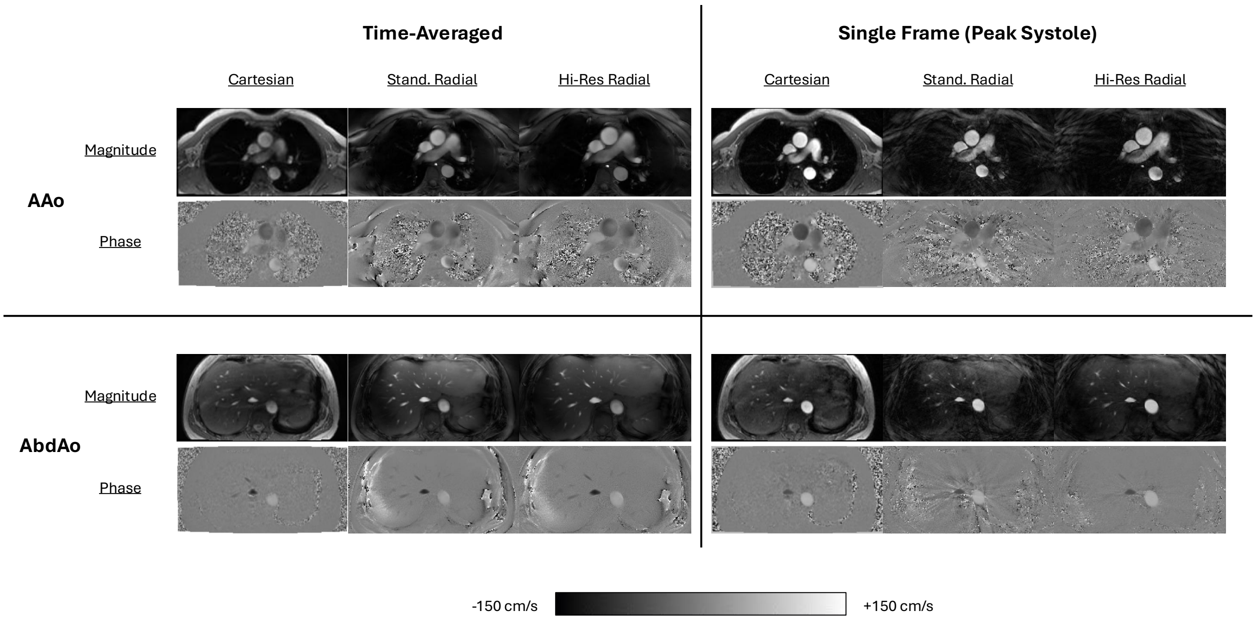

Figure 4.3: Examples of time-averaged and time-resolved (single frame) phase contrast images in the ascending aorta (AAo) and abdominal aorta (AbdAo) across a Cartesian product scan, a standard radial reconstruction, and a high-res (Locally Low Rank) radial reconstruction for a single subject. Time-averaged Cartesian image was constructed by averaging all reconstructed time-frames from the default vendor reconstruction. All images demonstrate sufficient image quality for pulse wave velocity analysis. Radial images contain some undersampling streak artifacts around the edges of the torso.

sitions, whereas comparisons between different transit time methods yielded lower correlations. Mean correlations averaged across transit time methods are displayed in Figure 4.5C. The Cartesian acquisition demonstrated the highest mean inter-method correlation (mean $r = 0.80$), although strong correlations were also observed for the standard radial (mean $r = 0.73$) and high temporal resolution radial (mean $r = 0.72$) acquisitions. While transit times between the two radial acquisitions were strongly correlated (mean $r = 0.67$), correlations between Cartesian and radial acquisitions were comparatively weak (mean $r < 0.28$).

Simple linear regression analysis revealed a significant association between aortic PWV and age for the TTP, TTF, and TTU methods in both standard and high-res radial acquisitions, whereas no significant relationship was observed for the XCOR method. In contrast, the Cartesian acquisition demonstrated a significant relationship between PWV and age for

Table 4.4: Aortic path lengths and pulse wave velocities (PWVs) stratified by age group and sex for three different 2D phase contrast image types: Cartesian, standard radial (Std. Radial), and high temporal resolution radial (HR Radial) from four different transit time algorithms (TTP = time-to-point; TTF = time-to-foot; TTU = time-to-upstroke; XCOR = cross-correlation). All values expressed as: mean $\pm$ standard deviation.

Parameter	**Overall** (N=45)	**50–59** (N=13)	**60–69** (N=17)	**70–79** (N=15)	**Male** (N=11)	**Female** (N=34)
Aortic Length	258.0 $\pm$ 23.6	252.1 $\pm$ 21.1	257.5 $\pm$ 24.6	263.6 $\pm$ 24.7	275.8 $\pm$ 18.6	252.2 $\pm$ 22.3
Cartesian						
TTP	7.2 ± 2.3	6.5 ± 1.2	6.6 ± 2.0	8.4 ± 2.7	8.1 ± 3.4	6.9 ± 1.7
TTF	7.1 ± 2.9	6.0 ± 0.9	6.3 ± 2.4	9.1 ± 3.6	7.7 ± 3.7	6.9 ± 2.6
TTU	7.3 ± 2.8	6.2 ± 1.0	6.5 ± 2.4	9.1 ± 3.4	8.1 ± 3.8	7.0 ± 2.3
XCOR	7.3 ± 1.8	7.4 ± 1.6	6.8 ± 2.2	7.8 ± 1.7	7.8 ± 2.3	7.2 ± 1.7
Std. Radial						
TTP	7.6 ± 1.4	7.0 ± 1.0	7.2 ± 1.3	8.6 ± 1.3	7.4 ± 1.4	7.7 ± 1.4
TTF	7.6 ± 1.9	6.8 ± 1.3	7.0 ± 1.6	9.0 ± 2.1	7.3 ± 2.4	7.7 ± 1.8
TTU	7.6 ± 1.9	6.7 ± 2.1	7.2 ± 1.6	8.7 ± 1.3	7.1 ± 2.5	7.7 ± 1.6
XCOR	7.8 ± 2.1	7.7 ± 1.9	7.7 ± 2.5	7.9 ± 2.0	7.4 ± 2.0	7.9 ± 2.1
HR Radial						
TTP	7.5 ± 1.3	7.1 ± 1.2	7.1 ± 1.2	8.2 ± 1.0	7.3 ± 1.4	7.5 ± 1.2
TTF	7.3 ± 1.6	6.6 ± 1.2	6.9 ± 1.5	8.4 ± 1.5	7.1 ± 2.0	7.4 ± 1.5
TTU	7.3 ± 1.7	6.4 ± 2.0	7.1 ± 1.4	8.3 ± 1.1	7.2 ± 1.7	7.3 ± 1.7
XCOR	7.6 ± 2.0	7.5 ± 2.1	7.4 ± 2.1	7.8 ± 1.9	7.2 ± 2.1	7.7 ± 2.0

the TTF and TTU methods only, with no significant associations for TTP or XCOR. Representative regression plots for the TTF method across all three acquisitions are provided in Figure 4.6. For the TTF method, regression coefficients for age (β_{age}) ranged from 0.06 to 0.11 m/s, with corresponding R^2 values between 0.16 and 0.22 and standard errors between 0.03 and 0.04. Among all approaches, the high temporal resolution radial acquisition exhibited the strongest relationship with age, with R^2 values ranging from 0.02 to 0.20 across methods.

4.4. Discussion

In this study, we implemented a radial, free-breathing two-dimensional phase contrast (2DPC) MRI sequence and assessed the feasibility of utilizing it for measuring aortic pulse wave velocity (PWV). The radial free-breathing approach was evaluated against a standard

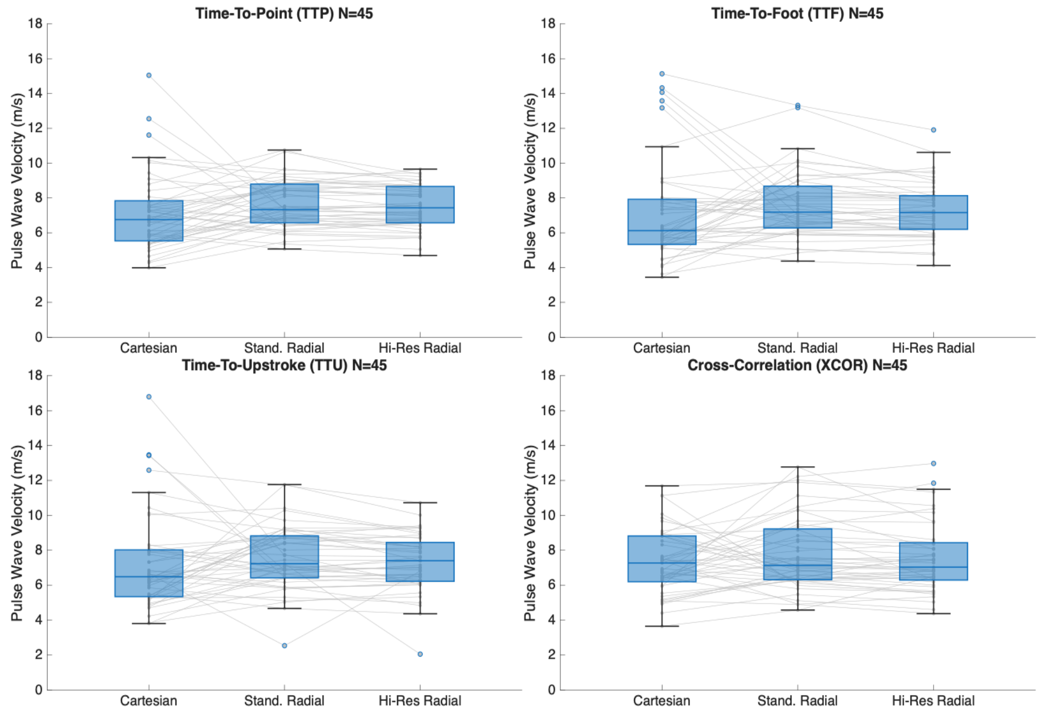

Figure 4.4: Distributions of pulse wave velocities calculated using four different transit time methods across a Cartesian product scan, a standard radial reconstruction, and a high-res (Locally Low Rank) radial reconstruction. Connecting lines between paired data points highlight the individual correlations on a participant level.

clinical breath-hold Cartesian acquisition. The radial data was used to reconstruct two different datasets: one with spatial and temporal resolution matched to the Cartesian acquisition, and a second with enhanced spatial and temporal resolution achieved with a constrained local low-rank (LLR) reconstruction. PWVs were calculated via 4 different transit time algorithms for each of the three acquisitions to compare the impact of acquisition type and transit time method on PWV estimates. The relationship between age and aortic PWV was also assessed for each acquisition and transit time method pair.

Radial data utilized in this study was heavily undersampled with approximately 250 projections per cardiac frame for the standard reconstruction. This resulted in some streaking artifacts around the edges of structures and vessels which could then manifest as noise within the derived flow waveforms. However, as PWV calculations rely on time-shifts between flow waveforms, it is likely that PWV analysis is robust to undersampling artifacts as long as the

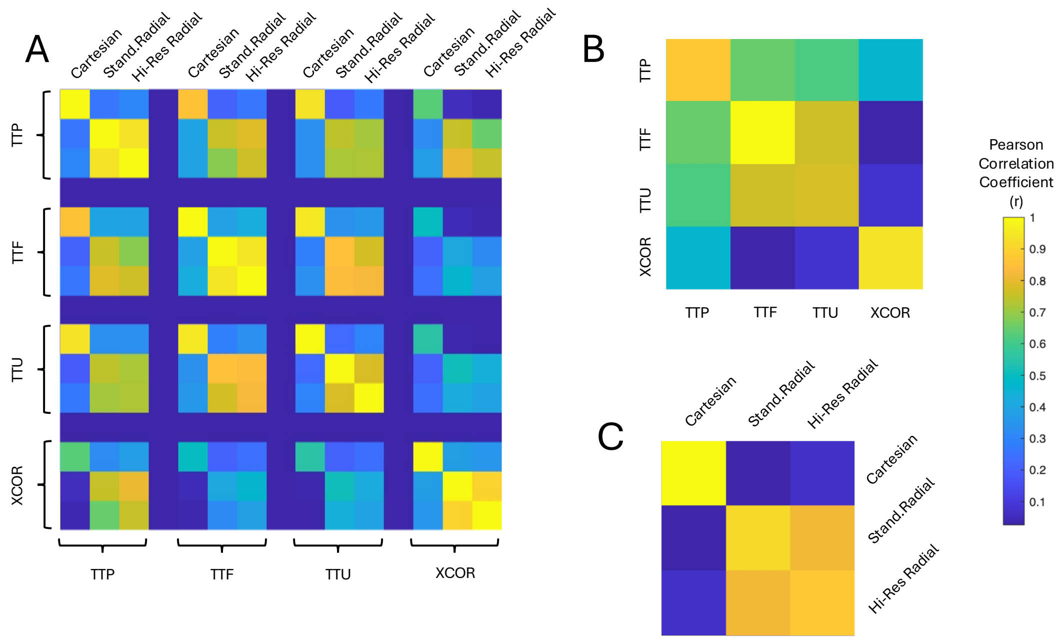

Figure 4.5: (A) Correlation matrix of Pearson correlation coefficients between individual pairs of transit time methods and acquisitions. The transit time methods include Time-To-Point (TTP), Time-To-Foot (TTF), Time-To-Upstroke (TTU), and Cross-Correlation (XCOR) while the acquisitions include a Cartesian product sequence, a standard radial reconstruction, and a high temporal resolution (Locally Low Rank) radial reconstruction. (B) Mean correlations averaged across acquisitions (C) Mean correlations averaged across transit time methods

upstroke portion of the flow waveform maintains its general shape between acquisitions. The LLR reconstruction doubled the temporal resolution by halving the number of projections per frame but still resulted in comparable image quality. It was hypothesized that higher temporal resolutions would enhance the accuracy of PWV measurements, but while the high temporal resolution dataset did demonstrate the strongest correlation with age, in most other aspects, the results were largely similar to the standard radial.

While the radial reconstructions showed strong agreement for all algorithms, there was less agreement between the Cartesian and radial reconstructions, as indicated by the difference in correlations (Figure 4.5) and paired line segments in the box plots (Figure 4.4). There are several possible contributors to this observation, such as gating timing issues resulting from separate acquisitions, respiratory effects of breath-holds vs free-breathing, or

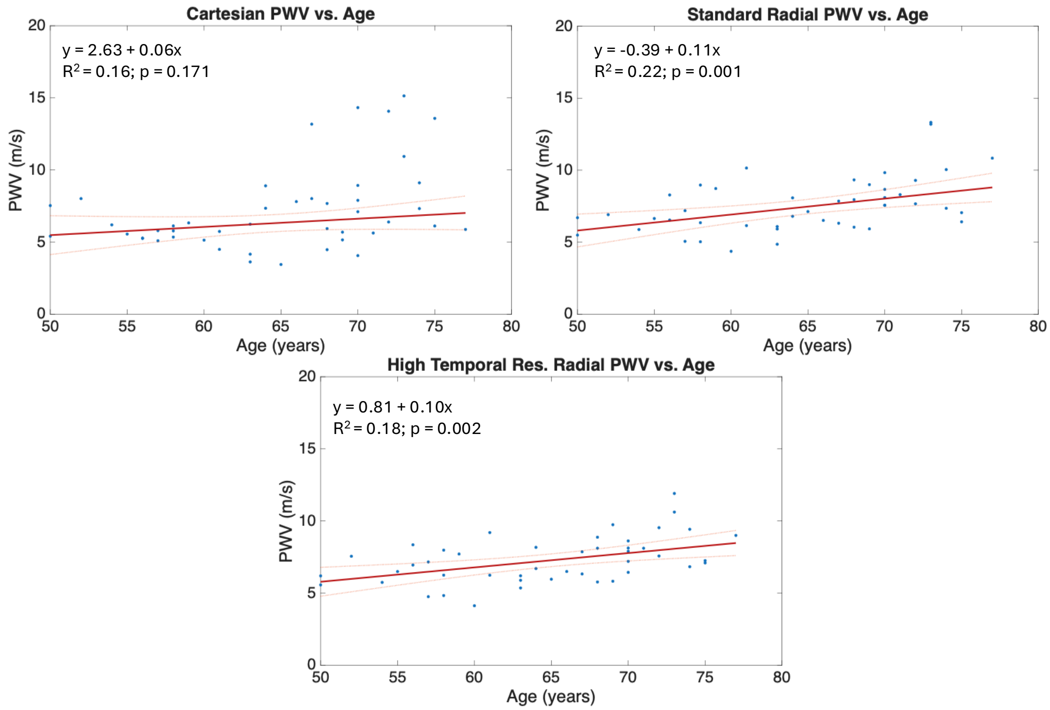

Figure 4.6: Linear regression models showcasing the positive relationship between age and pulse wave velocity calculated via the Time-to-Foot method in three different acquisitions: a Cartesian product sequence, a standard radial reconstruction, and a high-res (Locally Low Rank) radial reconstruction.

differences in the post-processing between automatically-reconstructed Cartesian images and the radial reconstructions. Additionally, heart rate variability between the two acquisitions may also be a contributing factor as breath-holds have been shown to have effects on heart rate [76].

All acquisitions and transit time methods (aside from XCOR) showed significant relationships with age. Specifically, we found that aortic PWV increases by around 1.1–1.4 m/s each decade of life. This agrees closely with a large study (N=397) performed by van Hout et al., in which they assessed aortic PWV in older adults (mean age 55 $\pm$ 6 years) using 2DPC MRI [77]. They observed an average PWV increase of 0.9 m/s over 10 years. Similar results have been demonstrated in other 2DPC MRI studies [78–80]. Furthermore, the study by van Hout et al. reported a mean PWV value of 6.8 m/s for the age range 60–65 years using the TTF method, which is similar to the results obtained here (mean $\mathrm{PWV}_{\mathrm{TTF}}$ = 6.3

$\pm$ 2.4 m/s) [77]. Another study by Hickson et al. also reported a mean PWV in a similar age bracket of approximately 6 m/s using the TTP method which again agrees with our results (mean $PWV_{TTP} = 6.6 \pm 2.4$ m/s) [80]. However, a large normative study (N=777) by Nethononda et al. reported a mean aortic PWV of 9.6 $\pm$ 4.1 m/s in the 60–69 year age range using the TTF method which is considerably higher than what our and the other referenced studies reported [81]. One possible explanation for this could be the number of slices acquired for PWV calculations. While our study and Nethononda et al. acquired 2 slices in the aorta (3 total ROIs), van Hout et al. acquired 3 slices (4 ROIs) and Hickson et al. acquired 4 slices (5 ROIs). It has been shown that the sampling density along the aorta for 2DPC based measurements can affect the accuracy of PWV calculations which may indicate that Hickson et al. is the most reliable value for comparison [82].

In prior work performed on this cohort analyzing PWV, there were considerable methodological complications that compromised the reliability of the PWV analysis [39]. For one, data acquired for this study were tagged onto a larger cranial study and performed at a location specializing in head and neck imaging. This resulted in a number of concessions including the use of a smaller chest coil (8-channel), cardiac gating performed with PPG instead of ECG signals, and an acquisition order in which free-breathing acquisitions were performed directly after breath-hold acquisitions without time for subjects to return to a resting state. To amend these issues, during follow-up visits, scheduling was gradually adjusted to transition subjects to another location better equipped for cardiac imaging, with MR technologists more experienced in cardiac imaging and a larger chest coil (30 channel) was available. Cardiac gating was still performed with PPG signals to remain consistent with prior data, however, the acquisition order was modified to perform all breath-hold acquisitions at the beginning of the session (Cartesian scans) followed by the free-breathing acquisitions (radial scans) with the intention of reducing the variability in heart rates between acquisitions. All data analyzed in this study were acquired after these adjustments were made, with the result that while we increased our confidence in the quality of analyzed data, we experience a considerable reduction in sample size (N=117 vs N=45) and thus statistical power [39]. However, the updated results presented here demonstrated an overall reduction in calculated PWVs across most categories and transit time algorithms. The previous analysis reported mean PWVs across the entire cohort ranging from 7.7–8.4 m/s and for comparisons to other studies, $\mathrm{PWV_{TTP}}$ and $\mathrm{PWV_{TTF}}$ of 7.9 $\pm$ 3.4 and 7.8 $\pm$ 3.7, respectively [39]. As stated previously, our results show closer agreement with these other studies, which may indicate that the methodological adjustments we made were successful in improving the reliability of our PWV measurements.

Despite the efforts made to improve the acquisitions, a considerable number of datasets still had to be excluded in analysis due to poor prescription, image quality, or gating (21 out of 83). One potential reason for plane placement issues could be due to the unconventional prescription needed for the radial acquisition. The sequence was implemented via modification of an existing 3D radial research sequence and as such, the prescription was still performed as a 3D acquisition, requiring some effort to correctly center the volume at the desired location for the 2D slice. Additionally, as the radial images had to be reconstructed offline, immediate feedback after acquisition was not available like it was for the automatically reconstructed product Cartesian sequence, so there was no possibility of a quick re-acquisition to correct issues. On the other hand, the majority of image quality issues were present in the Cartesian images (8 of 11) due to ghosting artifacts from respiratory motion or aliasing from parallel imaging that interfered with velocity measurements. Meanwhile, a few radial images (3) demonstrated artifacts similar to the ones shown in Figure 3.6 which may indicate issues with the sequence. Further investigation into the source of this artifact is needed. Only 2 datasets were excluded due to poor gating, which is a considerable decrease from the 17 excluded in the original analysis [39]. Finally, an additional 17 datasets were excluded due to physiologically impossible PWV values, an issue that was present in the original analysis as well but was accounted for by simply excluding individual values. The primary cause of these physiological impossible PWVs was likely due to changes in heart rates between acquisitions, resulting in changes to the abdominal aorta waveforms with a different period (R-R interval) and temporal resolution compared to waveforms measured in the aortic arch. When time-shifts are calculated between the waveforms, if the waveforms are too close together, it can result in very large PWVs (PWV $>$ 20 m/s), or if the waveforms overlap, it can result in negative PWVs. It should be noted also that different time-shift methods fail in different ways and that some methods may be more robust than others in different failure scenarios. For example, the TTP algorithm determines time shifts from only a single point and is thus more sensitive to outliers, but by the same token may be the least affected by heart rate variation. Meanwhile, the TTF method relies on the slopes of the waveforms and may produce negative time-shifts if later waveforms have flatter waveforms. The TTU method is dependent on identifying the point at the beginning of the upstroke through fitting of a sigmoid function but can be compromised by increased noise in the late diastolic/early systolic portion of the flow waveform which is common in 2DPC acquisitions due to the reduction in signal from reduced inflow. Lastly, the XCOR algorithm calculates time-shifts by perform correlations across the entire waveform and is thus more prone to errors from wave reflections that are present in diastole. Considering the variability in all these methods, it may be beneficial to acquire all time-shift results (which take relatively little time to compute) to provide backup measures in the event one of the algorithms fails. Wentland et al. also suggest that averaging multiple time-shift results together may provide the most robust PWV values [42].

Limitations

There were several limitations to this study. As discussed, changes in heart rate between acquisitions led to physiologically impossible PWV values in a number of subjects. This was present in both Cartesian and radial acquisitions and is limitation of sequential acquisitions. A 3D or interleaved acquisition that captures multiple slices simultaneously would alleviate this issue, but typically requires reduced temporal resolution or prolonged acquisition times.

The simultaneous multi-slice (SMS) acquisition described in Chapter 3 is therefore a promising alternative as it acquires multiple slices at the same time without sacrificing either of those.

Another limitation is the usage of PPG gating over more typical ECG gating for cardiac MRI. While ECG signals are more susceptible to noise due to magneto-hydrodynamic effects from the scanner, PPG signals could potentially have less accurate triggering due to a broader R wave peak and physiological delays from measuring the cardiac signal at the finger, which could have considerable impacts on estimated PWVs [83]. Further exploration in a study acquiring both ECG and PPG signals and comparing PWVs values calculated between them may be instructive.

Lastly, one factor that was not explored in this study was the physiological effects of respiratory phase (e.g., forced breath-hold compared to free-breathing state) in PWV. Other studies have investigated this and noted significant differences [84–86], however no exploration of the effects of free-breathing and respiratory phase has been done. With a free-breathing acquisition, it would be possible to retrospectively bin data according to respiratory phase and evaluate these effects on PWV. This avenue will be further explored in Chapter 5.

Future technical improvements could include the application of machine learning for automating aorta centerline tracing and 2DPC segmentation to decrease processing time, increase accuracy of vessel area measurements, and increase repeatability. This would also allow for an additional measure of local PWV ("QA" method), that requires accurate flow and area quantification [87].

4.5. Conclusion

In conclusion, the radial, free-breathing 2DPC MRI sequence offers a viable imaging technique for assessing aortic stiffness which may be particularly beneficial for subjects with breath-hold difficulty. The radial sequence offers a great deal of flexibility for physiological gating and for image reconstruction, which allowed us to generate a high temporal resolution dataset using local low rank reconstruction. Mean pulse wave velocities were comparable between the radial and Cartesian acquisitions and are consistent with other studies, however, the correlation between paired measures is relatively poor. Pulse wave velocities were tabulated in our cohort of 45 cognitively unimpaired older adults and showed strong associations with age. Future studies should seek to study intracranial hemodynamics alongside the obtained aortic pulse wave velocity data.

Acknowledgements

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Chapter 5: Respiratory Effects on Aortic Pulse Wave Velocity Measurements with a Radial Simultaneous Multi-Slice 2D Phase Contrast Acquisition

In this chapter, I investigate the effects of controlled respiration on pulse wave velocity via the radial simultaneous multi-slice 2D phase contrast acquisition described in Chapter 3. I acquire data in a small volunteer cohort to establish the feasibility of this technique and explore if differences in PWV can be observed across respiratory phases and maneuvers. Additionally, I test this in a controlled respiration exercise to evaluate its effects on pulse wave velocity. This work resulted in 3 first author abstracts [51, 52, 88].

5.1. Introduction

As described in Chapter 3, pulse wave velocity (PWV) is a well-established biomarker for arterial stiffness and an important predictor of cardiovascular morbidity and mortality [41, 42]. In Chapter 4, I explored the differences between PWVs acquired with a Cartesian acquisition during a breath-hold and with a radial acquisition during free-breathing, specifically in end-exhalation. Other studies have explored the effects of respiration on PWV measurements, particularly the effects of short-term breath-holds [84–86]; however, exploration of steady-state breathing phases on PWV with MRI is limited.

Respiration plays a fundamental role in modulating cardiovascular dynamics through complex mechanical interactions. During inhalation, the decrease in intrathoracic pressure enhances venous return to the right heart, leading to increased right ventricular preload and a transient reduction in left ventricular stroke volume [89, 90]. Concurrently, the expansion of the thoracic cavity reduces aortic pressure potentially altering wave propagation characteristics [89, 90]. In contrast, exhalation is associated with increased intrathoracic pressure, reduced venous return, and relatively increased left ventricular stroke volume, which may result in higher systolic pressures and pulse wave transmission [89, 90].

In addition to normal respiratory variation, pathology-induced breathing patterns can also have effects on cardiovascular function. Dynamic hyperinflation is a phenomenon common in patients with Chronic Obstructive Pulmonary Disease (COPD) where airflow obstructions can impair exhalation [91]. This leads to an increased accumulation of air in lungs which results in shallow, strained breathing and an inability to perform exercise [92]. Dynamic hyperinflation is thus associated with poor cardiovascular response and increased arterial stiffness, so there is strong clinical interest in evaluating these effects through PWV measurements [93–95]. Metronome-paced tachypnea (MPT) is a breathing technique characterized by rhythmic inhalation and exhalation according to a steady tempo (often 40 breaths/min) and is used for inducing dynamic hyperinflation in patients with COPD [96]. This has also previously been applied in pulmonary MRI studies [97].

In Chapter 3, we introduced and validated a radial 2DPC simultaneous multi-slice (SMS) acquisition for measuring PWV. The flexibility of the retrospectively gated acquisition allows us to acquire data under free-breathing conditions as well as reconstruct different respiratory phases beyond end-exhalation. It also enables the acquisition of flow data under other modes of respiration such as MPT. In this study, we propose the use of this SMS sequence to measure PWV under varying respiratory conditions. Specifically, we aim to compare PWV measurements obtained during two different respiratory modes: free-breathing (FB) and metronome-paced tachypnea (MPT). Additionally, we aim to compare PWV measurements between respiratory phases (exhalation and inhalation) for both modes of respiration.

5.2. Materials & Methods

MR Imaging Protocol

A cohort of 9 healthy volunteers (all male, 22–36 years old) were recruited for this HIPPA-compliant study under a University of Wisconsin-Madison IRB approved protocol and provided written informed consent. Subjects were scanned on 3T SIGNA Premier system (GE Healthcare, Waukesha, WI) with a 30-channel chest coil and 40-channel posterior table coil. For physiological gating, cardiac waveforms were recorded with an electrocardiogram (ECG) system while respiratory motion was recorded using respiratory bellows attached around the diaphragm. Ungated anatomical balanced steady state free precession (bSSFP) product sequences (axial, coronal, sagittal) were acquired for aortic centerline tracing and vessel path length calculation (see Section 3.2 for detailed acquisition parameters). For each subject, 2 radial SMS PC acquisitions with an SMS factor of 3 (3 slices) were acquired (see Section 3.2 for additional details on construction of the sequence). The spacing between slices was adjusted per subject such that the superior slice bisected the aortic arch at the bifurcation of the pulmonary artery and the inferior slice was located just above the renal arteries. This resulted in the intermediate slice falling equidistant between the other two at approximately the location of the thoracic aorta. Between the 3 slices, 4 total regions of interest (ROIs) were captured: the ascending aorta (AAo), descending aorta (DAo), thoracic aorta (ThoAo), and abdominal aorta (AbdAo). Scan parameters for the SMS acquisitions were: TR/TE = 8.4/5.0 ms; flip = 10°; venc = 150 cm/s; velocity encoding scheme = 2-point; golden-angle sampling; acquired FOV = 320 mm; slice thickness = 5 mm; acquired spatial resolution = 1.40×1.40 mm^2, reconstructed cardiac frames = 40.

The first SMS acquisition was acquired under FB conditions with 20,000 radial projections for a total scan time of 336.3 seconds. The second acquisition was acquired under MPT with 8,000 radial projections for a total scan time of 134.9 seconds. To pace the subject for the MPT acquisition, a metronome sound with a tempo of 80 beats/min was played over the scanner intercom system and subjects were instructed to breathe in and out on each beat resulting in a breathing rate of 40 breaths/min. Subjects began the acquisition breathing freely until the 30-second mark, at which point the metronome was started. MPT continued for the remaining 105 seconds until the end of the acquisition (see Figure 5.1).

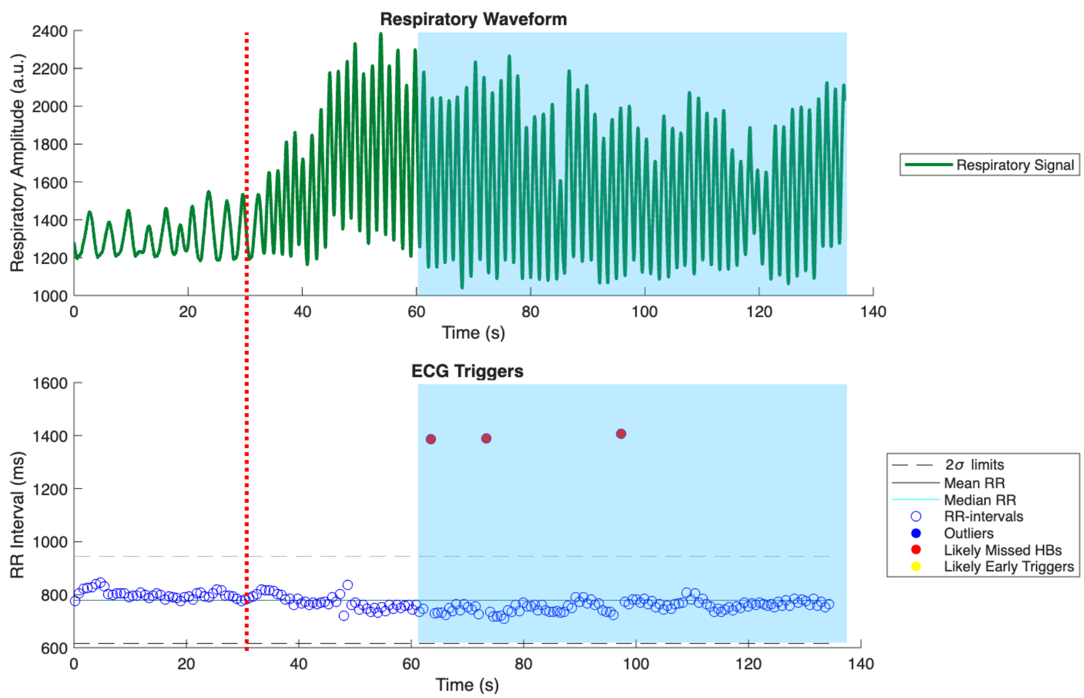

Figure 5.1: Physiological waveforms acquired during metronome-paced tachypnea. Acquisition was started free-breathing until the 30-second mark when a metronome with a tempo of 80 beats/min was played over the scanner intercom system (red dotted line). Subjects breathed in and out on every beat (40 breaths/min). MPT images were reconstructed using data from the 60-second mark onwards (shaded blue box, 75 seconds total) where heart rates appeared to stabilize.

The first 30 seconds after MPT breathing begins is dominated by transient effects in heart rate and respiration. As such the MPT reconstruction utilized data acquired only after reaching a steady-state heart rate (approximately around the 60-second mark). Thus, only the later 75 seconds of each MPT acquisition were reconstructed. Respiratory thresholding was performed with a 50% threshold of the respiratory signal with a sliding window to account for respiratory drift. Data above and below the respiratory threshold were utilized to reconstruct MPT images both inhalation (MPT-In) and exhalation (MPT-Ex). In

Chapter 4, a 10-second sliding window was utilized, however, here a 5-second window was used to account for the faster breathing rate. The FB acquisitions were similarly thresholded to reconstruct images in inhalation (FB-In) and exhalation (FB-Ex). Additionally, the FB acquisitions were retrospectively subsampled down to match the number of projections utilized in the MPT reconstructions to provide a fair comparison in image quality.

Both SMS acquisitions were reconstructed using a similar framework to the one described in Section 3.2, but with the addition of a constrained reconstruction approach to improve image quality for the highly undersampled images. The iterative method described in Equation 3.6 was modified by adding a sparsifying wavelet transform W and an L1 norm to perform a compressed sensing (CS) reconstruction [30].

$$\min_x \frac{1}{2}\|DCFSx - y\|_2^2 + \frac{\lambda}{2}\|Wx\|_1 \tag{5.1}$$

A gradient descent algorithm with 30 iterations was used to solve Equation 5.1 for the final images.

Analysis

Prior to analysis, reconstructed images and physiological gating signals were inspected for general image quality, proper plane placement, presence of velocity aliasing, and poor gating signals. Datasets that failed to meet quality standards for any of those reasons were excluded from analysis. The PWV analysis tool described in Sections 3.2 and 4.2 was used trace aortic centerlines from anatomical images, determine aortic path lengths between scan planes, and calculate flow curves from reconstructed SMS images. PWV was calculated with the 4 time-shift algorithms described in Section 3.2: time-to-point (TTP) [58, 59], time-to-foot (TTF) with baseline offset correction [60], time-to-upstroke (TTU) [59], and cross-correlation (XCOR) [61]. The resulting PWVs from each time-shift method were then averaged together to a combined PWV value as suggested by Wentland et al. to reduce the errors inherent to each individual method [42]. Mean inter-method standard deviation and coefficient of variation across methods for the combined PWVs were calculated. Two-way repeated measures ANOVA was performed to compare the statistical significance of FB vs MPT and exhalation vs inhalation, where statistical significance is defined as $p < 0.05$. Pairwise post-hoc tests were performed between each respiratory mode and phase to assess statistical differences. In addition to PWV analysis, analysis of the recorded heart rates was also performed to further assess the relationships between FB vs MPT. Heart rate was calculated from the mean of the recorded R-R intervals utilized in the reconstruction via the cardiac gating signal.

5.3. Results

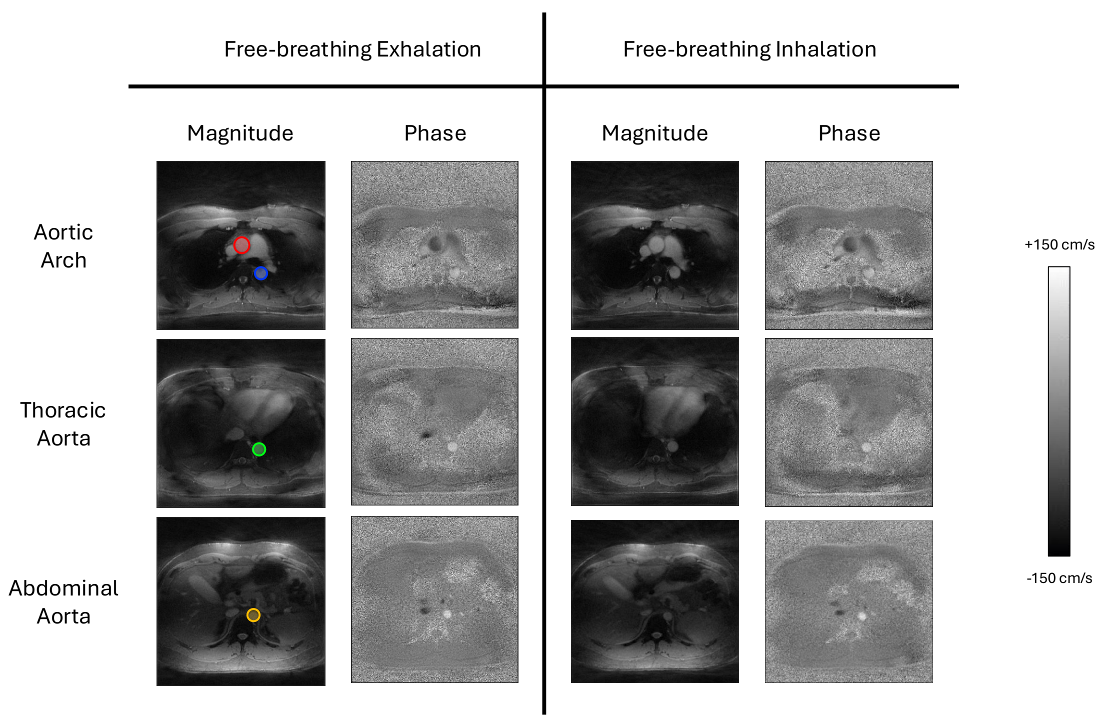

Figure 5.2: Representative images of time-averaged simultaneous multi-slice (SMS) 2D Phase Contrast (2DPC) images acquired during free-breathing. Images on the left were reconstructed using a 50% respiratory threshold for exhalation while images on the right used the remaining 50% for inhalation. Colored circles on the exhalation magnitude images indicate locations where flow curves were measured for pulse wave velocity calculation: red = ascending aorta, blue = descending aorta, green = thoracic aorta, and orange = abdominal aorta.

Respiratory-resolved SMS images were successfully reconstructed for all 9 subjects for both respiratory modes and phases. All images were found to be of sufficient image quality for analysis. Figure 5.2 depicts representative images of reconstructed time-averaged SMS phase contrast images under free-breathing conditions. Colored circles indicate the 4 ROIs where flow curves were measured for PWV analysis. PWVs from the 4 different time-shift methods were calculated via the MATLAB GUI tool and averaged together as a combined PWV for each respiratory phase. Mean inter-method standard deviation was 0.50 m/s while the coefficient of variation was 9.0% indicating relative variability was low and that averaging time-shift methods provides a representative value for subsequent analysis.

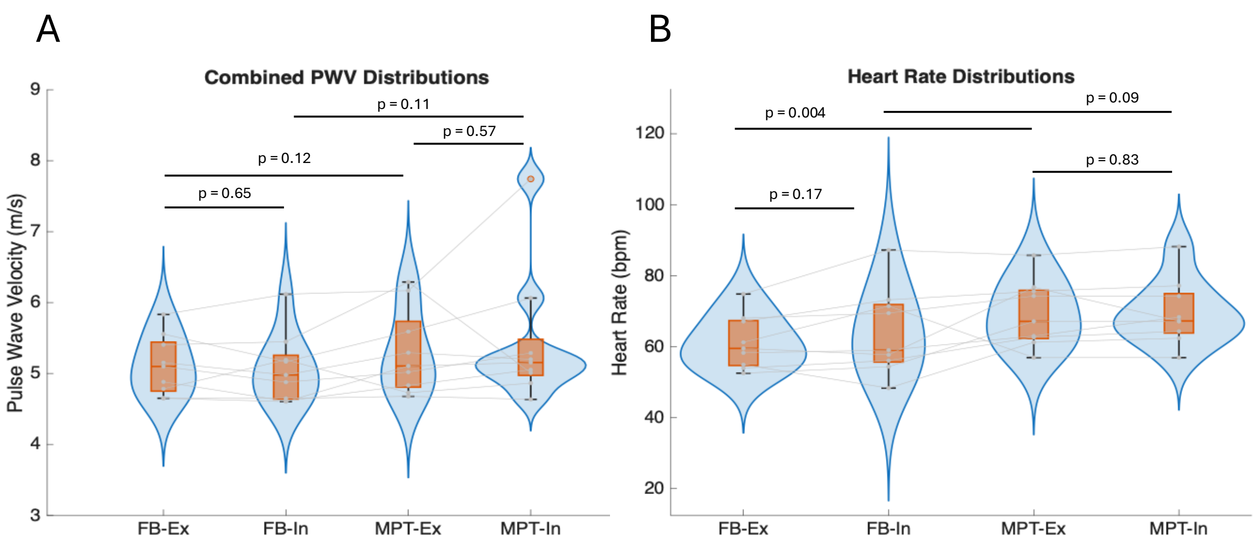

Figure 5.3: (A) Violin and box plots illustrating the distributions of combined pulse wave velocities (PWV) for free-breathing (FB) and metronome-paced tachypnea (MPT) under exhalation (Ex) and inhalation (In). (B) Violin and box plots for heart rate distributions across the same respiratory phases as in (A)

Figure 5.3 illustrates the distributions of combined PWVs as well as measured heart rates for both respiratory modes and phases. Average $\pm$ standard deviation for combined PWVs was 5.11 $\pm$ 0.4 m/s for FB-Ex, 5.07 $\pm$ 0.5 m/s for FB-In, 5.30 $\pm$ 0.6 m/s for MPT-Ex, and 5.45 $\pm$ 1.0 m/s for MPT-In. Qualitatively, it is apparent that MPT-In spans the largest range in PWVs with the most outliers. PWVs showed close agreements between respiratory phases with small increases between FB and MPT. ANOVA tests showed a statistically significant difference between respiratory modes ($p = 0.04$) but no significant differences between respiratory phases ($p = 0.71$). However, pairwise post-hoc tests showed no significant differences between modes ($p > 0.11$). Similarly, for heart rates ANOVA tests showed a significant difference between respiratory modes ($p = 0.02$) but not between respiratory phases ($p = 0.26$). Pairwise post-hoc tests did show a significant difference between modes ($p = 0.004$) for heart rate, but not between phases ($p = 0.09$).

5.4. Discussion

In this study, we demonstrated the feasibility of radial, free-breathing simultaneous multi-slice (SMS) phase contrast MRI for the assessment of aortic pulse wave velocity (PWV) under both free-breathing (FB) and metronome-paced tachypnea (MPT) conditions. Multiple simultaneously acquired slices combined with heavy undersampling and a compressed sensing (CS) reconstruction enabled the acquisition of cardiac-resolved images with sufficient temporal resolution for PWV analysis in a short time period (75 seconds) with no temporal or physiological misalignment. To our knowledge, we are the first group to assess PWV under MPT.

A primary objective of this work was to evaluate whether PWV varies across respiratory phases. While physiological mechanisms suggest that intrathoracic pressure changes during inhalation and exhalation could influence pulse wave propagation, there was no statistically significant difference in PWV between respiratory phases. This may suggest that respiratory phase may have a limited impact on global aortic stiffness in healthy subjects under steady-state conditions.

In contrast, a statistically significant effect of respiratory mode (FB vs MPT) was observed in the ANOVA analysis for both PWV and heart rate. However, the absence of significant differences in pairwise post-hoc comparisons for PWV suggests that differences in PWV may not reflect a systematic physiological difference. Mean PWV values across all respiratory phases (5.0–5.4 m/s) fall within the expected physiological range for healthy young adults, so the statistical difference may be the result of variability across subjects [98]. A larger cohort is necessary to improve the statistical power of the analysis.

One potential confounding factor in the difference between PWVs in different respiratory modes may be the effect of motion artifacts under MPT. MPT induces rapid, shallow breathing and due to the nature of radial acquisitions, this extra motion may introduce increased diffuse pseudo-noise artifacts to the overall image. These artifacts can compromise image quality and thus flow measurements which impacts PWV. An increased variability in PWVs was observed under MPT, particularly during inhalation, which supports this theory. Inhalation is typically more prone to residual motion artifacts with respiratory thresholding as it is less consistent at maintaining the same position over time than exhalation [99].

Heart rate analysis provided additional context for interpreting the PWV results. The observed increase in heart rate during MPT relative to FB is consistent with known physiological responses to tachypnea [100]. However, there were no significant differences in heart rate across respiratory phases which supports the validity of retrospective cardiac gating which relies on consistent heart rates.

The averaging of multiple time-shift algorithms (TTP, TTF, TTU, and XCOR) and simultaneous acquisition of multiple slices proved effective in reducing the effect of erroneous PWV calculations and improving the reliability of PWV measurements. The low inter-method variability indicated strong agreement and suggests that the combined PWV metric may provide a more robust representation of vascular stiffness than any single method alone [42].

Limitations

Several limitations of this study should be acknowledged. First, the sample size was relatively small and limited to healthy young male volunteers, which restricts the generalizability of the findings. Future studies should include a larger and more diverse cohort, including subjects with pathologies such as COPD, to evaluate the clinical utility of the proposed method. Second, while CS reconstruction produced acceptable image quality from highly undersampled data, considerable undersampling artifacts were still present. Further optimization of the CS spatial regularization parameter may be needed to reduce the effects of these image artifacts, however, PWV calculations are hypothesized to be robust to these undersampling artifacts as long as the general waveform shape is maintained. Applying a local low rank reconstruction as was applied in Chapter 4 may help improve temporal resolution, though it was not demonstrated to have a substantial effect in that study [31, 32]. Lastly, the effect of breath-holds and the comparison to free-breathing was not investigated here due to time constraints. As noted in the literature, there may be effects of breath-holding on PWV and this SMS sequence may be used to assess this effect with MR-derived PWV. Additionally, there exist other modes of controlled respiration, such as the Valsalva and Mueller maneuvers, which have complex physiological mechanics and effects on the cardiovascular system and have been explored previously in the context of PWV measurements [101, 102]. However, due to the changing heart rates involved, acquisitions with retrospective cardiac gating may not be feasible as they rely on steady heart rates; thus, a real-time acquisition may be necessary to explore these breathing modes.

5.5. Conclusion

In this work, we demonstrated the feasibility of radial simultaneous multi-slice (SMS) phase contrast MRI for the assessment of aortic pulse wave velocity (PWV) under varying respiratory conditions, including both free-breathing (FB) and metronome-paced tachypnea

(MPT). The proposed framework enabled respiratory-resolved reconstruction and flow analysis within a short acquisition window, while maintaining sufficient temporal fidelity for robust PWV estimation. The results indicate that PWV measurements in healthy subjects remain largely consistent across respiratory phases, suggesting that global aortic stiffness is relatively insensitive to steady-state inhalation and exhalation. While differences between respiratory modes were observed at the statistical level, these did not translate into consistent pairwise differences in PWV, and are likely influenced by increased variability under MPT conditions. Nevertheless, the ability to acquire PWV measurements during controlled rapid breathing demonstrates the flexibility of the SMS approach and its potential applicability to more complex physiological or pathological conditions.

Acknowledgements

Special thanks to Grant Roberts and Kevin Johnson for the contributions to the development of this sequence.

Chapter 6: Optimizing Venc in Radial Free-Breathing 4D Flow MRI for the Portal Venous System

In this chapter, I shift focus from the use of 2D phase contrast imaging in the aorta to accelerating 4D flow acquisitions in the liver. Namely, I apply radial free-breathing 4D flow for hemodynamic analysis of the portal vein and optimize velocity encoding settings to streamline acquisitions. This work resulted in 2 first author abstracts [103, 104] and a first author manuscript currently under preparation.

6.1. Introduction

Portal hypertension is a condition described by elevated pressure within the portal venous system which delivers blood to the liver from the gastrointestinal tract, spleen, pancreas, and gallbladder [105, 106]. Clinically, portal hypertension is defined by a hepatic venous pressure gradient (HVPG) exceeding 10 mm Hg and is strongly associated with the development of severe complications, including gastroesophageal varices (GEVs) [107, 108]. GEVs are portosystemic collateral vessels that arise as a compensatory mechanism to decompress the portal venous system. In this process, blood is diverted away from the liver through alternative vascular pathways, most notably via reversed flow in the left gastric vein (LGV) [109, 110]. This reversal of flow leads to progressive dilation of submucosal veins in the esophagus and stomach [109, 110]. Due to their thin walls and elevated intraluminal pressure, these varices are highly susceptible to rupture, which can result in life-threatening gastrointestinal hemorrhage [111]. Variceal bleeding is associated with high morbidity and mortality, underscoring the importance of early identification and risk stratification of high-risk varices [107, 111].

Current clinical guidelines recommend screening for GEVs using esophagogastroduodenoscopy (EGD), an invasive endoscopic procedure that allows direct visualization of the esophageal and gastric mucosa [111]. While EGD is considered the gold standard for diagnosis, it is associated with several limitations, including high cost, patient discomfort, and procedural risks such as sedation-related complications and esophageal injury [112]. Furthermore, only a limited number (approximately 7% yearly) of cirrhotic patients develop substantial varices requiring intervention [108]. Consequently, routine screening of all cirrhotic patients using EGD results in a suboptimal risk-to-benefit ratio, signifying the need for non-invasive diagnostic alternatives.

In recent years, four-dimensional flow Magnetic Resonance Imaging (4D flow MRI) has emerged as a promising non-invasive modality for the assessment of complex vascular hemodynamics, including in the portal venous system [105, 113]. This technique enables time-resolved, three-dimensional visualization and quantification of blood flow velocities within the portal venous system and its associated collateral pathways. By capturing comprehensive hemodynamic information, 4D flow MRI has the potential to identify biomarkers indicative of portal hypertension and the formation of GEVs such as reversed flow in the LGV, thereby facilitating early detection and improved patient stratification [114, 115].

Despite its advantages, widespread clinical adoption of 4D flow MRI is hindered by several technical challenges. One of these is acquiring sufficient velocity-to-noise ratio (VNR) for accurate hemodynamic assessment in clinically feasible scan times. As discussed in Section 2.4, optimal setting of the velocity encoding parameter (venc) can be challenging. If the venc is set too low, the phases can wrap around leading to incorrect flow quantification due to velocity aliasing. Alternatively, if the venc is set too high, the VNR is decreased which also impacts flow quantification. Typical rule of thumb for phase contrast acquisitions is to apply a venc approximately 10% above the peak velocity [36]. However, this requires knowing the peak velocity in the vessel of interest a priori. This is often solved by acquiring 2D phase contrast (2DPC) "scout scans" that are performed before the 4D flow acquisition, yet this requires additional time for prescription, scanning, and analysis and scan time is at a premium in the clinic [116]. Additionally, a single 2DPC scan can sometimes be insufficient to fully describe complex flow dynamics in subjects with complicated or diseased vasculature; there is a risk that the venc might be set too low or high even with a scout scan.

A number of methods have been proposed to solve this issue. One approach is the utilization of dual-venc methods, which apply two sets of velocity encoding gradients to acquire both low and high venc settings in a single acquisition [117, 118]. This then allows for unwrapping of the low venc data using fully registered high venc data, however, the trade-off is a considerable increase in scan time. As an alternative, multiple post-processing techniques have been proposed to correct aliased velocity data for single-venc acquisitions. Such approaches include gradient-based methods, graph-cut-based methods, and deep learning networks [117, 119, 120]. Another approach is a 4D Laplacian-based phase unwrapping method proposed by Loecher et al. that has demonstrated promise in resolving phase wraps by leveraging spatial and temporal continuity constraints [121]. This approach was found to outperform other phase unwrapping methods in a recent study by Löcke et al. focused on the aorta [122]. The main advantage of this method is that it is a single-step algorithm with zero user-input required; this method can therefore be easily be incorporated into a reconstruction pipeline and computed with little additional computational overhead.

In this work, we investigate the application of this 4D Laplacian phase unwrapping approach for correcting velocity aliasing in 4D flow MRI of the portal venous system in patients with GEVs. We acquire multiple 4D flow datasets with low, medium, and high venc settings and apply phase unwrapping to identify the effectiveness of the method, with the ultimate goal of identifying the lowest single venc setting that can be used with phase unwrapping to provide images conducive for GEV evaluation.

6.2. Materials & Methods

4D Laplacian Phase Unwrapping

The 4D Laplacian phase unwrapping algorithm was implemented in a C++ based reconstruction framework for radial 4D flow MRI according to Loecher et al [121]. In this approach, the true phase $\phi(r)$ is assumed to be related to the measured phase $\phi_w(r)$ according to

$$\phi(r) = \phi_w(r) + 2\pi n(r) \tag{6.1}$$

where $n(r)$ describes the number of phase wraps of the voxel at position r. The Laplacian of Equation 6.1 is then

$$\nabla^2 \phi(r) = \nabla^2 \left[\phi_w(r) + 2\pi n(r)\right] \tag{6.2}$$

which has the solution

$$n(r) = \frac{1}{2\pi} \nabla^{-2} \left[\nabla^2 \phi(r) - \nabla^2 \phi_w(r)\right] \tag{6.3}$$

where ∇^2 and ∇^{-2} are the forward and inverse Laplacian operators, constructed via convolution with a kernel of finite difference second derivatives. The kernel is scaled by the voxel size squared in each spatial dimension as well as an additional scaling factor in the temporal dimension. As the true phase is unknown, the Laplacian of the true phase must be estimated via

$$\nabla^2 \phi(r) = -ie^{-i\phi_w(r)\nabla^2 e^{i\phi_w(r)}} \tag{6.4}$$

which is then simplified to

$$\nabla^2 \phi(r) = \cos\left[\phi_w(r)\right] \nabla^2 \sin\left[\phi_w(r)\right] - \sin\left[\phi_w(r)\right] \nabla^2 \cos\left[\phi_w(r)\right] \tag{6.5}$$

MR Imaging Protocol

Nineteen subjects (7M/12F, age 34–73y/mean=54y) were recruited for this HIPPA-compliant study under a University of Wisconsin-Madison IRB approved protocol and provided written informed consent. Subjects were eligible for this study if they were: 18 years or older; had known cirrhosis; or a known GEV. Exclusion criteria included: contraindictions to MRI and/or contrast agent; recent treatment (< 1 year) for varices with embolization, transjugular intrahepatic portosystemic shunt (TIPS), or other endovascular treatment; active GEV bleeding; known occlusive thrombus in portal vein, splenic vein, or superior mesenteric vein; or a large hepatocellular carcinoma (HCC) with known portal vein involvement. Every subject who participated in the study underwent a 5-hour fast prior to visit (to avoid meal effects on abdominal flow [114, 123]), MRI safety screening, height and weight measurements, and intravenous administration of Ferumoxytol MRI contrast agent (4 mg/kg, off-label use)[124]. Additionally, all cirrhotic subjects underwent an EGD and were graded based on presence/rupture risk of GEVs (1=no varices to 3=large varices). This cohort included 14 subjects with grade 1 varices and 5 with grade 3 varices.

Three 4D flow scans, with low (15 cm/s), medium (30 cm/s), and high (60 cm/s) venc settings, were acquired for all 19 subjects on a clinical 3T SIGNA Premier scanner (GE Healthcare, Waukesha, WI) with a 30-channel chest coil and 40-channel posterior table coil. All 4D flow scans were acquired with a radial, free-breathing 4D flow sequence (PCVIPR) with 5 point flow encoding and Inner Volume Excitation (IVE) to enable fat suppression in obese subjects [10, 38, 125]. Different vencs naturally resulted in differing values of TE and TR which in turn led to different scan times. As such the number of acquired radial projections were adjusted for each acquisition to match scan times. Scan parameters are described in detail in Table 6.1. Acquisition order of each venc was randomized between subjects to prevent influence on the analysis. Electrocardiogram (ECG) and respiratory bellows signals were acquired during the scans for retrospective cardiac and respiratory gating.

Table 6.1: Acquisition parameters for radial, free-breathing 4D flow acquisitions with multiple velocity encoding parameters

Acquisition	**Low Venc**	**Medium Venc**	**High Venc**
Venc (cm/s)	15	30	60
TE (ms)	3.2	2.6	2.2
TR (ms)	8.6	8.0	7.6
Flip angle	14°	14°	14°
Acquired Field of View (cm^3)	48 × 48 × 24	48 × 48 × 24	48 × 48 × 24
Acquired spatial res. (mm isotropic)	1.25	1.25	1.25
Scan time (s)	718.9	716.8	718.4
Radial projections	16800	18000	19000
Velocity encoding	5 point	5 point	5 point

4D flow datasets were reconstructed for each of the 3 venc settings using an iterative CG-SENSE algorithm for 20 iterations with gradient non-linearity and background phase correction [15]. Data were reconstructed with a 50% respiratory threshold (end-exhalation) and a 10-second sliding window (to correct respiratory drift) to produce time-averaged and time-resolved outputs (14 cardiac frames, temporal resolution = 40–50 ms depending on heart rate). For each dataset, three velocity images for x, y, and z were produced, plus magnitude and a complex difference angiogram. A second reconstruction was performed for the low and medium venc datasets with identical parameters using the 4D Laplacian phase unwrapping. The high venc dataset was not reconstructed with phase unwrapping as it was assumed there would be no phase wraps and served as reference with the true velocity values.

Analysis

Velocity aliasing was identified via manual inspection of time-resolved images by a single observer (T.N. with 5 years experience in 4D flow MRI analysis). Flow analysis and hemodynamic visualization was performed in GTFlow (Gyrotools, Zurich, Switzerland). Time-averaged datasets were imported for the medium venc dataset and segmentation of the portal vein was performed on the complex difference angiogram. Additionally, 10 cross-sectional flow analysis planes were placed at each of the major branches of the portal vein accord-

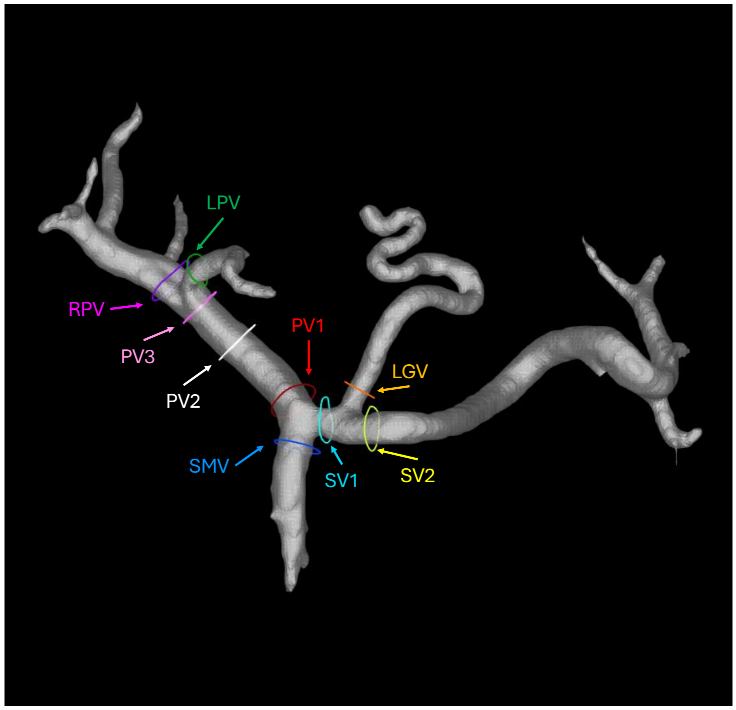

Figure 6.1: Locations of cross-sectional flow analysis plane placement in the portal vein. Planes were placed at the following locations: Superior Mesenteric Vein (SMV), Splenic Vein (SV) x2, Left Gastric Vein (LGV) (when present), Portal Vein (PV) x3, Left Portal Vein (LPV), Right Portal Vein (RPV).

ing to Figure 6.1: Superior Mesenteric Vein (SMV), Splenic Vein (SV) x 2, Left Gastric Vein (LGV) (when present), Portal Vein (PV) x 3, Left Portal Vein (LPV), and Right Portal Vein (RPV). Segmentations and plane placements were performed by a single observer (T.N.) blinded to clinical information and were verified by a cardiovascular radiologist with 6+ years of experience. Vessel segmentations and analysis plane locations were saved to reuse on the time-resolved data for the low, medium, and high venc datasets. In the case where subjects had considerable bulk motion between subsequent acquisitions, affine image registration using Advanced Normalization Tools (ANTs) was used to register the low and high venc datasets to the medium venc dataset [126]. For minor motion or shifting of vessels, contours were simply adjusted manually when reapplying to a new dataset.

Time-averaged streamlines and time-resolved vector fields were generated for each analyzed dataset using GTFlow. Flow and velocity measures including net flow, peak flow, and

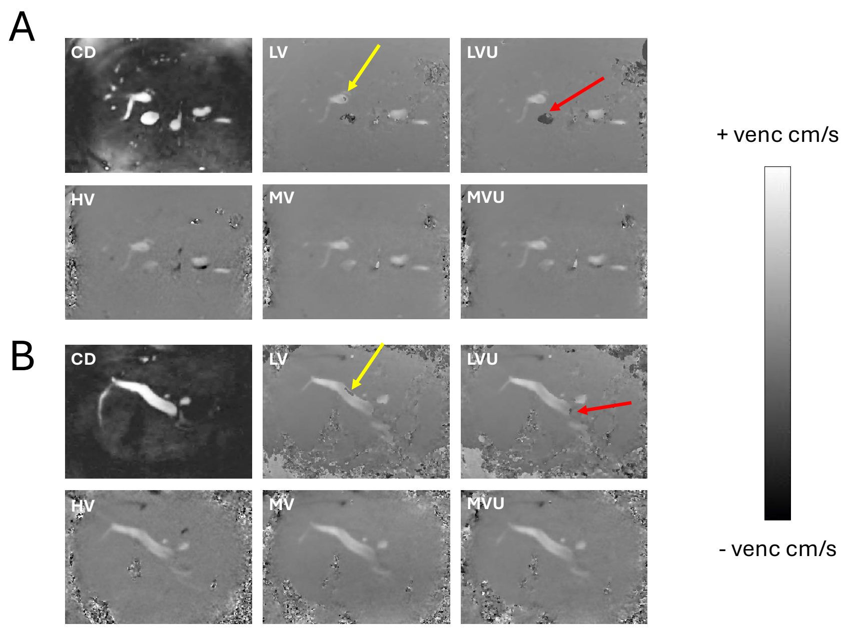

Figure 6.2: (A) Axial and (B) Sagittal views of reconstructed images for an example subject. Complex difference (CD) images provide anatomical visualization. LV = Low Venc (15 cm/s), LVU = Low Venc Unwrapped, MV = Medium Venc (30 cm/s), MVU = Medium Venc Unwrapped, HV = High Venc (60 cm/s). Yellow arrows indicate locations where 4D Laplacian algorithm successfully unwrapped aliased velocities. Red arrows indicate locations failed to unwrap aliased velocities.

max velocity were generated for each plane. Internal consistency of flow measurements was assessed by calculating conservation of mass at both confluence and bifurcation of the portal vein by summing the net flow Q into and out of each branch point according to

$$\begin{aligned} \text{PV Confluence}: Q_{SV} + Q_{SMV} &= Q_{PV1} \\ \text{PV Bifurcation}: Q_{PV3} &= Q_{LPV} + Q_{RPV} \end{aligned} \tag{6.6}$$

Linear regression, Pearson correlation (r^2), and Bland-Altman analysis was performed to evaluate conservation of mass at both branch points and sum of squares error (SSE) was calculated to assess agreement of the results.

To evaluate voxel-wise velocity agreements with the reference high venc datasets, the low, medium, unwrapped low, and unwrapped medium venc datasets were normalized by venc and linear regressions against the high venc datasets were plotted for each direction (x, y, z). Pearson correlation coefficients were calculated for each plot.

6.3. Results

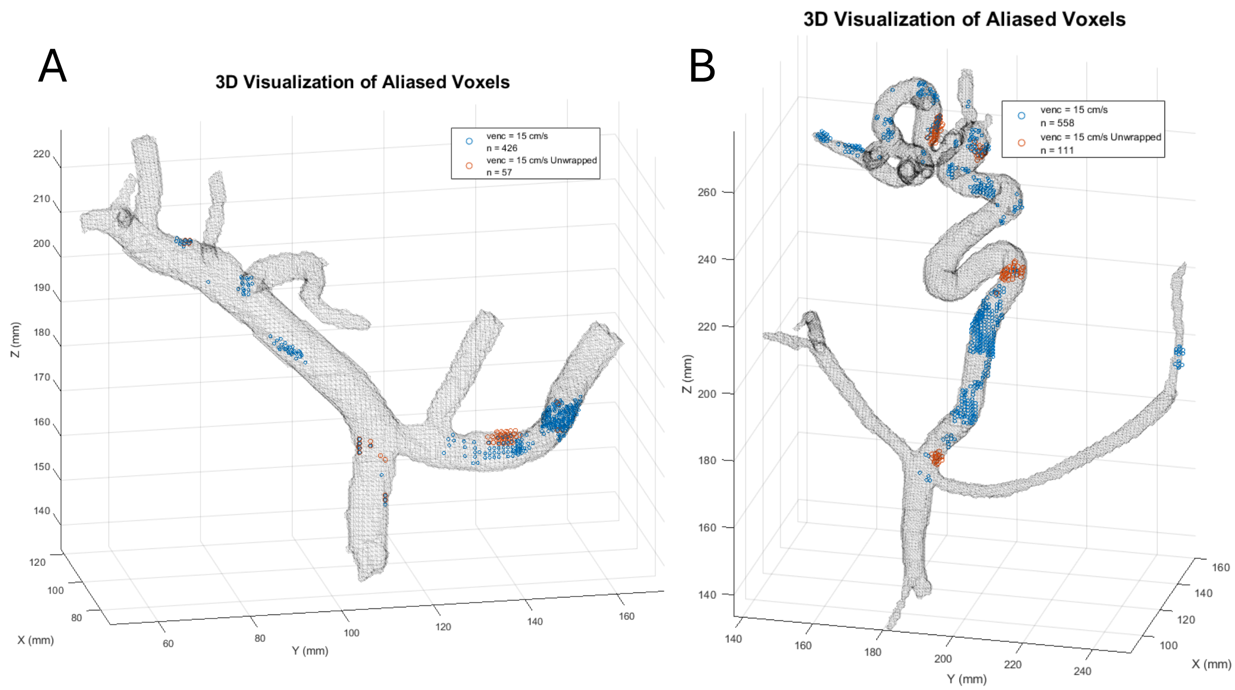

Figure 6.3: Three-dimensional visualizations of aliased voxels in (A) a subject with minimal gastroesophageal varices (GEVs) and (B) a subject with large GEVs. Blue circles indicate voxels with phase wrapping while orange circles indicate voxels with phase wrapping still present after 4D Laplacian phase unwrapping.

Data were successfully acquired for 18 out of 19 subjects. Poor coil signal during acquisition for one subject necessitated exclusion of the dataset from analysis. The 4D Laplacian phase unwrapping algorithm was successfully applied to all low (15 cm/s) and medium (30 cm/s) venc datasets. Qualitative assessment demonstrated that the algorithm effectively corrected a substantial portion of aliased voxels, as illustrated in Figure 6.2. However, residual aliasing persisted in other regions, especially near vessel edges and regions with collateral vessels (Figure 6.3).

Flow analysis showed improved streamline calculations for unwrapped low venc datasets

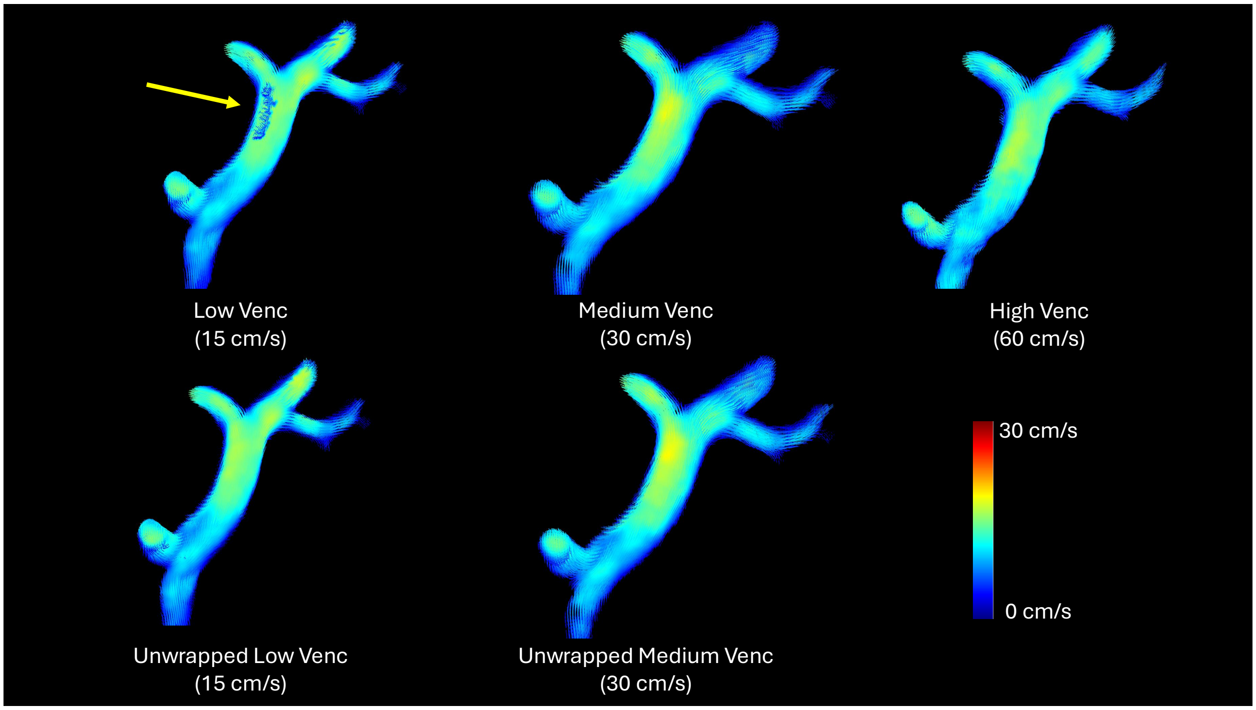

Figure 6.4: Time-averaged streamlines for an example subject computed in GTFlow (Gyrotools, Zurich, Switzerland). Streamlines were generated for low (15 cm/s), medium (30 cm/s), and high (60 cm/s) venc datasets along with 4D Laplacian unwrapped datasets for low and medium. Yellow arrow indicates region with aliased voxels that was successfully unwrapped.

compared to low venc datasets (Figure 6.4). Notably, the medium and unwrapped medium venc datasets demonstrated streamline patterns closely resembling those observed in the high venc reference datasets, likely indicating true velocities were below the medium venc.

Quantitative evaluation using conservation of mass demonstrated strong agreement between inflow and outflow measurements at both the portal vein confluence and bifurcation for the medium, unwrapped medium, and high venc datasets with high correlation ($r^2 = 0.94$, 0.93, and 0.93, respectively) and low dispersion (SSE = 76 mL/s, 88 mL/s, and 76 mL/s) indicating reliable flow quantification (Figure 6.5). Conversely, both the low venc and unwrapped low venc datasets exhibited reduced correlation ($r^2 = 0.70$ and 0.64) and greater dispersion (SSE = 500 mL/s and 640 mL/s, respectively), suggesting persistent inaccuracies in flow estimation.

Bland-Altman analysis further supported these findings (Figure 6.6). The medium, un-

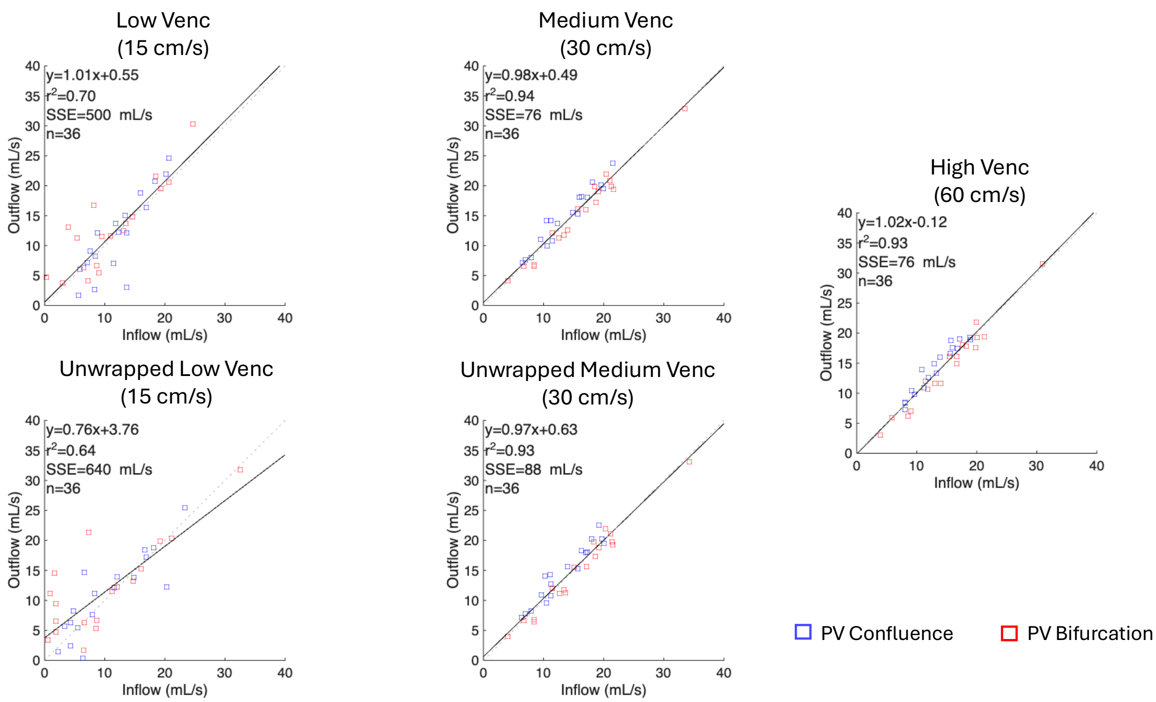

Figure 6.5: Linear regression plots indicating agreement in conservation of mass for the portal vein (PV) confluence and bifurcation for all subjects. Conservation of mass is calculated as $Q_{\text{inflow}} = Q_{\text{outflow}}$. For the confluence, $Q_{\text{Splenic Vein}} + Q_{\text{Superior Mesenteric Vein}} = Q_{\text{Portal Vein}}$ and for the bifurcation, $Q_{\text{Portal Vein}} = Q_{\text{Left Portal Vein}} + Q_{\text{Right Portal Vein}}$. There is strong correlation and agreement between the high, medium, and medium unwrapped vencs, but less so from the low and unwrapped low vencs.

wrapped medium, and high venc datasets demonstrated low standard deviation (SD = 1.5 mL/s, 1.6 mL/s, and 1.5 mL/s), narrow limits of agreement (LOA = 2.9 mL/s, 3.1 mL/s, and 2.9 mL/s), and low coefficients of variation (CV = 9.9%, 11%, and 10%), indicating strong agreement in flow conservation. In contrast, the low and unwrapped low venc datasets showed increased standard deviation (SD = 3.8 mL/s and 4.6 mL/s), wider LOA (LOA = 7.4 mL/s and 9.1 mL/s), and higher CV (CV = 32% and 43%), reflecting reduced reliability.

Voxel-wise velocity comparisons with the high venc reference dataset for an exampled subject (Figure 6.7) demonstrated that the unwrapped medium venc dataset exhibited strong linear correlation across all velocity components for the example dataset ($r_x^2 = 0.809$, $r_y^2 = 0.753$, $r_z^2 = 0.798$). The medium venc showed close agreement as well ($r_x^2 = 0.776$, $r_y^2 = 0.616$, $r_z^2 = 0.746$). Phase unwrapping provided marginal improvements in regions affected

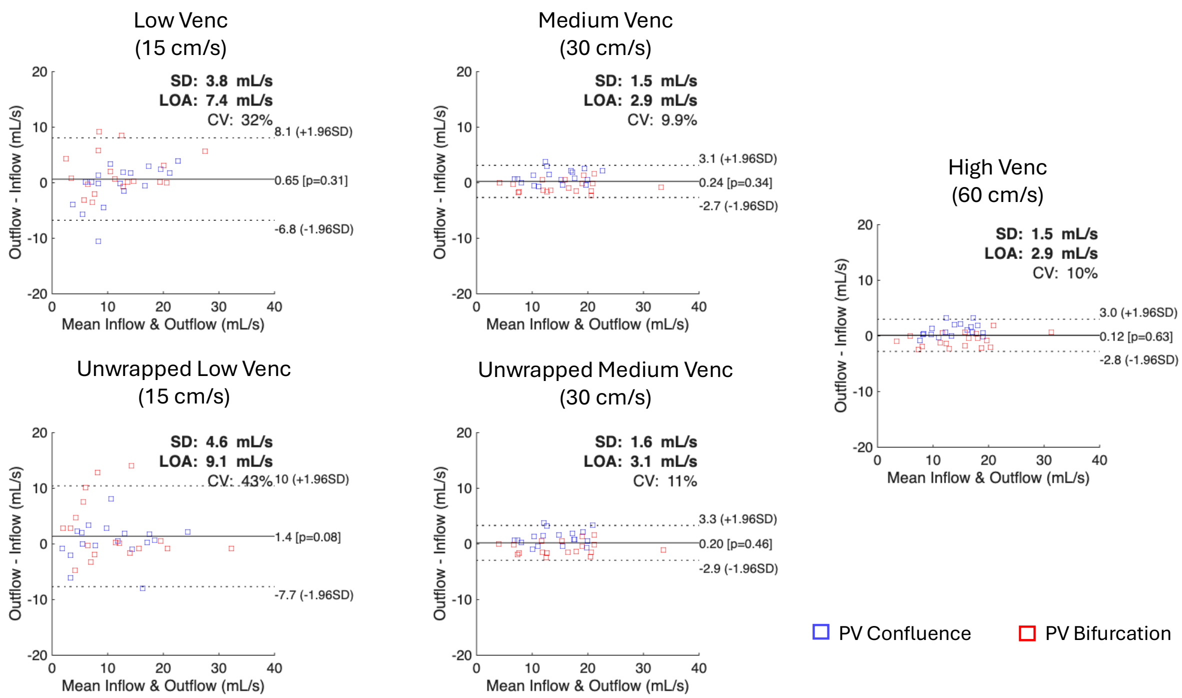

Figure 6.6: Bland-Altman plots indicating agreement in conservation of mass for the portal vein (PV) confluence and bifurcation for all subjects. Conservation of mass is calculated as $Q_{\text{inflow}} = Q_{\text{outflow}}$. For the confluence, $Q_{\text{Splenic Vein}} + Q_{\text{Superior Mesenteric Vein}} = Q_{\text{Portal Vein}}$ and for the bifurcation, $Q_{\text{Portal Vein}} = Q_{\text{Left Portal Vein}} + Q_{\text{Right Portal Vein}}$. The medium, unwrapped medium, and high venc datasets show low standard deviations (SD), close limits of agreement (LOA), and low coefficients of variation (CV) between the high, medium, and medium unwrapped venc datasets. Both low and unwrapped low vencs show poorer agreement.

by aliasing. In contrast, the low venc dataset showed poorer correlation after unwrapping ($r_x^2 = 0.479$, $r_y^2 = 0.062$, $r_z^2 = 0.385$) compared to the original low venc dataset ($r_x^2 = 0.650$, $r_y^2 = 0.202$, $r_z^2 = 0.664$).

6.4. Discussion

In this study, we evaluated the performance of a 4D Laplacian-based phase unwrapping method to correct velocity aliasing across multiple velocity encoding (venc) settings in 4D flow MRI of the portal venous system in patients with gastroesophageal varices. Qualitative and quantitative results showed that the medium venc dataset (30 cm/s), with and without phase unwrapping, had very close agreement with the high venc datasets (60 cm/s) which

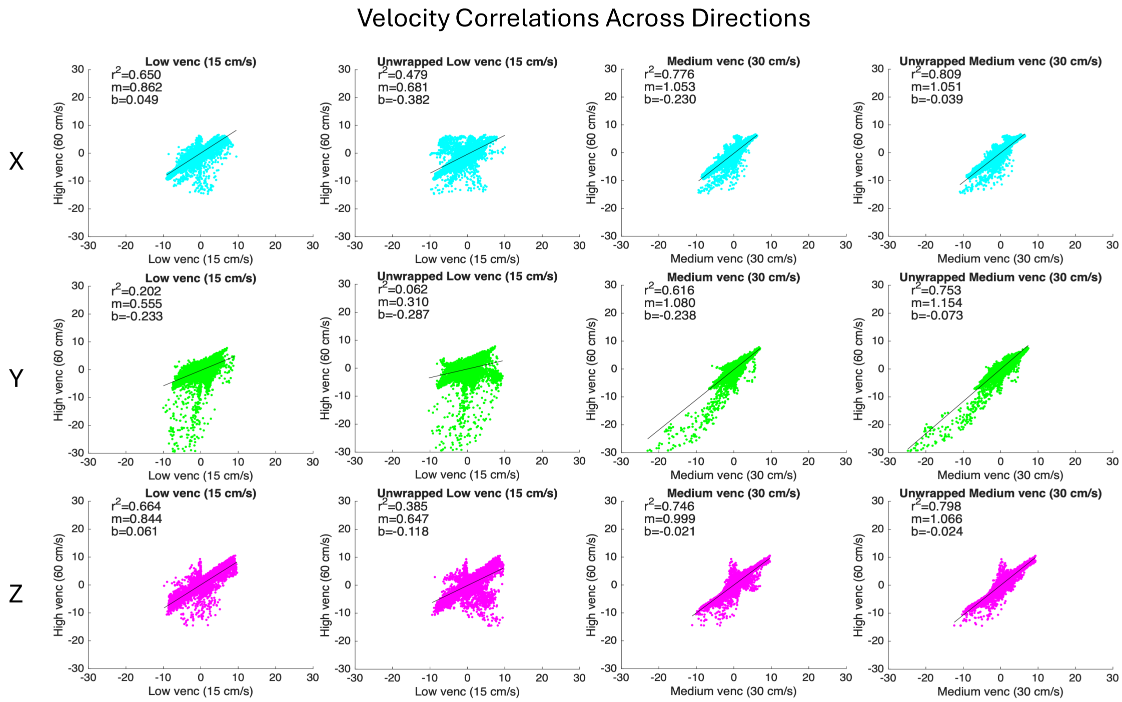

Figure 6.7: Correlation plots for an example subject indicating voxel-wise correlations between (from left to right) low (15 cm/s), phase unwrapped low (15 cm/s), medium (30 cm/s), and phase unwrapped medium venc (30 cm/s) datasets with the ground truth high venc (60 cm/s) dataset. Both medium and unwrapped medium venc data showed strong linear correlation with the high venc data. Low venc data showed poorer correlation while the unwrapped low venc data exhibited the poorest correlation.

were assumed to have the true unwrapped velocities. These results indicate that the medium venc setting is sufficient to capture the majority of physiologic velocities within the portal venous system in this cohort. Furthermore, the close agreement between the unwrapped and non-unwrapped medium venc datasets suggests that aliasing at this venc is relatively limited; no velocities above 30 cm/s were observed in this cohort which agrees with other studies that utilized venc settings of 25–35 cm/s [113, 127].

In contrast to the medium venc, the low venc (15 cm/s) datasets demonstrated considerably reduced accuracy, even after phase unwrapping. Although partial correction of aliased regions was observed in some subjects, quantitative metrics continued to show discrepancies. Conservation of mass analysis showed substantially lower correlation and markedly higher error, while Bland-Altman results indicated increased variability and wider limits of agreement. These findings may imply that the degree of aliasing present in the low venc datasets exceeds the capability of the Laplacian-based method to reliably recover the true velocity field. One reason for this may be due to the size of vessels of interest. As noted in Figure 6.2, the 4D Laplacian method successfully unwrapped a large aliased region in the center of the portal vein, but failed to unwrap a smaller region near the edge. Additionally, the 4D Laplacian also performed some phase unwrapping in the inferior vena cava as indicated by the red arrow in Figure 6.3, though it failed overall as the medium venc images reveal it was double wrapped. As noted in the original paper, the 4D Laplacian method relies on a Laplacian kernel to detect the edges of aliased region [121]. If the phase wrapping is extensive within the vessel, then the Laplacian kernel will fail to detect the aliased regions. Alternatively, if there are partially wrapped voxels near vessel boundaries due to partial volume effects, as was apparent in some collateral vessels (Figure 6.3B), the Laplacian kernel may again fail to unwrap them [121]. These results agree with the study performed by Löcke et al. where they concluded that the 4D Laplacian method outperformed other phase unwrapping methods, but was not perfect.

As mentioned previously, there are alternative methods to the 4D Laplacian-based phase unwrapping. One such promising method is the graph-cut method proposed by Justice et al. which leverages the max-flow min-cut theorem to unwrap voxels in an iterative method [120]. The authors reported marginally superior performance to the 4D Laplacian method in lower SNR conditions, but also noted their method was more computationally expensive. Another alternative is the application of physics-driven neural networks which can utilize the strength of deep learning approaches in solving complex problems [119, 128]. However, this approach would require a considerable amount of training data and computational resources to implement. The 4D Laplacian method by contrast is quick to implement and computationally cheap to run.

From a practical perspective, our results suggest that a single-venc acquisition at an intermediate venc around 25–30 cm/s with 4D Laplacian phase unwrapping provides a favorable balance between minimizing aliasing and mitigating the need for dual-venc approaches or individualized venc investigation for every subject. While a more aggressive venc of 15–20 cm/s would provide improved VNR, the risk of aliasing increases and the ability of the 4D Laplacian method to compensate for them decreases. As such, the 4D Laplacian might function best as a safeguard method, to catch aliasing in edge case subjects where velocities might exceed a standard, moderate venc setting for an entire cohort.

Several limitations of this study should be acknowledged. First, the multiple venc scans were not acquired in a single acquisition as they might be in a dual or multi-venc acquisition. Thus, there was room for bulk motion between scans and a considerable number of subjects exhibited this issue. As such, images were not necessarily perfectly registered to each other which required application of image-registration methods and manually adjustment in some cases. However, image-registration methods are imperfect and may not result in a perfect alignment. For that reason, a full voxel-to-voxel comparison of the low and medium venc datasets to the reference high venc dataset was infeasible and there was some motion correction bias to the correlation plots (Figure 6.7). Additionally, the imperfect registration meant that applying dual-venc unwrapping algorithms to determine the true ground truth velocity values in the lower venc datasets was difficult to implement. Lastly, the scan time of even a single venc acquisition, at 12-minutes, is prohibitively long for a standard clinical session. For that reason, future work will aim to reduce scan time so that these 4D flow acquisitions are suitable for routine clinical screening practice, which we estimate to be at 5 minutes or less.

6.5. Conclusion

In conclusion, the 4D Laplacian phase unwrapping method provides effective correction of moderate velocity aliasing in 4D flow MRI of the portal venous system. Our data supports the choice of a venc of 30 cm/s as the optimal venc to capture the velocity spectrum encountered in subjects with portal hypertension. This venc setting is high enough for velocity aliasing to be correctable with the 4D Laplacian method but low enough to benefit from the increased VNR over higher venc settings.

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Chapter 7: Iterative Motion Correction for Improved Efficiency of Radial Free-Breathing 4D Flow MRI

In this chapter, I describe the implementation of an iterative motion correction (iMoCo) framework for 4D flow MRI of the portal venous system and demonstrate its feasibility in a pilot study. This approach is based on a previously developed method for pulmonary imaging [129] and has been shown to be effective in improving image quality via increased data efficiency for retrospectively-gated acquisitions. I aim to apply it to 4D flow MRI and investigate whether it is possible for this approach to enable further acceleration of 4D flow acquisitions. I perform this analysis in a reduced cohort of subjects introduced in Chapter 6 and present preliminary results.

7.1. Introduction

As discussed in Chapter 6, there is high interest in 4D flow MRI acquisitions for imaging the portal venous system, including in patients at risk for development of gastroesophageal varices (GEVs). However, the long scan times necessary to acquire images of a diagnostic quality limit the clinical usability of these sequences. Many previous studies have explored potential avenues for accelerating acquisitions including the use of advanced view sharing and filtering, constrained reconstruction techniques, and even deep learning reconstructions [130–135]. One promising, yet understudied and underutilized approach is to improve the efficiency of respiratory gating. As discussed in Section 2.2, retrospective respiratory gating is commonly used to mitigate motion artifacts in MRI and enable free-breathing acquisitions. However, this approach is highly inefficient as it only utilizes data acquired during a specific respiratory phase, typically end-exhalation according to a 50% threshold of the respiratory waveform. With 50% efficiency, acquisitions must be twice as long to acquire the same amount of data as a non-gated acquisition. A few prior studies have suggested that respiratory gating can be neglected in 4D flow acquisitions with minimal effects on hemodynamic results, even in regions affected by respiratory motion such as the aorta [136, 137]; however, Dyverfeldt et al. concluded that respiratory motion can severely degrade image quality and quantitative values if voxel sizes are not smaller than the degree of respiratory motion [138, 139]. As such there has been growing interest in more advanced methods of respiratory motion management in 4D flow imaging that can mitigate motion artifacts while also increasing data efficiency.

One such approach is the implementation of soft-gating methods that utilize data acquired across all respiratory phases and weight them according to the respiratory signal. By decreasing the weighting of data acquired in respiratory phases more distant from the reference phase (typically end-exhalation), data efficiency can be increased while reducing residual motion artifacts. This approach has been demonstrated in several recent studies and when combined with constrained reconstruction techniques, can enable considerable acceleration of 4D flow acquisitions [131, 133, 140]. However, this approach is limited by residual motion artifacts due to the inclusion of data from all respiratory phases. Motion Correction (MoCo) methods are a promising solution to this problem. By utilizing image-based registration, they can align multiple respiratory resolved images into a single image, increasing SNR, reducing motion artifacts, and thereby improving data efficiency. MoCo strategies have previously been applied to structural imaging applications [141, 142], and more recently to cardiac-resolved imaging of the coronary arteries [143, 144]. Recently, Zhu et al. developed an iterative MoCo (iMoCo) framework for pulmonary MRI [129] that utilizes motion-resolved imaging with an XD-GRASP style reconstruction [9], deformable image registration, and compressed sensing [30] to generate high-quality, motion compensated images. This method was shown to improve image quality and data efficiency in pulmonary MRI applications, suggesting its potential utility in other regions affected by respiratory motion such as the abdomen.

In this chapter, an iMoCo framework is adapted and implemented for 4D flow MRI of the portal venous system. The primary objective is to investigate whether a motion-compensated reconstruction can enable further acceleration of 4D flow acquisitions, thereby reducing the required scan time necessary to generate datasets suitable for hemodynamic characterization of the portal vein. The method is evaluated in a subset of subjects from the cohort described in Chapter 6 by retrospectively sub-sampling 12-minute acquisitions to simulate a 3-minute scan time and applying the iMoCo method to generate motion-compensated reconstructions. Image quality and motion artifacts are assessed in the iMoCo reconstructions compared to standard reconstructions with both the full 12-minute scan time and the simulated 3-minute scan time. Quantitative image quality metrics are calculated to compare between reconstructions, and statistical analysis is performed to evaluate whether the iMoCo method can compensate for a reduced scan time.

7.2. Materials & Methods

Iterative Motion Correction Framework

The iMoCo method described by Zhu et al. was implemented into a Python-based 4D flow reconstruction framework backed by the SigPy library (available at`https://github.com/uwmri/flow_recon/tree/imoco`) [55, 129]. It consists of three main steps: (1) a respiratory-resolved reconstruction that reconstructs R distinct respiratory phases, (2) deformable image registration of each motion phase to a reference phase to derive deformation fields, and (3) a motion-compensated reconstruction utilizing all acquired data and the derived deformation fields to reconstruct a single image at the reference phase.

In the first step, an XD-GRASP reconstruction was performed to generate N_r distinct motion states based on the respiratory signal [9] (Figure 7.1). Respiratory drift in the bellows signal was corrected for by identifying local minima and maxima to generate a signal envelope and identifying a drift curve from the baseline which was then subtracted from the original

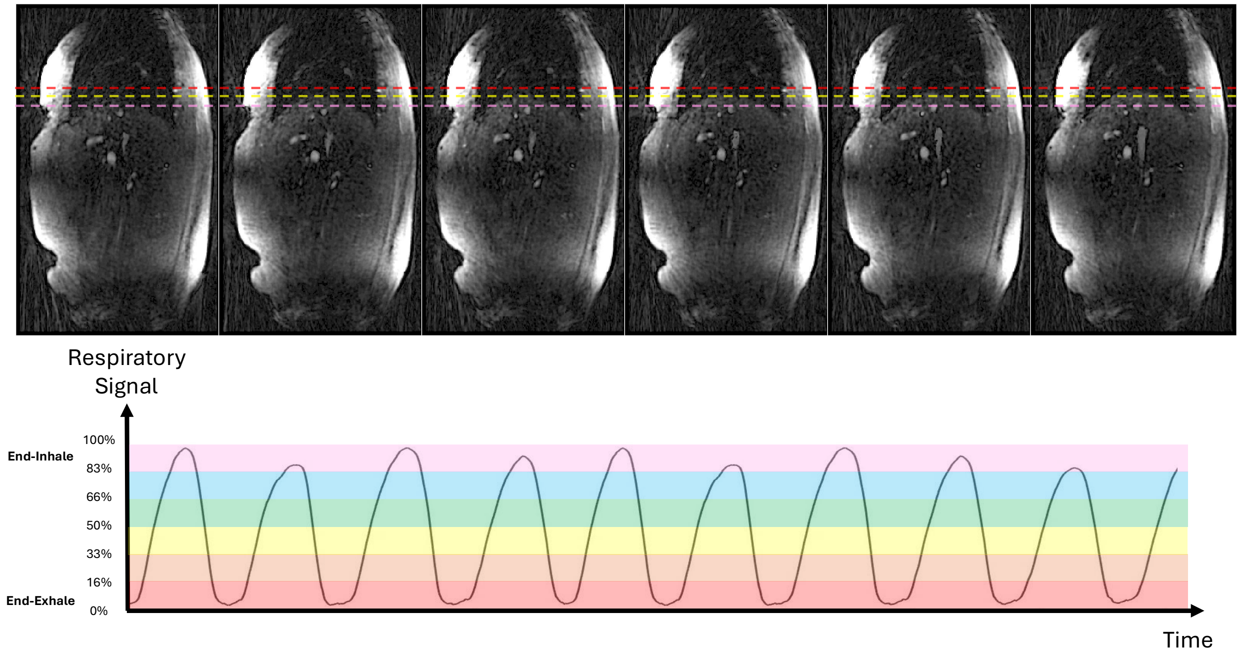

Figure 7.1: Respiratory motion-resolved reconstruction performed using an XD-GRASP style framework [9]. Data are binned into distinct phases based on the amplitude of the respiratory signal illustrated by colored bars (Bottom). For this study, 6 volumes representing the 6 respiratory phases were reconstructed, shown here as a sagittal slice through the right lung (Top). Dashed lines indicate position of the superior edge of the diaphragm at end-exhalation (Red), end-inhalation (Pink), and midway between (Yellow).

signal. A sliding threshold operation with a 10-second window was performed to bin data into R phases based on the amplitude of the bellows signal while ensuring that all bins contained an equal number of radial projections. Cardiac gating was not performed at this stage as only bulk motion of the abdomen was of interest. Each respiratory phase was then reconstructed separately using an iterative compressed sensing reconstruction according to Equation 7.1

$$\min_x \frac{1}{2} \sum_{r=0}^{R-1} \|DFSx_r - y_r\|_2^2 + \frac{\lambda}{2} \|Wx_r\|_1 \tag{7.1}$$

where D is the density compensation factor, F is the nonuniform Fourier transform, S are multichannel coil sensitivity maps generated by reconstructing low-pass filtered, low-resolution images, x_r is the estimated image for respiratory phase r, y_r is the corresponding acquired k-space data for phase r, λ is a regularization parameter, and W is a sparsifying wavelet transform with an L1 norm [30]. Gradient descent was used for 20 iterations to solve Equation 7.1 with a regularization parameter of $\lambda = 0.001$. Reconstructed images were then solved for velocity to generate magnitude, velocity, and complex difference images for each respiratory phase.

In the second step, deformable image registration was performed using the Advanced Normalization Tools (ANTs) software package to derive motion deformation fields [126]. Each respiratory phase was registered to a reference phase, selected to be end-exhalation, using the non-rigid Demons algorithm with the following parameters: 5 pyramid levels (coarse to fine) with a maximum of 200 iterations at the lowest level and a gradient step of 0.3 [145]. To emphasize the vascular structures, registration was performed on the complex difference images which are weighted to contain the most signal in the vessels. Thus, forward and inverse deformation fields describing the motion of the abdominal vessels in all 3 spatial dimensions were generated (Figure 7.2).

In the final step, an iterative motion-compensated reconstruction with compressed sensing was performed to reconstruct a single respiratory frame at end-exhalation [30]. This reconstruction is described by

$$\min_x \frac{1}{2} \sum_{r=0}^{R-1} \|DFSM_r x_0 - y_r\|_2^2 + \frac{\lambda}{2} \|W x_0\|_1 \tag{7.2}$$

where M_r is a motion warping operator that applies the previously derived deformation field for the respiratory phase r to the image at the reference phase x_0. Similar to the motion-resolved reconstruction, gradient descent was performed for 20 iterations to solve Equation 7.2 with a regularization parameter of $\lambda = 0.001$. The resulting motion-compensated reconstruction was then solved for velocity to generate magnitude, velocity, and complex difference images.

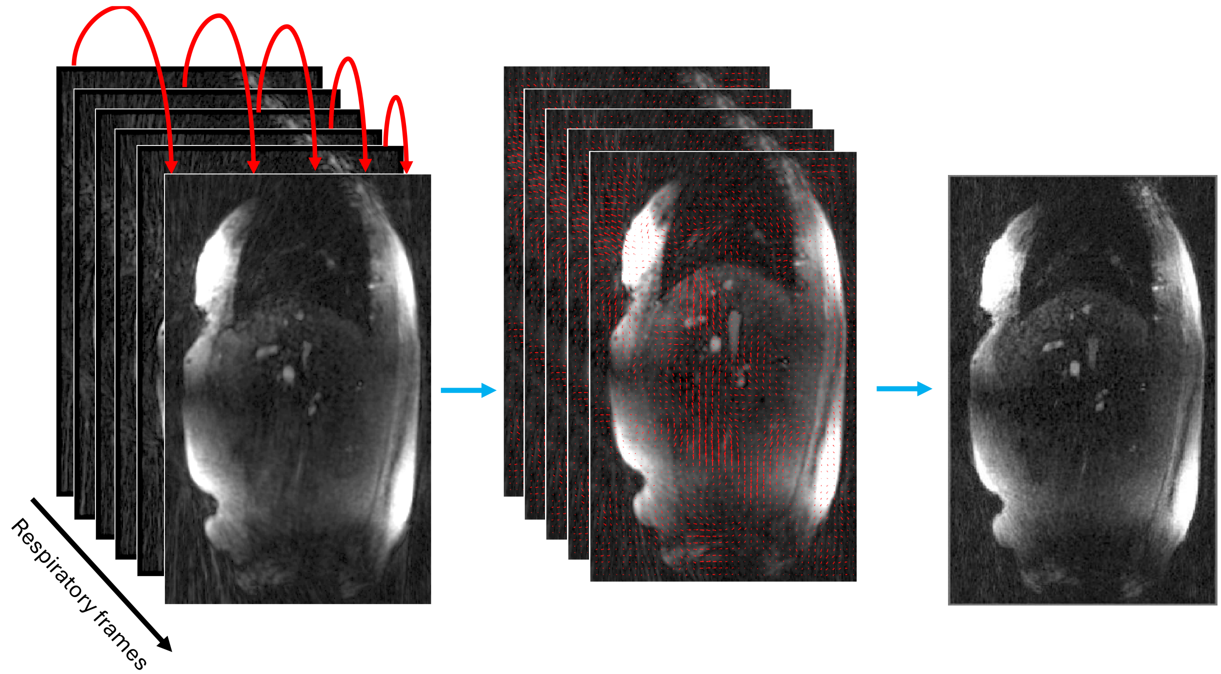

Figure 7.2: (Left) Motion resolved complex difference volumes are registered to a reference phase in end-exhale. (Center) 3D motion vector fields for x, y, and z directions are generated for each registration. (Right) Motion fields are incorporated into the iterative reconstruction to solve for a single volume at the reference phase of end-exhalation.

MR Imaging Protocol

This retrospective study was performed using data acquired from the same cohort utilized in Chapter 6. Full details of the study cohort and sequence parameters can be found in Section 6.2. In brief, free-breathing, radial 4D flow MRI data with 5-point velocity encoding [38] was acquired with three different velocity encoding (venc) parameters of 15 cm/s, 30 cm/s, and 60 cm/s. In this study, only the data with a venc of 30 cm/s were utilized as they were identified to provide the best combination of image quality and velocity-to-noise ratio (VNR) with minimal aliasing errors. Data were acquired with 18,000 radial projections with a scan time of approximately 12 minutes. Physiological signals were recorded during acquisition for retrospective gating, including electrocardiogram (ECG) and respiratory bellows signals.

Three sets of reconstructions were performed for each subject: (1) a standard CG-SENSE reconstruction matching the reconstructions performed in Chapter 6 using an exhalation respiratory threshold of 50% (designated Standard-12), (2) another standard exhalation reconstruction but retrospectively sub-sampled to simulate a reduced scan time of 3 minutes (Standard-3), and (3) a motion-compensated reconstruction with the simulated 3-minute scan time using the iMoCo method described above (iMoCo-3) [15]. For the iMoCo-3 reconstruction, $R = 6$ respiratory phases were utilized to strike a balance between minimizing streaking artifacts due to undersampling and minimizing residual motion from respiratory-binning. All reconstructions were performed on a high-performance computing cluster with an NVIDIA Tesla V100 GPU.

Analysis

To evaluate the effectiveness of the iMoCo method, image quality and motion artifacts were assessed in the iMoCo-3 reconstruction compared to the Standard-12 and Standard-3 reconstructions. Image quality was evaluated qualitatively by visual inspection of magnitude and complex difference images for residual motion artifacts. Quantitative analysis was performed by calculating several image quality metrics on the complex difference images in a region of interest in the portal vein and comparing between reconstructions. These metrics include: apparent signal-to-noise ratio (SNR), contrast-to-noise ratio (CNR), and vessel sharpness. SNR was calculated as

$$SNR = \frac{\bar{I}_{ROI}}{\sigma_{BG}} \tag{7.3}$$

where $\bar{I}$ is the mean signal intensity, σ is the standard deviation of the signal, and ROI and BG refer to the region of interest in the vessel and background, respectively. CNR was calculated as

$$CNR = \frac{\bar{I}_{ROI} - \bar{I}_{BG}}{\sigma_{BG}} \tag{7.4}$$

Vessel sharpness was assessed by calculating the relative maximum derivative (RMD) defined as

$$RMD = \frac{\max|\frac{dL(x)}{dx}|}{\bar{I}_{liver}} \tag{7.5}$$

where $L(x)$ a line intensity profile across the vessel-liver interface and These metrics were compared between reconstructions to evaluate whether the iMoCo method can improve image quality compared to a standard reconstruction as well as compensate for a reduced scan time. Statistical analysis was performed using paired t-tests between reconstructions to compare quantitative measurements with a significance level of $p < 0.05$.

7.3. Results

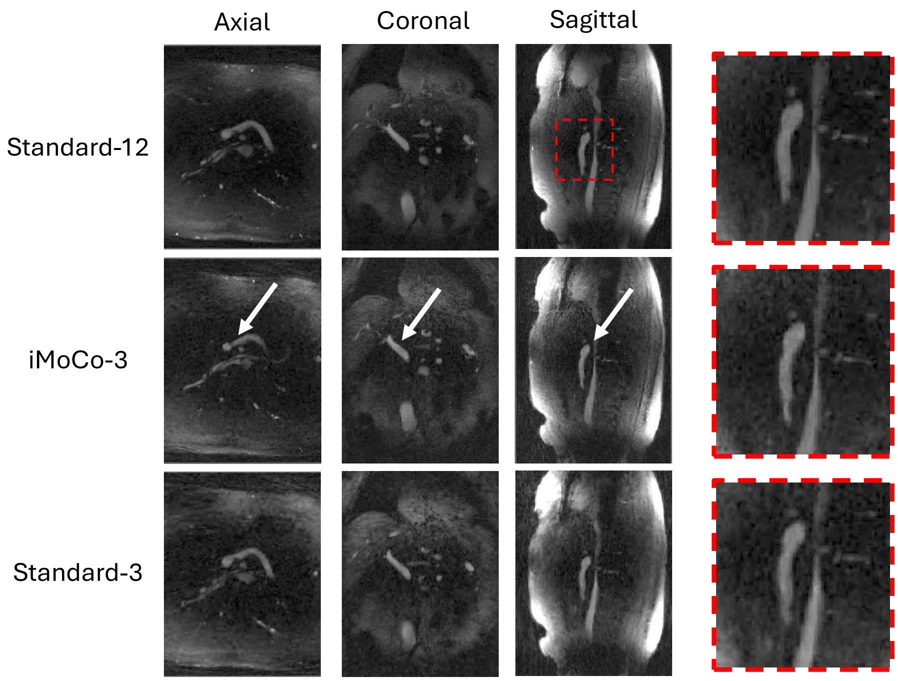

Figure 7.3: Representative complex difference angiogram images reconstructed with (Top) a standard end-exhalation reconstruction of a 12-minute acquisition with a 50% respiratory threshold, effectively using half of the acquired projections (Standard-12), (Middle) the iterative Motion Correction using only 3 minutes of acquired data, effectively using $1/3^{rd}$ of acquired data (iMoCo-3), (Bottom) and a standard end-exhalation reconstruction using only 3 minutes of acquired data with a 50% respiratory threshold, effectively using only $1/6^{th}$ of acquired data (Standard-3). White arrows indicate location of portal vein. Red dashed boxes are zoomed in on the sagittal view of the portal vein. Qualitatively, the Standard-12 reconstruction has the sharpest image but the iMoCo-3 reconstruction is similarly sharp. Standard-3 reconstruction exhibits blurred vessel edges and increased background noise.

Data for 5 subjects were successfully reconstructed with the Standard-12, Standard-3, and iMoCo-3 reconstructions. Figure 7.3 shows representative complex difference images for each reconstruction type in a representative subject. Qualitatively, the Standard-12 reconstruction has the highest image quality with clear delineation of the portal vein. The iMoCo-3 reconstruction has similar image quality to the Standard-12 reconstruction with some blurring at the lower superior mesenteric vein (SMV). The Standard-3 reconstruction has substantially reduced image quality with increased noise around the edges of the vessel lumen. These observations were consistent across the 5 subjects.

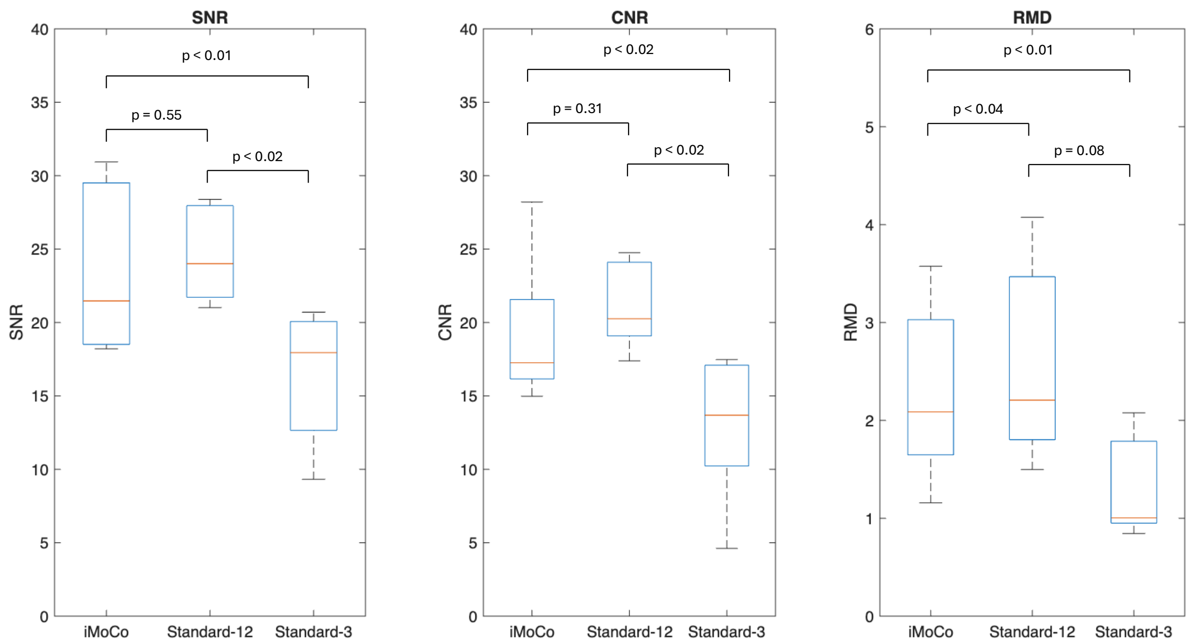

Figure 7.4: Box plots summarizing distributions of apparent signal-to-noise ratio (SNR), contrast-to-noise ratio (CNR), and relative maximum derivative (RMD) (calculated via maximum derivative of a line intensity plot across the vessel edge) for the iterative Motion Correction reconstruction using only 3 minutes of acquired data (iMoCo-3), the standard end-exhalation reconstruction with a 50% respiratory threshold using all 12 minutes of acquired data (Standard-12), and standard end-exhalation reconstruction with a 50% respiratory threshold using only 3 minutes of acquired data (Standard-3). Standard-12 has the highest mean SNR, CNR, and RMD indicating overall best image quality, followed closely by iMoCo-3, and then Standard-3.

The results of the quantitative SNR, CNR, and RMD distributions are shown in individual box plots in Figure 7.4. Standard-12 had the highest mean $\pm$ standard deviation for

SNR (24.6 ± 3.4), CNR (21.2 ± 3.1), and RMD (2.6 ± 1.1) indicating overall best image quality. It was followed closely by iMoCo-3 with SNR (23.7 ± 6.0), CNR (19.3 ± 5.2), and RMD (2.3 ± 0.9), and then Standard-3 SNR (16.3 ± 4.7), CNR (13.0 ± 5.2), and RMD (1.3 ± 0.5). Paired t-tests showed significant differences between iMoCo-3 and Standard-3 for all metrics. Significant differences were observed between Standard-12 and Standard-3 for SNR and CNR but not RMD. Lastly, significant differences were observed between iMoCo-3 and Standard-12 for RMD only.

7.4. Discussion

In this study, an iterative motion correction (iMoCo) framework was successfully adapted and applied to 4D flow MRI of the portal venous system, with the goal of improving data sampling efficiency and thus enabling accelerated acquisitions. The results demonstrate that a motion-compensated reconstruction using only 3 minutes of scan time (iMoCo-3) yields image quality that approaches that of a standard 12-minute reconstruction gated in end-exhalation with a 50% respiratory threshold (Standard-12), and outperforms a standard reconstruction with equivalent scan time also gated in end-exhalation (Standard-3).

Qualitative assessment indicates that the iMoCo-3 reconstructions preserved vessel delineation and sharpness with only minor residual blurring compared to Standard-12. In contrast, Standard-3 reconstructions exhibit pronounced residual blurring. These observations are supported by quantitative metrics, where iMoCo-3 demonstrates improved SNR, CNR, and vessel sharpness relative to Standard-3, and values that are comparable to Standard-12. Although not all comparisons reached statistical significance, likely due to the small sample size, the overall trends consistently favor the motion-compensated approach. It should also be noted that the reference motion state for the iMoCo reconstruction was based on a 16.67% threshold of the respiratory waveform ($1/6^{\text{th}}$ of the respiratory amplitude). A larger number of respiratory states R could have been utilized to further reduce residual motion within each bin, however, it was observed empirically that increasing R led to increased undersampling artifacts for each motion state, especially at a highly accelerated simulated 3-minute acquisition. The compressed sensing constraint utilized in this reconstruction for the time-resolved reconstruction step could not compensate for values of R greater than 6 and the resulting undersampling artifacts compromised the deformable registration step. However, there is potential here for alternative constrained reconstruction strategies to push the limits of R further. For example, a local low rank constraint could be applied to take advantage of both spatial and temporal redundancies in the data [32, 146].

While iMoCo approaches have been applied in other contexts, to our knowledge, this is the first application of this method to 4D flow MRI in the abdomen. One study by Andersen et al. applied a similar iMoCo framework to cerebrovascular angiography performed with navigator-gated Time of Flight (TOF) MRI and reported considerable image quality improvement [147]. Additionally, they tested various bulk motions such as head jerks and nodding and reported their iMoCo implementation was robust to these types of motion. In our study, only respiratory motion was corrected for, however, with long acquisitions, it is possible that other types of motion may occur and the iMoCo framework may be able to correct for these as well.

Despite these promising results, several limitations should be considered. First, this study was conducted on a small cohort of 5 subjects, limiting statistical power and generalizability. Second, the implemented reconstruction framework does not incorporate cardiac gating. A cardiac-resolved reconstruction was attempted during testing stages, but the further division of data into even smaller bins proved impossible for the constrained reconstruction to recover high quality images from. However, Ramasawmy et al. recently demonstrated a 4D iMoCo framework, performing a cardiac-resolved acquisition with respiratory motion correction [148]. In their approach, they utilized total variation terms in both spatial and temporal dimensions to regularize the reconstruction and were able to successfully reconstruct cine cardiac images. This approach could potentially be adapted in this framework, however, it would necessitate simultaneous solving of both cardiac and respiratory dimensions. Due to the large reconstructed matrix size and GPU memory limitations, this was not currently feasible for this study, but it is a possibility that should be explored in the future.

Lastly, flow analysis was not performed in these datasets due to time constraints. While the lack of a cardiac-resolved data suggested that flow analysis may not be possible, a study by Landgraf et al. showed that due to portal venous flow being non-pulsatile, flow analysis on a highly undersampled time-averaged dataset may be sufficient for hemodynamic assessment [149]. Thus, the iMoCo reconstruction framework implemented here could provide high-quality images for measurements of average flow and could be added to a comprehensive liver screening protocol.

7.5. Conclusion

In conclusion, this work demonstrates the feasibility of applying an iterative motion correction framework to 4D flow MRI of the portal venous system. The iMoCo approach enables improved data utilization by incorporating information from all respiratory phases while compensating for motion, resulting in image quality comparable to a standard 12-minute acquisition using only 3 minutes of data. These findings suggest that motion-compensated reconstruction has the potential to substantially reduce scan times in 4D flow MRI without substantially compromising image quality which could enable utilization of this sequence in standard clinical screening sessions for GEVs. Future work should aim to extend this analysis to the full cohort, perform comprehensive hemodynamic assessments, and extend the method to incorporate cardiac gating.

Acknowledgements

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Chapter 8: A Flexible 5D Cardiac MRI Reconstruction Framework for Motion Characterization in Radiation Therapy

In this final chapter, I shift focus again from flow-related acquisitions to explore motion assessment with radial, free-breathing MRI. I extend this approach to acquire and reconstruct five-dimensional images of the heart that resolve both cardiac and respiratory-induced motion. This work resulted in a poster presentation at ISMRM and 2 manuscripts, a first-author publication currently under review and a second-author publication recently accepted to *Physics of Medicine and Biology* [150].

8.1. Introduction

Radiation therapy (RT) is a widely used technique for both palliative and curative cancer treatment; however, it can also damage healthy tissue outside the treated target volume [151]. For thoracic cancers, the heart is a vital organ at risk for inadvertent radiation exposure that can result in complications such as congestive heart failure, pericardial effusion, and coronary artery disease [152–154]. Lung and esophageal cancer patients have shown high rates of radiation-induced cardiotoxicities (RICTs) [155]. Yet current radiation therapy practices do not account for the heart's complex anatomy when developing treatment plans. Current dose limits, such as QUANTEC recommendations that $<10\%$ of the heart receive > 25 Gy [156, 157], are based on single-organ metrics and do not account for local dose distributions among cardiac substructures and their specific radiosensitivities [155]; Recent evidence has emerged that links dose to cardiac substructures to RICTs and the development of future coronary events [155, 158, 159]. In particular, dose to the left ventricle has been associated with increased diastolic dysfunction [160]. Additionally, intra-fraction motion due to respiration has been shown to be substructure-specific and anisotropic [155, 161], yet it is currently not incorporated into standard treatment planning procedures. The incorporation of appropriate substructure-specific margins into radiation treatment planning would improve sparing of radiosensitive cardiac substructures and enable cardiorespiratory motion-robust planning. However, this is currently challenged by the limitations of radiation therapy imaging techniques in managing complex cardio-respiratory motion.

MRI is considered the gold standard for cardiac imaging due to its 3D capabilities and superior soft-tissue contrast to other modalities and is routinely used for evaluation of cardiac function and diagnosis of pathologies such as congenital heart disease and cardiomyopathy [162, 163]. While MRI enables comprehensive anatomical and functional imaging, standard exam times can be long, and volumetric coverage is typically achieved through serial 2D acquisitions performed over multiple breath-holds. However, recent advances in MR hardware, computational power, sequence design, and reconstruction techniques have enabled higher-dimensional cardiac imaging, progressing from 2D to 3D and to 4D (cardiac-resolved) and 5D (cardiac and respiratory-resolved) imaging [164–168]. Prior work in this field includes the development of frameworks such as XD-GRASP, which enables motion-resolved imaging across multiple dimensions and has been applied to 5D whole-heart cardiac functional imaging [9]. In this study, we implement and test a non-contrast-enhanced, flexible, retrospective cardiac MRI acquisition and XD-GRASP-style reconstruction framework to isolate and quantify cardiac substructure motion from the cardiac and respiratory cycles to facilitate development of improved cardiac substructure-sparing margins in radiation therapy treatment planning. Additionally, we explore the limits of accelerating the acquisition via constrained reconstruction techniques to reduce the scan time without compromising image quality.

8.2. Materials and Methods

MR Imaging Protocol

The central building block of our 5D MRI framework is a highly undersampled 3D radial balanced Steady State Free Precession (bSSFP) acquisition. BSSFP sequences are widely used in cardiac imaging due to their high SNR and superior blood-pool-to-myocardium contrast [169]. Additionally, radial sampling offers advantages for 5D acquisitions including: (1) motion robustness [7, 9], (2) flexible retrospective reconstructions [7, 170], and (3) strong support for constrained reconstruction [7, 8, 30]. Radial projections were acquired with a 2D bit-reversed view ordering, a pseudo-random method derived from the reversal of the binary representation of natural view ordering, to improve acquisition sparsity [8, 171]. Fractional echoes were used to acquire only 80% of each projection, shortening the TE and TR [172].

Ten healthy volunteers (9M/1F, 23–65 years old) were recruited for this study under a University of Wisconsin-Madison IRB approved protocol and provided written informed consent. One subject had an MR-conditional stent (Medtronic Resolute Onyx) in their left anterior descending artery (LADA) and was scanned in a reduced Specific Absorption Rate (SAR) mode. Subjects were scanned using a 40-channel chest coil on a 1.5T SIGNA Artist clinical simulator (GE Healthcare, Waukesha, WI). Scan parameters were optimized in feasibility experiments in human volunteers to maximize image contrast and spatial resolution while maintaining a large field of view and staying within allowable Specific Absorption Rate (SAR) limitations. The optimized parameters are as follows: TR = 3.4 ms, TE = 0.9 ms, flip angle = 35°, image volume = 40 cm × 40 cm × 40 cm, acquired spatial resolution = 1.56 mm isotropic. For each study subject, an acquisition with 250,000 radial projections was acquired over a 14.8-minute scan time. Cardiac and respiratory signals were recorded simultaneously during the acquisition using photoplethysmogram (PPG) and respiratory bellows, respectively, to enable dual gating.

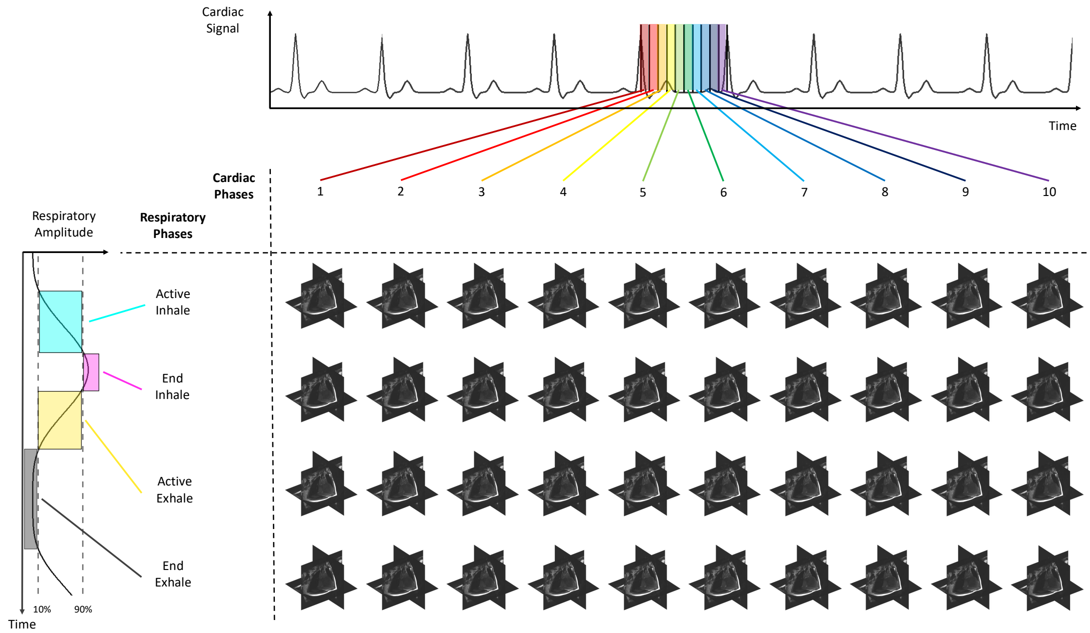

Figure 8.1: The retrospective 5D dual-gating process used to separate cardiac and respiratory motion states. (A) Data were acquired radially in a pseudo-random 2D bit-reverse view ordering. Radial projections were timestamped and matched with physiological waveforms (cardiac and respiratory) to sort accordingly. Here projections are colored to symbolize different respiratory phases and numbered according to cardiac phase. (B) Cardiac binning was performed by determining the mean R-R interval of the cardiac signal and dividing each R-R interval into 10 evenly spaced bins. Respiratory gating used a 5-second sliding window threshold to classify the top and bottom 10% of the respiratory signal into End Inhale and End Exhale. The remaining 80% were classified as Active Inhale or Active Exhale based on the slope of the waveform (positive and negative respectively). 10 cardiac phases and 4 respiratory phases were resolved for a total of 40 motion states.

Reconstruction Framework

A custom C++ reconstruction framework for radial reconstructions (available at: `https://github.com/uwmri/mri_recon`) was used for retrospective dual-gated binning (Figure 8.1). During scanning, each acquired radial projection was timestamped and matched with the respective points in the cardiac and respiratory waveforms (Figure 8.2). From the collected PPG signal, the mean R-R interval was calculated, and projections were divided into 10 cardiac phases of equal duration. The cardiac waveform was shifted by 300 ms to account for the transit-time delay introduced by measuring the cardiac signal at the left index finger instead of at the heart, resulting in a standard representation that starts shortly after the R-wave, just before end-diastole. Respiratory binning was performed using a 5-second sliding-window threshold (to account for respiratory drift), classifying the bellows waveform into 4 respiratory phases: active-exhale (AEx), end-exhale (EEx), active-inhale (AIn), and end-inhale (EIn). EIn and EEx are determined by taking the top and bottom 10% of the bellows waveform, while AIn and AEx utilize the middle 80% of the bellows waveform, differentiated by positive and negative slope, respectively.

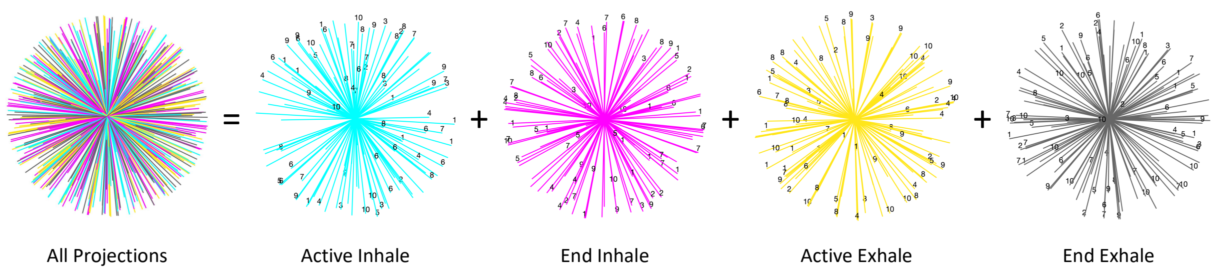

Figure 8.2: Data were acquired radially in a pseudo-random 2D bit-reverse view ordering to increase the incoherence of undersampling artifacts for constrained reconstruction. Projections were timestamped and matched with physiological waveforms (cardiac and respiratory) to bin accordingly. Here projections are colored to symbolize different respiratory phases and numbered according to cardiac phase.

Analysis

The acquired 250,000-projection dataset was retrospectively subsampled to 120,000, 80,000, and 40,000 projections to simulate scan durations of approximately 7, 5, and 2.5 minutes, respectively, to determine the trade-off between scan duration and image quality for substructure segmentation.

Three reconstructions were performed: (1) a standard 3D radial reconstruction utilizing parallel imaging [16] and iterative constrained reconstructions with either (2) compressed sensing (CS) [30] or (3) local low rank (LLR) [31, 146]. The CS regularization parameter was increased with increasing undersampling factor to mitigate overfitting, while the LLR regularization parameter remained constant. Iterative reconstructions utilized the Fast Iterative Shrinkage-Thresholding Algorithm (FISTA) and were run for a maximum of 100 iterations with an early stopping condition if the standard deviation of the data consistency error term fell below 0.01% for 5 iterations [173]. To identify the optimal reconstruction schema for motion analysis to use for all subjects in the study, the segmentation quality of each permutation of undersampling factor and reconstruction type was evaluated. This analysis was performed in a small, representative subset of subjects due to the substantial time investment segmentation of multiple reconstruction permutations for all 10 subjects would require. Segmentation of the left ventricle (LV) at end-systole in EEx was performed by a single observer in 3 subjects for 8 different reconstruction permutations. The Dice Similarity Coefficient (DSC) was calculated between LV segmentations across reconstructions, with the 250,000-projection LLR reconstruction serving as the baseline. Scan time and DSC performance were used to determine the final reconstruction schema, which was applied to all subsequent motion analyses for all study subjects.

For quantitative motion estimates, the LV was segmented at end-systole (ESys) and end-diastole (EDia) in all 4 respiratory phases for all subjects using the previously determined reconstruction type. Segmentations were performed by a single second observer in MIM Maestro (MIM Software, OH) and verified by a trained radiologist with 10+ years of experience. LV displacement due to cardiac motion was calculated from EEx, EDia to EEx, ESys, while LV displacement due to respiratory motion was calculated from EEx, EDia to EIn, EDia. Averages and standard deviations of displacements across the cardiac and respiratory cycles in each cardinal axis were calculated. Line pixel intensity measurements across the intraventricular septum were performed for all motion states in MATLAB.

8.3. Results

Datasets representing 40 motion states were successfully generated for each of the 10 volunteers. Figure 8.3 illustrates the simulated shortened scan times in an example subject and compares the effects of the two types of constrained reconstruction employed. The standard reconstruction demonstrated severe undersampling artifacts at all scan times, while the CS and LLR recons proved more robust at all scan times. However, spatial smoothing became more evident with decreasing scan time in the CS recons. Quantitative comparisons via DSC between the 250k projection LLR reconstruction and all other reconstructions are shown in Table 8.1. Based on visual assessment of the images and a demonstrated decrease in average DSC of only 1% with a 5-minute scan time from the 15-minute scan time, the 5-minute LLR reconstruction was chosen to be the final reconstruction approach, providing a favorable balance between scan duration and image quality. Average reconstruction time for the 5-minute LLR scan for all subjects was 6.2 ± 0.7 hours performed on a computing cluster with a 64-core Intel Xeon (Skylake) CPU.

Table 8.1: DICE coefficient scores for left ventricle (LV) segmentations at end-systole and end-exhale in multiple reconstruction schema. This preliminary analysis was performed in a representative subset of the full cohort (3 subjects) to determine an optimal reconstruction schema to use for the entire cohort for motion analysis. Eight reconstruction schemes were analyzed using the 250,000 projection Local Low Rank (LLR) reconstruction as a baseline. Compressed Sensing (CS) reconstructions show poorer agreement with the LLR reconstructions at all simulated scan times.

Reconstruction	**Projections**	**Scan Time (min)**	**Subj. 1**	**Subj. 2**	**Subj. 3**	**Average**
LLR	250,000	14.8	1.00	1.00	1.00	1.00
LLR	120,000	7.5	0.99	0.99	0.99	0.99
LLR	80,000	4.7	0.99	0.98	0.99	0.99
LLR	40,000	2.3	0.95	0.93	0.94	0.94
CS	250,000	14.8	0.98	0.97	0.97	0.97
CS	120,000	7.5	0.94	0.96	0.98	0.96
CS	80,000	4.7	0.77	0.75	0.65	0.72
CS	40,000	2.3	0.53	0.42	0.56	0.59

Figure 8.4 demonstrates qualitative motion assessments at the four chosen motion states for an example subject: End-Exhale, End-Systole (EEx, ESys); End-Exhale, End-Diastole (EEx, EDia); End-Inhale, End-Systole (EIn, ESys); and End-Inhale, End-Diastole (EIn, EDia). Difference images between ESys and EDia at EEx show intensity differences in the shape of the LV, indicating differentiation in cardiac phase. The differences between EEx and EIn are also apparent from the changes in the intensities of the diaphragm and the bottom edge of the right ventricle. Figure 8.5 illustrates motion profiles via pixel intensity measurements for another example subject through the (B) respiratory and (C) cardiac cycles. In the respiratory cycle, motion is most apparent during the AIn and AEx periods, compared to EEx and EIn, during which the septum moves most in the superior-inferior direction. AEx shows bulk motion in both the left and right sides of the heart, while AIn, EEx, and EIn show motion mainly in the left side of the heart. During the cardiac cycle, the widening dip in intensity around cardiac phase 4 corresponds to end-systole when the LV contracts and then narrows from phases 7–9 during diastole as the LV expands to fill with blood. Additionally, the left side of the heart shows contraction between respiratory phases that is not apparent on the right side, indicating anisotropic motion of different substructures.

Tables 8.2 and 8.3 show LV centroid displacements in each direction for each subject as well as the magnitude of the overall vector displacement for cardiac and respiratory motion, respectively. The quantitative motion analysis revealed prominent displacement patterns across subjects, with the largest LV displacements occurring in the superior-inferior direction (cardiac: -0.5 $\pm$ 0.7 mm, respiratory: -5.7 $\pm$ 1.3 mm), followed by the anterior-posterior (cardiac: 0.0 $\pm$ 0.3, respiratory: 1.3 $\pm$ 1.2), and right-left (cardiac: 0.2 $\pm$ 0.4 mm, respiratory: 1.6 $\pm$ 1.0 mm) directions. Figure 8.6 further visualizes the trajectories of LV centroid motion across the 4 respiratory states at both ESys and EDia, normalized to EEx and EDia. Respiratory trajectories and displacement magnitudes were shown to be highly subject specific.

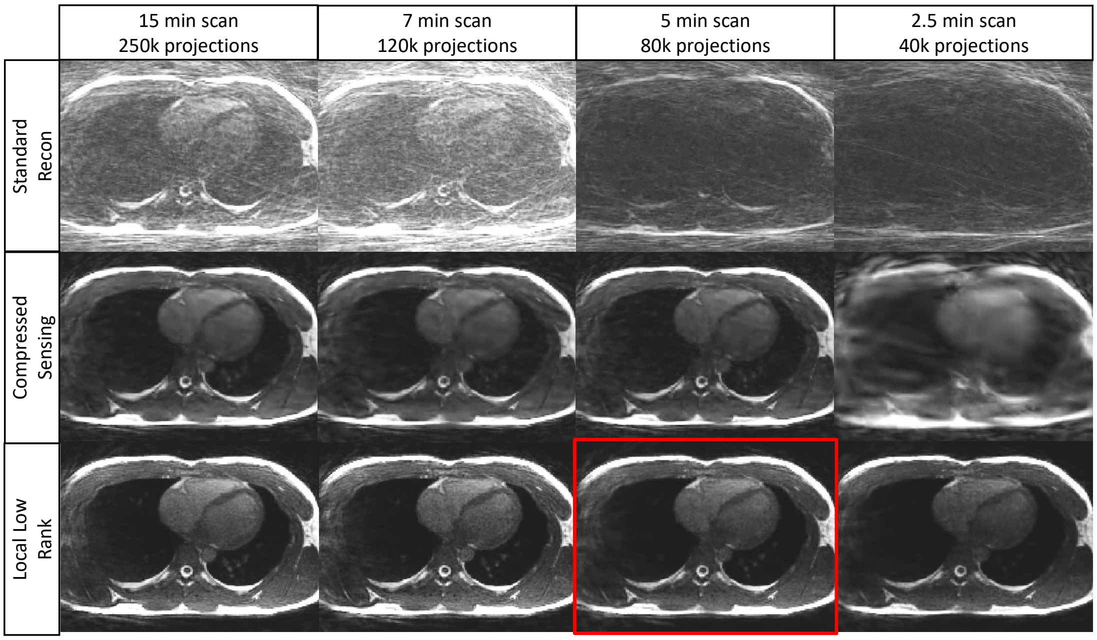

Figure 8.3: Representative images for a healthy volunteer reconstructed with decreasing numbers of radial projections simulating decreased scan time. Standard reconstruction with only parallel imaging (Top Row) leads to substantial undersampling artifacts in form of streaks, while compressed sensing (Center Row) leads to excess smoothing. The local low rank reconstruction (Bottom Row) provided the cleanest images, remaining robust even at 1/8th of the scan time. The red border indicates the reconstruction utilized for quantitative substructure motion analysis based on evaluation of segmentation quality.

8.4. Discussion

In this study, a flexible 5D cardiac MRI framework was demonstrated that combines dual cardiac and respiratory gating with highly undersampled 3D radial bSSFP acquisitions to isolate cardiac and respiratory motion of the heart. By leveraging a radial trajectory, retrospective binning, and iterative constrained reconstructions, we decoupled cardiac and respiratory motion and produced 40 distinct motion states per subject suitable for motion tracking within approximately 5 minutes of scan time. To our knowledge, this is the first study to utilize 5D cardiac MRI for motion assessment in radiation therapy. Respiratory-gated 4D CT is commonly utilized in radiation therapy for treatment planning and has previously been used for assessing respiration-induced motion of thoracic tumors [174, 175]. Other studies have

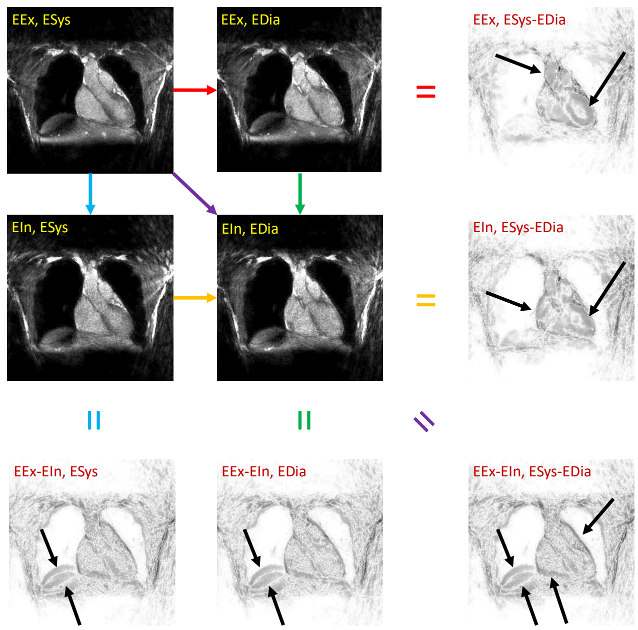

Figure 8.4: (Dark images) Representative images (central coronal slice) reconstructed with the 5D-MRI framework. Each image belongs to a unique cardiac and respiratory phase: End-Exhale (EEx), End-Inhale (EIn), End-Systole (ESys), End-Diastole (EDia). Colored arrows are difference operations that result in the bright difference images between two images on the same row, column or diagonal, e.g., (EEx-EIn, EDia) shows the difference map between (EEx, EDia) and (EIn, EDia). Black arrows indicate prominent cardiac and respiratory motion differences between phases.

utilized cardiac-gated 4D CT for analyzing cardiac-induced motion of the substructures [176, 177] or used image registration to apply motion fields derived from separate cardiac-resolved datasets to respiratory-resolved 4D CT for motion assessment [178–180], but our framework is the first to isolate cardiac and respiratory motion simultaneously from a single MR acquisition in the context of radiation therapy.

Exploration of 5D cardiac MRI acquisitions has been limited by the enormous data burden required for resolving two different types of motion. As the number of motion states to be resolved increases, the amount of data per motion state is decreased resulting in substantial reductions in image quality. As seen in Figure 8.3, a traditional 5D reconstruction, even with a clinically infeasible 15-minute scan time, is not viable due to the prominent streaking artifacts caused by undersampling. However, constrained reconstruction techniques enable

Table 8.2: Left ventricle centroid displacements for 10 subjects for cardiac motion in the R-L, A-P, S-I directions plus the magnitude of the overall displacement vector. Cardiac motion was defined as End-Exhale, End-Diastole → End-Exhale, End-Systole. Most substantial displacements were seen in the superior-inferior direction, followed by the anterior-posterior direction, and the right-left direction for both cardiac motion.

Subject	R-L (mm)	A-P (mm)	S-I (mm)	Magnitude (mm)
1	-0.3	-0.2	1.1	1.1
2	0.0	0.2	-0.6	0.6
3	-0.5	0.4	-0.8	1.0
4	0.1	-0.1	0.1	0.2
5	-0.2	-0.2	-0.3	0.4
6	0.1	0.6	-1.0	1.1
7	-0.5	0.0	-1.1	1.2
8	0.6	0.0	-0.6	0.8
9	0.4	0.4	-0.9	1.0
10	0.1	0.9	-1.1	1.4
Average	0.0 ± 0.3	0.2 ± 0.4	-0.5 ± 0.7	0.9 ± 0.4

reconstructing clinically acceptable images with much fewer samples than indicated by the Nyquist criterion by taking advantage of the inherent data redundancy in the acquired images. That proved sufficient for both CS and LLR reconstructions for longer scan times, yet the LLR reconstructions remained sharper at lower scan times than the CS reconstructions. This is likely due to the dynamic nature of the LLR reconstruction, which optimizes local regions of the dataset piecemeal across both space and time whereas the CS approach considers the whole volume for each time frame independently. As there is considerable information redundancy across time frames, it is straightforward to find a sufficiently low-rank basis to represent all the data. By leveraging the LLR reconstruction, we were able to reconstruct 40 motion states with sufficient image quality for substructure motion tracking within a clinically feasible scan time.

Regarding substructure motion assessment, the greatest LV displacement was observed in the superior-inferior direction for both cardiac and respiratory motion. This is consistent with prior studies assessing cardiac substructure displacement [177, 178]. However, both cardiac and respiratory motion exhibited clear anisotropy, with non-uniform displacement

Table 8.3: Left ventricle centroid displacements for 10 subjects for respiratory motion in the R-L, A-P, S-I directions plus the magnitude of the overall displacement vector. Respiratory motion was defined as End-Exhale, End-Diastole → End-Inhale, End-Diastole. Most substantial displacements were seen in the superior-inferior direction, followed by the anterior-posterior direction, and the right-left direction for respiratory motion.

Subject	**R-L (mm)**	**A-P (mm)**	**S-I (mm)**	**Magnitude (mm)**
1	0.3	0.9	-5.0	5.1
2	0.3	0.6	-7.5	7.5
3	3.5	2.7	-4.8	6.5
4	3.3	3.7	-7.4	8.9
5	1.6	1.4	-5.7	6.1
6	1.2	2.0	-7.0	7.4
7	0.4	0.6	-4.0	4.1
8	0.8	1.5	-4.2	4.5
9	0.6	1.5	-6.3	6.5
10	0.9	1.0	-5.3	5.4
Average	1.3 ± 1.2	1.6 ± 1.0	-5.7 ± 1.3	6.2 ± 1.5

across spatial directions and between phases of the motion cycles. While general trends (e.g., SI dominance of respiratory motion) were consistent across the cohort, the exact displacement vectors and motion trajectories differed substantially between subjects, as illustrated in Figure 8.6. This variability likely arises from differences in anatomy, breathing patterns, cardiac function, and thoracic geometry. In the context of radiation therapy, these results may suggest that developing subject and target-specific margins are necessary for sparing organs at risk.

Limitations

While our 5D datasets were sufficient for preliminary motion analysis and left ventricular segmentation, multiple complicating factors affected image contrast in this study. For one, these scans were acquired with a large volumetric FOV, which can precipitate washout effects leading to reduced signal from the saturated spins flowing out of the excited volume before readout [181]. While a large imaging FOV was desired to capture diaphragm and lung motion as needed for radiotherapy planning, reducing the excitation volume may improve

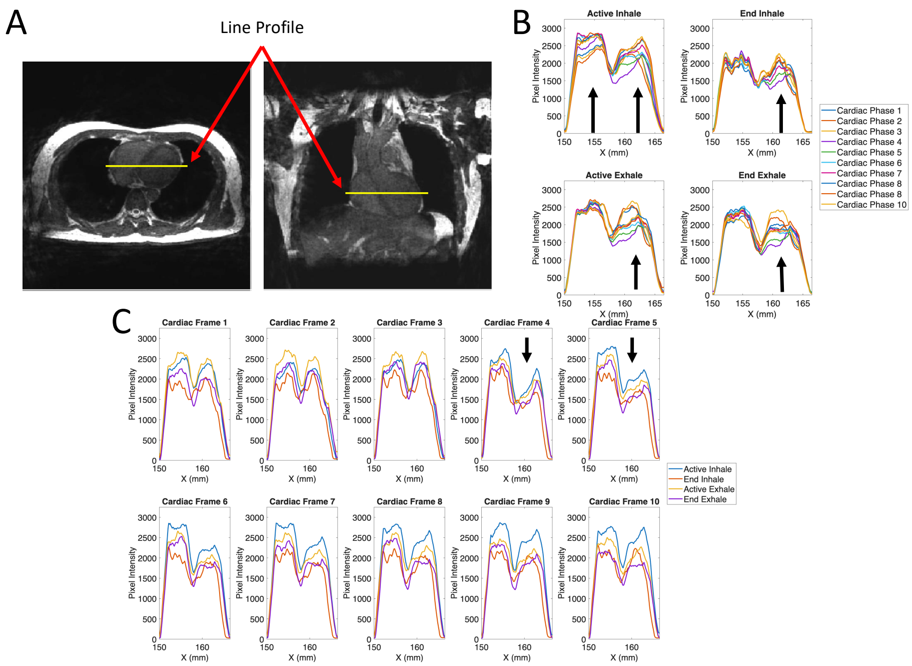

Figure 8.5: (A) Pixel intensities were measured along a line segment (yellow) through the intraventricular (IV) septum to analyze motion across the respiratory and cardiac cycles for a representative subject. Pixel intensity curves were smoothed with a Gaussian filter. The dip in the plateau of the pixel intensity corresponds to the reduction in signal intensity at the location of the IV septum. (B) Motion profiles of the heart through the respiratory cycle. Black arrows indicate bulk motion of the left and right sides of the heart across the cardiac cycle. (C) Motion profiles of the heart throughout the cardiac cycle. A reduction in intensity amplitude can be distinguished in cardiac frames 4 and 5 indicating end-systole.

blood-myocardium contrast through increased flow-related enhancement. A larger flip angle would also improve SNR and CNR; however, the short TR and frequent RF pulses in this bSSFP sequence limits the maximum allowable flip angle due to SAR limits. Lastly, bSSFP sequences are known to have bright fat signal, which can lead to it dominating the image contrast [169]. Fat saturation pulses are often used to combat this issue, however deploying fat-sat pulses necessitates repeatedly interrupting the steady state and then restoring it, which can substantially increase the acquisition time, especially in large acquisition volumes [7]. Also, alternative radial trajectories, such as spiral phyllotaxis, could improve acquisition efficiency by reducing eddy currents as well as enabling self-navigation, thereby eliminating the need for physical gating devices [182].

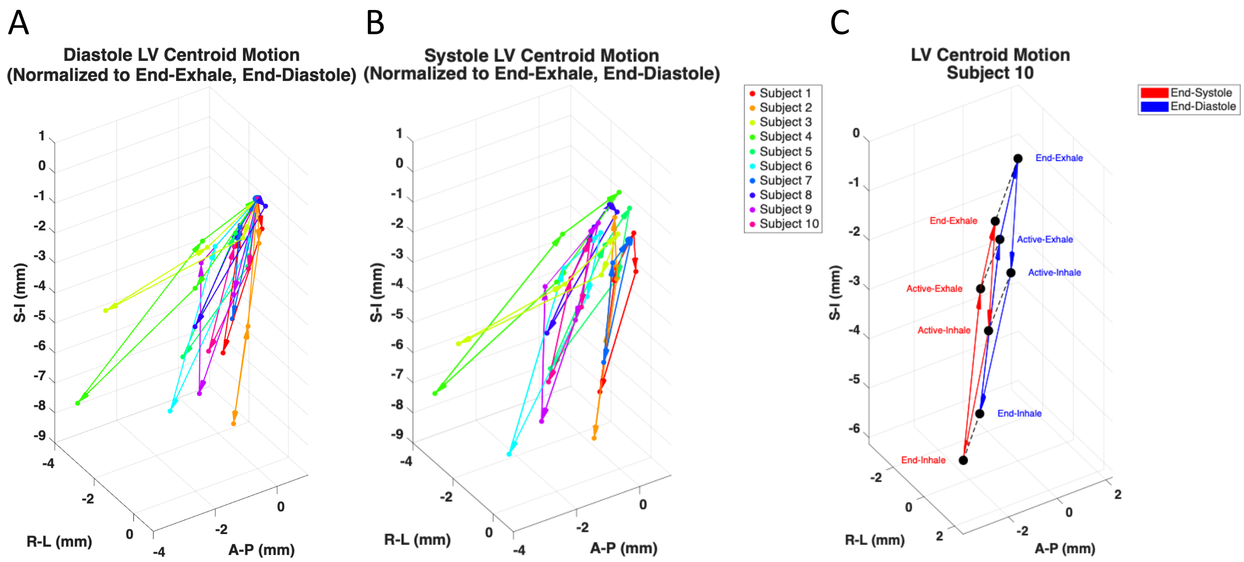

Figure 8.6: Centroid trajectories illustrating the motion of the left ventricle over the 4 respiratory phases. (A) and (B) show the trajectories for all subjects normalized to End-Exhale, End-Diastole for Diastole and Systole respectively while (C) shows all trajectories for an example subject. Respiratory trajectories vary considerably between subjects.

The 5D datasets were reconstructed with 10 cardiac and 4 respiratory phases. This is below the number of motion bins typically utilized in radiology for cine cardiac imaging (20 cardiac phases) and radiation oncology (up to 10 respiratory phases) for characterizing cardiac and respiratory motion respectively [183, 184]. While increasing the number of cardiac and respiratory phases would increase the temporal resolution in both dimensions and provide more detailed motion information, it would also increase the amount of reconstructed data, the computational demands of the reconstruction, and the time required for post-processing and analysis. In our setup, the 5D image reconstructions were computationally intensive, requiring an average of 6.2 $\pm$ 0.7 hours for all cardiac and respiratory frames. Additionally, memory constraints necessitated separate reconstructions of each respiratory phase per subject. A higher-performance compute system or GPU acceleration could improve reconstruction times and memory efficiency, allowing for simultaneous solving of both cardiac and respiratory frames and enforcement of regularization constraints in both dimensions to improve image quality. Furthermore, it could also enable a more systematic investigation of optimal regularization parameters for the constrained reconstructions, which were determined empirically.

Lastly, this study was performed with a limited cohort of predominantly healthy males. Only two subjects were over 50, and only one presented a complication in the form of a LADA stent. Patients with cancer in the thoracic region are likely to have more advanced age with potential complications such as arrhythmias or irregular breathing patterns, which could affect image quality. Further, studies will include expansion to a representative cohort for future applications to radiotherapy.

8.5. Conclusion

In conclusion, we presented a 5D MRI framework that resolves patient-specific cardiac and respiratory motion with a single 5-minute acquisition. The retrospective nature of the framework allows for flexible subsampling and reconstruction strategies, and constrained reconstructions improve image quality even at accelerated acquisition times. Image quality was sufficient for left ventricular segmentation and motion analysis. Future work will explore implementing fat saturation methods to improve image quality, as well as implement GPU acceleration to reduce the computational burden of image reconstruction.

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Chapter 9: Conclusion

9.1. Summary

In this dissertation, I optimized, developed, and implemented novel approaches for handling respiratory motion in cardiovascular MRI. Moving beyond just the mitigation of respiration-induced artifacts, I also investigated the utilization and characterization of respiratory motion in the context of multidimensional hemodynamic imaging. I performed this in three different clinical contexts, the measurement of pulse wave velocity in the aorta, the assessment of portal venous flow in the liver, and the characterization of substructure motion in the heart.

A major contribution of this work is the optimization and validation of radial, free-breathing 2D Phase Contrast (2DPC) MRI techniques for assessing aortic pulse wave velocity (PWV) [39]. Through development of an aortic flow phantom to simulate physiologically realistic flow, I established baseline comparisons between MRI-derived PWVs and ground-truth ultrasound values. These techniques were subsequently translated to a clinical research setting in the LIFE study, where PWV was assessed in a large cohort of elderly participants. While population-level agreement with established methods was observed, variability in paired measurements indicated methodological challenges in PWV estimation and the need for further improvement in PWV analysis. This work resulted in 2 first author abstracts [40, 53] and a first author manuscript currently under preparation.

Another contribution of this work was the extension of a radial simultaneous multi-slice (SMS) approach for more reliable, respiratory-resolved PWV imaging [39]. By increasing the number of acquirable slices and implementing iterative, constrained reconstructions, it became possible to measure PWV under more varied respiratory conditions, including free-breathing and metronome-paced tachypnea [96]. Additionally, the SMS approach improved data consistency due to the mitigation of heart rate variability from sequential acquisitions, enabling more reliable measurements. This work resulted in 3 first author abstracts [51, 52, 88].

In the next clinical context, I investigated 4D flow MRI of the portal venous system, exploring limitations related to velocity encoding and phase wrapping of velocity data, for acquisitions in patients with gastroesophageal varices (GEVs). A 4D Laplacian phase unwrapping method was shown to effectively correct moderate velocity aliasing, enabling more efficient use of 4D flow acquisitions without requiring individualized investigation and testing of optimal velocity encoding settings [121]. This work resulted in 2 first author abstracts [103, 104] and a first author manuscript currently under preparation. In parallel, I implemented an iterative motion correction (iMoCo) framework to increase data efficiency and demonstrated that high-quality reconstructions could be achieved using substantially reduced scan times by incorporating data across all respiratory phases [129].

Finally, I implemented a 5D MRI framework capable of resolving both cardiac and respiratory motion within a single free-breathing acquisition. By leveraging retrospective binning and constrained reconstruction techniques, this approach enabled a clinically feasible scan time while maintaining sufficient image quality for preliminary motion analysis and cardiac segmentation. This work resulted in 1 first author abstract [185], 2 second author abstracts [186, 187], a second author publication [150], and a first author manuscript currently under review.

9.2. Recommendations for Future Work

While the work presented in this dissertation represents substantial advancements in the handling of respiratory motion in cardiovascular MRI, there are several avenues for future research that can build upon these findings and further enhance the capabilities and clinical applicability of these techniques.

First, the observed variability between time-shift methods for PWV in Chapters 3 and 4 suggests that greater investigation into the precision and accuracy of these methods is needed. There is no standard consensus on the optimal time-shift algorithm to use for PWV calculation, so additional exploration is needed of each methods' reproducibility and repeatability. Moreover, alternative transit time methods such as the flow-area method could be explored [87].

Second, the respiratory-resolved imaging strategies developed in Chapter 5 should be expanded to more clinically relevant populations. The effects of metronome-paced tachypnea on PWV were investigated in healthy volunteers, however, clinical evidence suggests that tachypnea may have more pronounced effects on PWV in patients with underlying cardiovascular or pulmonary diseases [93]. Future studies should focus on the application of the developed SMS sequence to measure PWVs in patient populations, such as those with chronic obstructive pulmonary disease (COPD), to better understand the connection between respiratory dynamics and cardiovascular function.

Third, the acceleration of 4D flow MRI acquisitions remains an important priority for inclusion in clinical workflows. The demonstrated success of 4D Laplacian phase unwrapping and iterative motion-compensated reconstruction in 4D flow MRI suggests that a highly accelerated 3-minute acquisition with a pre-determined velocity encoding setting could be feasibly implemented in clinical practice for better screening of gastroesophageal varices [121, 129]. An ongoing patient study with 4D Flow and invasive measurements will provide further information on the clinical suitability of full duration and accelerated 4D Flow MRI scans in GEV risk assessment.

Lastly, for the developed 5D MRI framework, further improvements to the highly undersampled 3D radial sequence to bolster image quality could be made. This includes the incorporation of fat saturation techniques and self-navigation to enhance the robustness of acquisition and segmentation. Furthermore, implementation of GPU-accelerated reconstruction could reduce the computational burden of the 5D reconstruction. Another ongoing patient study utilizing the developed 5D framework could demonstrate the reliability of substructure motion estimates derived from 5D cardiac data in patients with thoracic cancers.

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